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JAMB Principles of Accounts: The Complete Course

Concepts, worked examples and exercises for the UTME

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Chapter 1: Foundations of Accounting and the Double-Entry System

Every business, from a petty trader in Ariaria Market to a multinational bank in Lagos, must be able to answer a few simple questions. How much money does the business have? How much does it owe, and how much is owed to it? Did it make a profit this year, and if so, how much? Accounting is the discipline that answers these questions in a reliable and orderly way. Without it the owner of a business would be guessing, lenders would not know whom to trust, and the tax authorities would have no basis for assessment.

This opening chapter lays the groundwork for the entire course. It explains what accounting is and who uses it, introduces the agreed principles (the concepts and conventions) on which all records are built, and then develops the accounting equation and the double-entry system. If you master the ideas in this chapter, every later topic, from the trial balance to company accounts, becomes an extension of the same logic. If you are uncertain here, the later chapters will feel like a collection of unrelated rules. Read slowly, and work every example with pencil and paper.

1.1 The nature and purpose of accounting

Accounting is the process of identifying, measuring, recording, classifying, summarising and communicating financial information about an organisation so that users can make informed decisions. It is often called the language of business because it converts thousands of separate transactions into a few statements that a reader can understand.

The word **bookkeeping** is sometimes used as a synonym, but the two are not identical. Bookkeeping is the routine, mechanical part of the work: recording each transaction, in the proper book and in the proper account, as it occurs. Accounting begins where bookkeeping ends. The accountant takes the recorded data, checks it, summarises it into a trial balance, prepares the final accounts and balance sheet, analyses the results and advises the owner. Bookkeeping can be learnt in weeks; accounting judgement takes years.

Objectives of accounting

- To keep a systematic and permanent record of all financial transactions.
- To determine the profit or loss of the business for a given period.
- To show the financial position of the business (what it owns and owes) at a given date.

- To provide information to owners, managers, lenders, the government and other interested parties.
- To help prevent and detect fraud and errors, because a proper record makes unusual items stand out.
- To provide evidence in disputes, for example when a customer disagrees with the amount he owes.

Users of accounting information

Different people read accounts for different reasons. **Owners and proprietors** want to know whether their investment is earning a satisfactory return. **Managers** need detailed figures to plan, control and take decisions. **Lenders**, such as banks, wish to know whether interest and capital will be repaid. **Suppliers** (creditors) want assurance that they will be paid on time. **Employees** are interested in job security and in the ability of the firm to pay wages and bonuses. **Government agencies**, including the Federal Inland Revenue Service and state revenue boards, use accounts to assess tax. **Prospective investors** and **customers**, and the general **public**, may also have an interest, especially in large companies.

1.2 Concepts and conventions of accounting

Accountants everywhere follow a set of agreed ideas so that the accounts of different businesses can be compared and trusted. The ideas that express the basic assumptions are called **concepts**; those that express widely accepted ways of dealing with uncertain matters are called **conventions**. In practice the words are often used interchangeably, and the syllabus does not penalise you for confusing them. What matters is that you can state each one and apply it to a short situation.

The main concepts

Business entity concept. The business is treated as a separate person from its owner. The owner's private money, house and car are not recorded in the business books. If the owner takes cash or goods for personal use, this is recorded as *drawings*, a reduction of what the business owes to the owner.

Going concern concept. It is assumed that the business will continue to operate for the foreseeable future and will not be sold or closed. For this reason, a delivery van is shown at its cost less depreciation, not at what it could fetch if sold tomorrow in a forced sale.

Money measurement concept. Only facts that can be expressed in money are recorded. The skill of the staff, the loyalty of customers and the quarrel between two managers are important, but they do not appear in the books because they cannot be given an objective naira value.

Historical cost concept. Assets are recorded at the price actually paid for them, not at their current market value. A shop bought for ₦8,000,000 in 2015 remains in the books at that cost (less any depreciation) even if it would now sell for much more. This makes the figures objective and verifiable by invoices.

Duality (dual aspect) concept. Every transaction has two effects, and both must be recorded. When a business buys a machine for cash, it gains a machine and loses cash. This concept is the foundation of double-entry bookkeeping.

Accounting period concept. The life of a business is divided into equal periods, usually one year, so that profit can be measured and reported regularly rather than only when the business closes.

Realisation concept. Revenue is recognised when goods are delivered or services are performed and the right to payment is established, not when the order is received or when the cash is collected. Revenue is therefore recorded at the time of sale.

Accruals (matching) concept. Expenses are matched with the revenue they helped to earn, whether or not cash was paid in that period. Rent owed for December is an expense of the year even if it is paid in January.

The main conventions

Prudence (conservatism). Profits are not anticipated, but losses are provided for as soon as they are foreseen. When in doubt about two possible values, choose the one that gives the lower profit or the lower asset value. Inventory is valued at the lower of cost and net realisable value for this reason.

Consistency. Once a method of accounting has been chosen, such as straight-line depreciation, it should be applied in the same way from year to year, so that results can be compared. A change is allowed only for good reason, and must be disclosed.

Materiality. Only items large enough to influence the decisions of users need to be shown separately. A box of staples bought for ₦500 is treated as an expense, not an asset, even though it will last for several months.

Objectivity. Records should be based on verifiable evidence, such as receipts, invoices and bank statements, and not on the personal opinion of the bookkeeper.

> **Key point.** In exam questions, the situation usually describes an action and asks which concept it breaks or supports. For example: "A trader includes his personal car among the assets of his business." This breaks the business entity concept. "A trader records a sale when the customer places an order but has not yet been supplied." This breaks the realisation concept.

1.3 Assets, liabilities and capital

All double-entry bookkeeping rests on three words that you must understand precisely.

An **asset** is a resource owned or controlled by the business, from which future economic benefit is expected. Assets are classified into two groups. **Non-current (fixed) assets** are held for long-term use in the business and not for resale: land, buildings, plant and machinery, motor vehicles, furniture and fittings. **Current assets** are cash or items expected to be turned into cash within a year: inventory (stock), trade receivables (debtors), prepaid expenses, bank balance and cash in hand.

A **liability** is an amount the business owes to outsiders. **Non-current liabilities** are repayable after more than one year, for example a bank loan of five years. **Current liabilities** are due within one year: trade payables (creditors), bank overdraft, accrued expenses and short-term loans.

Capital is the amount the owner has invested in the business, and it represents what the business owes to the owner. Because the business is a separate entity, it owes this amount to its owner just as it owes money to a supplier. Capital is also called **owner's equity** or **net worth**, and it is calculated as total assets minus total liabilities.

1.4 The accounting equation

The relationship among the three elements is expressed by the **accounting equation**:

Key point. Assets = Capital + Liabilities. Equivalently, Capital = Assets − Liabilities, and Liabilities = Assets − Capital.

The equation says that everything the business owns has been provided by someone: either by the owner (capital) or by outsiders (liabilities). The equation must balance after every transaction. If it does not, an error has been made. A transaction can change the equation in only a few ways: it can increase one asset and decrease another; increase an asset and a liability together; increase an asset and capital together; decrease an asset and a liability together; and so on. Total assets always equal total capital plus liabilities.

WORKED EXAMPLE 1.1

Mr Adewale Bakare starts a provisions shop in Ibadan and the following transactions occur. Show the effect of each on the accounting equation and state the position at the end.

1. He pays ₦500,000 cash into the business as capital.
2. He buys furniture for ₦80,000 cash.
3. He buys goods for ₦150,000 on credit from Alhaji Musa.
4. He sells goods which cost ₦60,000 for ₦90,000 cash.
5. He pays Alhaji Musa ₦100,000.

Solution:

Transaction	Cash (₦)	Furniture (₦)	Inventory (₦)	Creditors (₦)	Capital (₦)
1. Capital introduced	500,000	0	0	0	500,000
2. Furniture bought	−80,000	+80,000	0	0	0
3. Goods on credit	0	0	+150,000	+150,000	0
4. Goods sold	+90,000	0	−60,000	0	+30,000
5. Creditor paid	−100,000	0	0	−100,000	0
Balances	**410,000**	**80,000**	**90,000**	**50,000**	**530,000**

Check on the cash column: 500,000 − 80,000 + 90,000 − 100,000 = 410,000. Inventory: 150,000 − 60,000 = 90,000. Creditors: 150,000 − 100,000 = 50,000. Capital: 500,000 + 30,000 = 530,000 (the ₦30,000 is the profit on the sale, 90,000 − 60,000).

Total assets = 410,000 + 80,000 + 90,000 = ₦580,000. Capital plus liabilities = 530,000 + 50,000 = ₦580,000. The equation balances.

WORKED EXAMPLE 1.2

On 31 December, Mrs Funke Adeyemi's business has: land ₦2,000,000; equipment ₦600,000; inventory ₦350,000; trade receivables ₦150,000; cash ₦100,000; a bank loan (repayable in four years) ₦1,000,000; trade payables ₦400,000. Classify the items and find her capital.

Solution: Non-current assets: land ₦2,000,000 + equipment ₦600,000 = ₦2,600,000. Current assets: inventory 350,000 + receivables 150,000 + cash 100,000 = ₦600,000. Total assets = 2,600,000 + 600,000 = ₦3,200,000.

Non-current liability: bank loan ₦1,000,000. Current liability: trade payables ₦400,000. Total liabilities = ₦1,400,000.

Capital = Assets − Liabilities = 3,200,000 − 1,400,000 = **₦1,800,000**.

Drawings, profit and the changing capital

Capital does not stay constant. It is increased by further investment by the owner and by profit earned by the business. It is decreased by **drawings** (cash or goods taken by the owner for private use) and by losses. The relationship for a period is:

> **Key point.** Closing capital = Opening capital + Additional capital introduced + Net profit − Drawings (or + Net loss, if a loss was made).

Suppose a trader begins the year with capital of ₦800,000, introduces a further ₦200,000, earns a profit of ₦350,000 and draws ₦150,000. Closing capital is 800,000 + 200,000 + 350,000 − 150,000 = ₦1,200,000. Notice that drawings are not an expense of the business. They are a withdrawal of the owner's own investment and therefore do not enter the calculation of profit.

1.5 The double-entry system

The duality concept tells us that each transaction has two sides. The **double-entry system**, developed by Italian merchants and described by Luca Pacioli in 1494, turns this into a rule: *for every debit entry there must be a credit entry of equal amount*. The records are kept in **accounts**, one for every kind of asset, liability, capital, expense and income. A collection of accounts is called a **ledger**.

Each account has two sides. The left-hand side is the **debit** side (abbreviated Dr) and the right-hand side is the **credit** side (abbreviated Cr). These words do not mean "good" or "bad", and they do not mean that money is owed or received. They only indicate left and right. An account is traditionally drawn in the shape of the letter T, so it is sometimes called a **T-account**.

The rules of debit and credit

Type of account	To increase	To decrease	Normal balance
Asset (cash, inventory, equipment, receivables)	Debit	Credit	Debit
Expense (rent, wages, purchases)	Debit	Credit	Debit
Drawings	Debit	Credit	Debit
Liability (creditors, loans)	Credit	Debit	Credit
Capital	Credit	Debit	Credit
Income (sales, rent received, discount received)	Credit	Debit	Credit

These rules follow logically from the accounting equation. Assets stand on the left of the equation, so they increase on the left (debit). Capital and liabilities stand on the right, so they increase on the right (credit). Expenses reduce capital, so they are the opposite of capital: they are debited. Income increases capital, so it is credited. A memory aid used by many students is **"DEAD CLIC"**: Debit Expenses, Assets, Drawings; Credit Liabilities, Income, Capital.

The steps in analysing a transaction

1. Identify the two accounts affected.
2. Decide the type of each account (asset, liability, capital, expense or income).
3. Decide whether each account is increasing or decreasing.
4. Apply the rules to decide which account is debited and which is credited.

For example, the owner pays ₦400,000 into the business bank account. The accounts affected are Bank (an asset, increasing, so debit) and Capital (increasing, so credit). When rent of ₦30,000 is paid in cash, the accounts affected are Rent (an expense, increasing, so debit) and Cash (an asset, decreasing, so credit).

Key point. The account that *receives* value is debited; the account that *gives* value is credited. When goods are bought for cash, Purchases receives value (debit) and Cash gives value (credit).

1.6 Recording transactions in ledger accounts

In a ledger, each entry shows the date, the **details** (the name of the other account in which the double entry is completed) and the amount. When the account is balanced at the end of a period, the difference between the two sides is the **balance**. To balance an account, add up each side, find the difference, and insert it on the smaller side as **balance carried down (c/d)**, so that both totals agree. The same figure is then written on the opposite side beneath the totals as **balance brought down (b/d)**, the opening figure for the next period.

WORKED EXAMPLE 1.3

Mrs Ngozi Okafor opens a business in Onitsha. In January: (1 Jan) she pays ₦400,000 cash as capital; (3 Jan) buys goods for cash ₦150,000; (5 Jan) buys shop equipment on credit from Bello Supplies for ₦90,000; (8 Jan) sells goods for cash ₦200,000; (10 Jan) pays rent ₦30,000 in cash; (15 Jan) pays Bello Supplies ₦60,000 in cash. Write up the accounts, balance them and state the nature of each balance.

Solution: The entries are analysed first.

Date	Debit	Credit
1 Jan	Cash	Capital
3 Jan	Purchases	Cash
5 Jan	Equipment	Bello Supplies
8 Jan	Cash	Sales
10 Jan	Rent	Cash
15 Jan	Bello Supplies	Cash

The Cash account:

Debit			Credit		
Date	**Details**	**₦**	**Date**	**Details**	**₦**
1 Jan	Capital	400,000	3 Jan	Purchases	150,000
8 Jan	Sales	200,000	10 Jan	Rent	30,000
			15 Jan	Bello Supplies	60,000
			31 Jan	Balance c/d	360,000
	Total	**600,000**		**Total**	**600,000**
1 Feb	Balance b/d	360,000			

Check: debits 400,000 + 200,000 = 600,000. Credits before balancing 150,000 + 30,000 + 60,000 = 240,000. The balance is 600,000 − 240,000 = 360,000, a debit balance, which is correct for an asset.

The Bello Supplies account:

Debit			Credit		
Date	**Details**	**₦**	**Date**	**Details**	**₦**
15 Jan	Cash	60,000	5 Jan	Equipment	90,000
31 Jan	Balance c/d	30,000			
	Total	**90,000**		**Total**	**90,000**
			1 Feb	Balance b/d	30,000

The credit balance of ₦30,000 is a liability: the business still owes Bello Supplies ₦30,000. The other accounts have a single entry each and carry the following balances: Capital ₦400,000 credit; Purchases ₦150,000 debit; Equipment ₦90,000 debit; Sales ₦200,000 credit; Rent ₦30,000 debit.

Key point. If an account is balanced and the larger side is the debit, the balance is a *debit balance* (asset, expense or amount owed by a customer). If the larger side is the credit, it is a *credit balance* (liability, income or capital).

Personal, real and nominal accounts

Ledger accounts are also grouped by their nature. **Personal accounts** relate to persons or organisations: customers (debtors), suppliers (creditors), the owner (capital and drawings), banks and lenders. **Real accounts** relate to things the business owns: cash, inventory, equipment, land. **Nominal accounts** relate to expenses, losses, incomes and gains: wages, rent, sales, discount allowed. Real and nominal accounts are not about persons; nominal accounts are closed at the end of the year by transferring their balances to the trading and profit and loss accounts, whereas real and personal accounts are carried forward to the next period.

WORKED EXAMPLE 1.4

Mr Emeka Eze's account in the books of a wholesaler on 1 March shows that he owes ₦120,000. During March he buys further goods on credit for ₦80,000 and pays ₦150,000 by cash. What is the balance on his account and what does it mean?

Solution: Eze is a customer, so his account is an asset (a receivable). Debit side: opening balance 120,000 + sales on credit 80,000 = 200,000. Credit side: cash received 150,000. Balance = 200,000 − 150,000 = **₦50,000 debit**, meaning Mr Eze still owes the business ₦50,000. In the books of the wholesaler, the double entry for the cash received is Debit Cash ₦150,000; Credit Eze ₦150,000.

WORKED EXAMPLE 1.5

State the accounts to be debited and credited for each transaction: (a) Mr Hassan buys a motor van for ₦2,500,000 and pays by cheque; (b) the trader takes goods worth ₦25,000 for his family's use; (c) goods worth ₦70,000 are sold on credit to Mrs Bisi; (d) a loan of ₦500,000 is received from a bank into the business bank account.

Solution:

1. (a) Debit Motor Van ₦2,500,000 (asset increases); Credit Bank ₦2,500,000 (asset decreases).
2. (b) Debit Drawings ₦25,000 (owner's withdrawal); Credit Purchases ₦25,000 (the goods leave the business).
3. (c) Debit Bisi ₦70,000 (a receivable arises); Credit Sales ₦70,000 (income).
4. (d) Debit Bank ₦500,000 (asset increases); Credit Loan ₦500,000 (liability increases).

1.7 Extending the system: all transactions affect the equation

A student sometimes wonders how profit and loss fit into double entry. The answer is that income and expenses are simply temporary sub-divisions of capital. When the owner sells goods for more than they cost, capital increases; when the business pays rent, capital decreases. By recording income and expenses in separate accounts instead of changing the capital account every time, the owner can see *where* the profit came from. At the end of the year the totals are transferred to the profit and loss account, and the net profit is transferred to capital. This is the reason why the final accounts, studied in Chapter 5, can be derived directly from the ledger.

Exercise 1.1

1. A trader has total assets of ₦900,000 and total liabilities of ₦350,000. Calculate his capital.
2. The capital of a business is ₦700,000 and its liabilities are ₦260,000. What is the value of its total assets?
3. Which concept is broken if a trader records his friend's wedding gift of ₦50,000 as sales revenue of the business? Name the concept and explain.
4. Name the account to be debited and the account to be credited when a business buys machinery for ₦1,200,000 by paying cash.
5. State the effect on assets and liabilities of paying a creditor ₦40,000 in cash.
6. At the start of the year a trader's assets were ₦1,250,000 and his liabilities ₦450,000. During the year he made a profit of ₦300,000 and took drawings of ₦120,000, and introduced no new capital. Calculate his closing capital.
7. A cash account has debit entries of ₦800,000 and ₦150,000 and credit entries of ₦420,000 and ₦95,000. State the balance and whether it is debit or credit.
8. Mr Okoro buys goods for ₦60,000 on credit from Madam Chika. Name the account debited and the account credited in his books, and say what kind of balance appears on Chika's account.
9. Classify each of the following as non-current asset, current asset, non-current liability or current liability: (a) a five-year bank loan; (b) trade receivables; (c) delivery van; (d) bank overdraft.
10. A business begins with cash ₦300,000 and capital ₦300,000. It then buys goods for ₦100,000 cash and sells them for ₦140,000 cash. Show total assets and capital after the sale.
11. Explain why the going concern concept justifies showing a building at cost less depreciation instead of its break-up value.

Answers to Exercise 1.1

1. Capital = Assets − Liabilities = 900,000 − 350,000 = ₦550,000.
2. Assets = Capital + Liabilities = 700,000 + 260,000 = ₦960,000.
3. The business entity concept: a personal gift belongs to the owner, not to the business, and in any case it is not revenue from trading.

4. Debit Machinery ₦1,200,000; credit Cash ₦1,200,000.
5. Assets (cash) decrease by ₦40,000 and liabilities (creditors) decrease by ₦40,000; the equation still balances.
6. Opening capital = 1,250,000 − 450,000 = 800,000. Closing capital = 800,000 + 300,000 − 120,000 = ₦980,000.
7. Debits = 950,000; credits = 515,000; balance = 435,000, a debit balance.
8. Debit Purchases ₦60,000; credit Madam Chika's account. Her account shows a credit balance of ₦60,000 because it is a liability.
9. (a) Non-current liability; (b) current asset; (c) non-current asset; (d) current liability.
10. Cash after purchase = 200,000; after sale = 340,000; no other assets remain. Total assets = ₦340,000; capital = 300,000 + 40,000 profit = ₦340,000.
11. The business is expected to continue operating, so the building will be used rather than sold. Its break-up value is therefore irrelevant, and its cost is spread over its useful life through depreciation.

Chapter summary

- Accounting identifies, records, summarises and communicates financial information; bookkeeping is the recording stage within it.
- Users include owners, managers, lenders, suppliers, employees, government and the public, each with different needs.
- Key concepts are business entity, going concern, money measurement, historical cost, duality, accounting period, realisation and accruals. Key conventions are prudence, consistency, materiality and objectivity.
- Assets are what the business owns, liabilities are what it owes outsiders, and capital is what it owes the owner: Assets = Capital + Liabilities.
- Closing capital = opening capital + additional capital + profit − drawings.
- In double entry every transaction is recorded as a debit in one account and an equal credit in another. Assets, expenses and drawings increase by debit; liabilities, capital and income increase by credit.
- Accounts are balanced by totalling both sides, inserting the difference as balance c/d on the smaller side and balance b/d on the opposite side below the totals.

- Accounts are classed as personal, real or nominal; nominal accounts are closed to the final accounts at the end of the year.

Chapter 2: Source Documents, Books of Original Entry and the Ledgers

In Chapter 1 every transaction was written directly into ledger accounts. That is fine for a handful of entries, but a real business in Aba or Kano may record hundreds of sales and purchases in a single week. If each of them were posted one by one into the ledger, the ledger would become crowded, errors would be hard to trace, and one clerk could not easily share the work with another. Accountants solve this problem by recording transactions first in specialised books, called **books of original entry** or **books of prime entry**, and then transferring summarised totals into the ledgers.

This chapter follows a transaction from the moment a piece of paper is created to the moment it appears in a ledger account. You will learn the main source documents, the sales and purchases day books, the returns books, the general journal and the petty cash book, and how the ledgers are divided. The cash book, which is the most important of the books of original entry, has a chapter of its own (Chapter 3).

2.1 Source documents

No entry should enter the books without evidence. The paper (or electronic) evidence that a transaction has taken place is called a **source document**. The accountant, the auditor and the tax officer can all return to the document later and confirm that the entry is genuine. The most important source documents are listed below.

Document	Prepared by	Purpose	Recorded in
Invoice	Seller	Bill sent to a buyer for goods supplied on credit, showing quantity, price, trade discount and total	Sales day book (seller); purchases day book (buyer)
Credit note	Seller	Reduces the amount a customer owes, for example when goods are returned or overcharged	Sales returns book (seller); purchases returns book (buyer)
Debit note	Buyer	Sent to a supplier to inform them that the buyer is returning goods or claims an allowance	Purchases returns book (buyer)
Receipt	Person receiving money	Acknowledges that cash or a cheque has been received	Cash book
Cheque counterfoil / bank slip	Payer / bank	Evidence of payment out of the bank or deposit into it	Cash book
Petty cash voucher	Person spending the money	Authorises and describes a small cash payment	Petty cash book
Statement of account	Supplier	Periodic summary of invoices, credits and payments on an account	Used to check the ledger; not itself a book entry

An **invoice** deserves special attention. When a supplier sells goods on credit, the invoice states the list price of the goods, any **trade discount** and the net amount payable. A trade discount is a reduction from the catalogue price allowed to a trader who buys in bulk or who is a regular customer. It is deducted on the invoice itself and is *never entered in the ledger*, because it is a matter of fixing the selling price and not a separate transaction. Only the net amount after trade discount is recorded in the books.

> **Key point.** Trade discount is deducted on the invoice and is not recorded in the books. Cash discount (discount allowed and discount received) is an inducement to pay promptly and *is* recorded in the books; it is studied in Chapter 3. Do not confuse the two. If a question says "goods list price ₦80,000 less 10% trade discount", record ₦72,000.

2.2 The sales day book and the purchases day book

The **sales day book** (also called the sales journal) is a list, in date order, of all *credit* sales made by the business. Each line shows the date, the name of the customer, the invoice number and the net invoice amount. Cash sales are not entered here, because they go straight into the cash book. Likewise, the **purchases day book** lists all goods bought *on credit* for resale. It does not contain the purchase of assets such as machinery or vehicles, even on credit; those go through the general journal, because the day books are reserved for the goods in which the firm trades.

At the end of each week or month the invoice column is added to give a total. This total is posted to the ledger in one entry: the total of the sales day book is credited to the Sales account, and the total of the purchases day book is debited to the Purchases account. In addition, each individual invoice is posted to the personal account of the customer (debit) or supplier (credit). The total debits and credits therefore balance, and the double entry is complete.

The returns books

Goods are sometimes returned because they are faulty, damaged or not as ordered. When a customer returns goods to the business, these are **sales returns** (also called returns inward). They are recorded in the **sales returns book** from the credit notes issued. When the business returns goods to its own supplier, these are **purchases returns** (returns outward), recorded in the **purchases returns book**. The posting rules are as follows.

Book	Individual entries posted to	Total posted to
Sales day book	Debit the customer's personal account	Credit Sales account
Sales returns book	Credit the customer's personal account	Debit Sales Returns account
Purchases day book	Credit the supplier's personal account	Debit Purchases account
Purchases returns book	Debit the supplier's personal account	Credit Purchases Returns account

WORKED EXAMPLE 2.1

Mrs Chinwe Okoro runs a clothing wholesale business in Kano. During January the following credit sales were made, after which one return was received. Prepare the sales day book and sales returns book, post the entries to the ledger accounts and find the amount owed by each customer.

- 3 Jan: Adamu Bala, goods at list price ₦80,000 less 10% trade discount.
- 9 Jan: Chika Obi, goods at list price ₦50,000 less 20% trade discount.
- 17 Jan: Musa Danjuma, goods at list price ₦120,000 less 10% trade discount.
- 24 Jan: Adamu Bala, goods at list price ₦30,000 less 10% trade discount.
- 27 Jan: Chika Obi returns goods with a net invoice value of ₦8,000 (credit note issued).

Solution: First compute the net invoice amounts. Bala (3 Jan): 80,000 − 10% (8,000) = 72,000. Obi: 50,000 − 20% (10,000) = 40,000. Danjuma: 120,000 − 10% (12,000) = 108,000. Bala (24 Jan): 30,000 − 10% (3,000) = 27,000.

Sales day book

Date	Customer	Invoice no.	₦
3 Jan	Adamu Bala	001	72,000
9 Jan	Chika Obi	002	40,000
17 Jan	Musa Danjuma	003	108,000
24 Jan	Adamu Bala	004	27,000
	Total, to Sales account		**247,000**

Total: 72,000 + 40,000 + 108,000 + 27,000 = 247,000.

Sales returns book

Date	Customer	Credit note no.	₦
27 Jan	Chika Obi	C01	8,000
	Total, to Sales Returns account		**8,000**

Ledger postings. In the general ledger: Credit Sales account ₦247,000; Debit Sales Returns account ₦8,000. In the sales ledger, the personal accounts are:

Debit			Credit		
Date	**Adamu Bala**	**₦**	**Date**	**Details**	**₦**
3 Jan	Sales	72,000	31 Jan	Balance c/d	99,000
24 Jan	Sales	27,000			
	Total	**99,000**		**Total**	**99,000**

Obi: debit 40,000 (9 Jan), credit 8,000 (return), so Obi owes 40,000 − 8,000 = ₦32,000. Danjuma owes ₦108,000. Bala owes 72,000 + 27,000 = ₦99,000.

Check: the debtors' balances total 99,000 + 32,000 + 108,000 = 239,000, which equals net credit sales 247,000 − 8,000 returns = 239,000. The books agree.

WORKED EXAMPLE 2.2

Mr Tunde Adeyemi, a hardware dealer in Ibadan, records these credit purchases and returns in March. Prepare the purchases day book and returns outward book, post to the supplier accounts, and state the total owed to suppliers.

- 2 Mar: from Ibrahim Traders, goods ₦150,000 (net of trade discount).
- 11 Mar: from Eze & Sons, goods ₦90,000 (net).
- 20 Mar: from Ibrahim Traders, goods ₦60,000 (net).
- 25 Mar: returned damaged goods worth ₦10,000 to Eze & Sons.

Solution: Purchases day book total = 150,000 + 90,000 + 60,000 = ₦300,000, which is debited to the Purchases account. The returns outward book total is ₦10,000, credited to the Purchases Returns account. In the purchases ledger: Ibrahim Traders is credited 150,000 + 60,000 = ₦210,000, so Mr Adeyemi owes him ₦210,000. Eze & Sons is credited ₦90,000 and debited ₦10,000 for the return, a balance of 90,000 − 10,000 = ₦80,000 credit. Total owed to suppliers = 210,000 + 80,000 = **₦290,000**, which agrees with 300,000 − 10,000 = 290,000.

Exam tip. The words "returns inward" and "sales returns" mean the same thing, and so do "returns outward" and "purchases returns". Remember that *inward* goods come back to you from a customer; *outward* goods go back from you to a supplier. Sales returns are deducted from sales, and purchases returns are deducted from purchases, in the trading account.

2.3 The three divisions of the ledger

As a business grows, a single ledger book is no longer practical. The ledger is therefore divided into three parts, each of which can be handled by a different person.

- The **sales ledger** (or debtors' ledger) contains the personal accounts of customers who owe money to the business. Its balances are almost always debit balances.
- The **purchases ledger** (or creditors' ledger) contains the personal accounts of suppliers to whom the business owes money. Its balances are almost always credit balances.
- The **general ledger** (or nominal ledger) contains all the other accounts: real accounts such as equipment, nominal accounts such as rent and wages, and the capital and drawings accounts. It also holds the Sales, Purchases, Sales Returns and Purchases Returns accounts, which receive the totals from the day books.

The cash book is a book of original entry *and* is itself part of the ledger system, since it holds the Cash and Bank accounts. Dividing the ledger also supports control: if the sales ledger is kept by one clerk and the cash book by another, it becomes harder for one person to conceal a fraud. In Chapter 7 you will see how a control account checks the accuracy of an entire ledger division.

2.4 The general journal

Not every transaction fits the day books or the cash book. Transactions that are unusual or that occur infrequently are first recorded in the **general journal** (often called simply the journal). It is the book of original entry for:

- the purchase or sale of non-current assets on credit;
- opening entries when a business starts or when a new set of books is opened;
- the correction of errors (Chapter 4);
- bad debts written off;

- adjustments at the end of the year, such as depreciation and provisions;
- closing entries that transfer balances to the final accounts;
- transactions with the owner other than ordinary cash receipts, for example when the owner brings a personal laptop into the business as capital.

The journal has a standard layout. Each entry gives the date, the account to be debited (written first), the account to be credited (written below it and indented), the amounts in the debit and credit columns, and a short explanation called the **narration**. A line is usually drawn across after each entry to separate it from the next. The journal itself is not an account: after an entry has been journalised, it must still be posted to the ledger accounts named in it.

WORKED EXAMPLE 2.3

Write up the journal entries for the following transactions in the books of Mrs Funke Balogun, a caterer in Abeokuta.

- (a) 4 May: bought a delivery van on credit from Prime Motors for ₦3,200,000.
- (b) 9 May: Mr Ojo, a debtor, is declared bankrupt and his debt of ₦45,000 is to be written off.
- (c) 12 May: the owner brings a personal laptop valued at ₦280,000 into the business.
- (d) 20 May: the owner takes goods costing ₦35,000 for private use.

Solution:

Date	Particulars	Dr ₦	Cr ₦
4 May	Motor Van account (Dr)	3,200,000	
	To Prime Motors		3,200,000
	Being purchase of van on credit		
9 May	Bad Debts account (Dr)	45,000	
	To Ojo		45,000
	Being debt written off as irrecoverable		
12 May	Office Equipment account (Dr)	280,000	
	To Capital		280,000
	Being laptop introduced by the owner		
20 May	Drawings account (Dr)	35,000	
	To Purchases		35,000
	Being goods taken by the owner for personal use		

Reasoning: in (a) an asset increases (debit) and a liability to Prime Motors arises (credit). In (b) a loss is incurred (debit Bad Debts) and the asset owed by Ojo disappears (credit his account). In (c) an asset increases and the owner's capital rises. In (d) the owner's drawings rise (debit) and inventory leaves the business (credit Purchases at cost). Total debits and credits are both 3,200,000 + 45,000 + 280,000 + 35,000 = ₦3,560,000.

The opening journal entry

When a business starts its books, or when a set of books is opened for a business that has been trading without proper records, an **opening entry** records every asset and liability. Each asset is debited, each liability is credited, and the difference between the two is the owner's capital, which is credited. This is the accounting equation (Assets = Capital + Liabilities) written as a journal entry.

WORKED EXAMPLE 2.4

On 1 January Alhaji Sule Garba, a grain merchant in Kaduna, decides to keep proper books. His records show: cash in hand ₦120,000; balance at bank ₦650,000; inventory ₦480,000; equipment ₦900,000; Mr Yusuf owes him ₦150,000; he owes Mrs Ada ₦220,000; and he has a bank loan of ₦500,000. Prepare the opening journal entry and find his capital.

Solution: Total assets = 120,000 + 650,000 + 480,000 + 900,000 + 150,000 = ₦2,300,000. Total liabilities = 220,000 + 500,000 = ₦720,000. Capital = 2,300,000 − 720,000 = **₦1,580,000**.

Date	Particulars	Dr ₦	Cr ₦
1 Jan	Cash (Dr)	120,000	
	Bank (Dr)	650,000	
	Inventory (Dr)	480,000	
	Equipment (Dr)	900,000	
	Yusuf (Dr)	150,000	
	To Ada		220,000
	To Bank Loan		500,000
	To Capital		1,580,000
	Being assets, liabilities and capital at the start	**2,300,000**	**2,300,000**

2.5 Discounts and value added tax in the day books

Two further points arise frequently. First, trade discount, as already stated, is deducted on the invoice, and the day books show only the net amount. Second, in Nigeria many suppliers add **value added tax (VAT)** to the invoice at the standard rate of 7.5%. A business that is registered for VAT collects it on its sales on behalf of the government and may set off the VAT it paid on purchases. At UTME level the VAT is normally ignored unless a question gives it explicitly; if it does, the VAT is calculated on the net price after trade discount. For example, a net invoice price of ₦200,000 carries VAT of 7.5% × 200,000 = ₦15,000, so the invoice total is ₦215,000.

2.6 The petty cash book and the imprest system

Every business makes many small payments for postage, bus fares, stationery, cleaning materials and refreshments. Recording each in the main cash book would clutter it, so these are dealt with by a junior employee, the **petty cashier**, who keeps the **petty cash book**. Most firms use the **imprest system**. At the start of the period the cashier is given a fixed sum of money, the **float** or imprest. Each payment is supported by a petty cash voucher and entered in the book. At the end of the period (a week or a month) the cashier totals the spending, and the main cashier reimburses exactly the amount spent, so that the float is restored to its original size.

The petty cash book is ruled in columns. On the left is the receipts column, which shows the float and the reimbursement. Then come the date, details, a total payments column and several **analysis columns** in which each payment is also entered under its type of expense. The analysis columns are totalled at the end of the period, and each total is debited to the relevant expense account in the ledger, while the credit entry is made in the main cash book for the reimbursement. In this way the petty cash book is both a book of original entry and a part of the ledger.

WORKED EXAMPLE 2.5

The petty cashier of Okeke Stores, Enugu, starts the week with an imprest of ₦20,000. The payments for the week were: Monday, postage ₦2,500; Tuesday, taxi fare ₦3,000; Wednesday, stationery ₦4,500; Thursday, cleaning ₦2,000; Friday, postage ₦1,500; Saturday, taxi fare ₦2,500. Write up the petty cash book, balance it and show the reimbursement.

Solution: Total payments = 2,500 + 3,000 + 4,500 + 2,000 + 1,500 + 2,500 = ₦16,000. The analysis totals are postage 2,500 + 1,500 = 4,000; travel 3,000 + 2,500 = 5,500; stationery 4,500; cleaning 2,000. These add up to 4,000 + 5,500 + 4,500 + 2,000 = 16,000, agreeing with the total column. The cash remaining is 20,000 − 16,000 = ₦4,000, and the reimbursement needed is ₦16,000.

Receipts ₦	Day	Details	Total ₦	Postage ₦	Travel ₦	Stationery ₦	Cleaning ₦
20,000		Cash float					
	Mon	Postage	2,500	2,500			
	Tue	Taxi fare	3,000		3,000		
	Wed	Stationery	4,500			4,500	
	Thu	Cleaning	2,000				2,000
	Fri	Postage	1,500	1,500			
	Sat	Taxi fare	2,500		2,500		
		Totals	**16,000**	**4,000**	**5,500**	**4,500**	**2,000**
		Balance c/d	4,000				
20,000			20,000				
4,000		Balance b/d					
16,000		Cash received (reimbursement)					

In the ledger, Postage is debited ₦4,000, Travelling ₦5,500, Stationery ₦4,500 and Cleaning ₦2,000, with the credit of ₦16,000 in the main cash book. The float is back to 4,000 + 16,000 = ₦20,000.

Key point. Under the imprest system the amount reimbursed always equals the total spent in the period, and the opening balance of the next period always equals the original imprest. If the imprest is ₦20,000 and the cashier has ₦6,500 left, the reimbursement must be ₦13,500.

2.7 Posting: the full cycle from document to ledger

The sequence for a credit sale in a business that uses all the books described above is a useful summary. First, the seller prepares an invoice (the source document). Second, the invoice is entered in the sales day book. Third, each invoice amount is posted to the debit of the customer's account in the sales ledger. Fourth, at the end of the month the total of the sales day book is posted to the credit of the Sales account in the general ledger. Fifth, when the customer pays, a receipt is issued, the money is entered in the cash book, and the credit is posted to the customer's account. If the customer returns goods, a credit note is issued and the return is entered in the sales returns book, with posting to the credit of the customer and, at the end of the month, the debit of the Sales Returns account.

WORKED EXAMPLE 2.6

An invoice from a supplier shows goods with a list price of ₦250,000, trade discount 12% and VAT at 7.5% on the net price. Calculate the invoice total and the amount debited to the Purchases account (assuming VAT is recorded separately).

Solution: Trade discount = 12% × 250,000 = ₦30,000, so net price = 250,000 − 30,000 = ₦220,000. VAT = 7.5% × 220,000 = ₦16,500. Invoice total = 220,000 + 16,500 = **₦236,500**. The Purchases account is debited ₦220,000, the VAT account is debited ₦16,500 and the supplier is credited ₦236,500. Debits 220,000 + 16,500 = 236,500 = credits.

Exercise 2.1

1. Name the book of original entry in which each of the following is first recorded: (a) credit sale of goods; (b) purchase of a machine on credit; (c) return of goods by a customer; (d) payment of a small bus fare; (e) correction of an error.

2. A supplier's catalogue price for goods is ₦160,000. A trade discount of 15% is allowed. What amount is entered in the purchases day book?
3. State, for each of the following, the account debited and the account credited: (a) goods returned by Mr Bello to the business; (b) goods returned by the business to Eze & Sons.
4. The sales day book for a month shows invoices of ₦45,000, ₦72,500, ₦38,000 and ₦64,500. There was a sales return of ₦9,500. Calculate the total posted to the Sales account and the net increase in debtors.
5. A business buys office furniture on credit from Prime Offices for ₦540,000. Write the journal entry (without narration).
6. A petty cashier is given an imprest of ₦30,000. At the end of the week the vouchers total ₦21,800. How much cash should the main cashier give to restore the imprest, and how much cash does the petty cashier hold just before the reimbursement?
7. List the three divisions of the ledger and state which type of account each contains.
8. A trader's assets on 1 July are: cash ₦85,000, bank ₦420,000, inventory ₦310,000, equipment ₦640,000, debtors ₦95,000. His liabilities are creditors ₦180,000 and a loan of ₦300,000. Find his capital.
9. Purchases day book total ₦520,000; returns outward book total ₦35,000; sales day book total ₦880,000; returns inward book total ₦48,000. Compute net purchases and net sales.
10. Goods with list price ₦400,000 are bought with 20% trade discount and 7.5% VAT is added on the net price. Find the total on the invoice.
11. Explain why the purchase of a delivery van on credit is not recorded in the purchases day book.

Answers to Exercise 2.1

1. (a) Sales day book; (b) general journal; (c) sales returns book; (d) petty cash book; (e) general journal.
2. 15% × 160,000 = 24,000; 160,000 − 24,000 = ₦136,000.
3. (a) Debit Sales Returns, credit Bello's account; (b) Debit Eze & Sons' account, credit Purchases Returns.
4. 45,000 + 72,500 + 38,000 + 64,500 = ₦220,000 to Sales; net increase in debtors = 220,000 − 9,500 = ₦210,500.
5. Dr Furniture ₦540,000; Cr Prime Offices ₦540,000.

6. Reimbursement = ₦21,800; cash held before reimbursement = 30,000 − 21,800 = ₦8,200.
7. Sales ledger (customers' accounts), purchases ledger (suppliers' accounts), general ledger (all other accounts, real and nominal, including capital and drawings).
8. Assets = 85,000 + 420,000 + 310,000 + 640,000 + 95,000 = 1,550,000; liabilities = 180,000 + 300,000 = 480,000; capital = 1,550,000 − 480,000 = ₦1,070,000.
9. Net purchases = 520,000 − 35,000 = ₦485,000; net sales = 880,000 − 48,000 = ₦832,000.
10. Trade discount = 80,000; net = 320,000; VAT = 7.5% × 320,000 = 24,000; total = ₦344,000.
11. The purchases day book is only for goods bought for resale; a van is a non-current asset, so it is recorded in the general journal and debited to the Motor Van account.

Chapter summary

- Every entry rests on a source document: invoice, credit note, debit note, receipt, cheque counterfoil or petty cash voucher.
- Books of original entry are the sales, purchases and returns day books, the cash book, the petty cash book and the general journal.
- Trade discount is deducted on the invoice and never recorded in the books; only the net amount is entered.
- Day book totals are posted to the Sales, Purchases and returns accounts in the general ledger, and each invoice is posted individually to the customer's or supplier's account.
- The ledger is divided into the sales ledger, the purchases ledger and the general ledger.
- The general journal records non-routine items: credit purchases of assets, opening entries, bad debts, corrections and adjustments, each with a narration.
- The opening entry debits all assets, credits all liabilities and credits capital as the balancing figure.
- Under the imprest system the petty cashier receives a fixed float and is reimbursed exactly the amount spent; analysis columns are totalled and posted to expense accounts.
- VAT, where given, is calculated on the net price after trade discount.

Chapter 3: The Cash Book and Bank Reconciliation

Money is the lifeblood of a business. Customers pay in cash or by cheque and transfer; the owner pays suppliers, staff and landlords; excess money is banked and is sometimes withdrawn again. Because cash and bank transactions are so frequent and so important, they receive special treatment: they are recorded in a **cash book**, which is at once a book of original entry and the ledger account for cash and bank. Every other business record may be delayed for a day or two, but the cash book is kept up to date, entry by entry.

In this chapter you will learn how the cash book is ruled and written, how discounts are recorded in it, how it deals with cash paid into the bank and withdrawn from it, and, finally, how the business checks its own bank record against the statement sent by the bank. This last check, the **bank reconciliation statement**, is a favourite UTME topic, and the arithmetic must be handled with care.

3.1 Types of cash book

The cash book has the same two-sided layout as any ledger account. Receipts of money are entered on the debit side and payments on the credit side. There are three common forms.

- The **single-column cash book** has one money column on each side and records cash only. It is used by very small businesses that keep their money in a drawer and do not use a bank.
- The **two-column cash book** has a cash column and a bank column on each side. It is the normal form for any business that banks its money.
- The **three-column cash book** adds a third column, called the discount column, to each side. The debit side column records *discount allowed* and the credit side column records *discount received*.

When the cash book is a two-column or three-column book, the business keeps two accounts side by side in the same book: the Cash account, which represents the notes and coins physically held, and the Bank account, which represents the business's money in the bank current account. A debit balance means there is cash or money in the bank. A credit balance in the bank column means an **overdraft**: the business has paid out more than it has put in and owes the bank the

difference. The cash column can never be overdrawn, because a business cannot pay out notes and coins it does not have; a credit balance in the cash column therefore always points to an error.

3.2 Discounts

As Chapter 2 explained, **trade discount** is deducted on the invoice and is never recorded. **Cash discount** is different. It is a reduction in the amount due that a supplier allows if the customer pays within an agreed time, for example "2.5% if paid within 10 days". It is recorded in the books, because the business that receives the discount has really paid less than the invoice amount, and the business that grants it has really received less.

- **Discount allowed** is the discount a business gives to its customers (debtors) for prompt payment. It is an *expense* of the business, since it reduces the money received.
- **Discount received** is the discount a business obtains from its suppliers (creditors) for prompt payment. It is an *income* of the business, since it reduces the money paid.

Suppose Mrs Bisi owes ₦80,000 and pays ₦78,000 in settlement. The business debits the Bank account ₦78,000 and the Discount Allowed column ₦2,000, and credits Bisi's account with the full ₦80,000 (the cash ₦78,000 and the discount ₦2,000, which is entered as two items). Bisi's account is then completely cleared. In the same way, if the business owes Mr Chidi ₦100,000 and pays ₦95,000 by cheque, it records credit Bank ₦95,000 and credit Discount Received ₦5,000, and debits Chidi's account ₦100,000.

> **Key point.** The discount columns are *memorandum columns*. They are not part of the cash balance. Their totals are posted at the end of the period: the total of the debit-side column is **debited** to the Discount Allowed account, and the total of the credit-side column is **credited** to the Discount Received account. Never add the discount column into the cash or bank balances. Each discount is also posted individually: a discount allowed is credited to the customer's account, and a discount received is debited to the supplier's account.

A useful way to remember: the discount appears on the *same side* of the cash book as the money it accompanies, but is posted to the *opposite* side in the personal account. A discount allowed sits on the debit side of the cash book (together with the receipt) but is credited to the debtor, because the debtor's account needs a credit to cancel the debt.

3.3 Contra entries

Sometimes a transaction involves the cash and bank columns both, because money moves from one to the other. Such an entry is called a **contra entry** (from the Latin for "against") and is marked with the letter **C** in the folio column. The whole double entry is made inside the cash book itself, with no posting to any other ledger.

- When cash is paid into the bank: debit Bank, credit Cash. The amount appears on the debit side of the bank column and on the credit side of the cash column.
- When cash is withdrawn from the bank for use in the office: debit Cash, credit Bank. The amount appears on the debit side of the cash column and on the credit side of the bank column.

WORKED EXAMPLE 3.1

Mr Kunle Adebayo is a provision seller in Ibadan. On 1 June he had cash ₦50,000 and a bank balance of ₦400,000. During June: (3 Jun) he received a cheque for ₦78,000 from Bisi, who owed him ₦80,000, and allowed her a discount of ₦2,000; (5 Jun) he paid Chidi by cheque ₦95,000 in full settlement of ₦100,000; (8 Jun) cash sales ₦60,000; (10 Jun) paid wages in cash ₦25,000; (12 Jun) banked ₦40,000 cash; (15 Jun) withdrew ₦30,000 from the bank for office use; (20 Jun) paid rent by cheque ₦50,000. Write up the three-column cash book and balance it.

Solution: Cash column: receipts 50,000 + 60,000 + 30,000 = 140,000. Payments 25,000 + 40,000 = 65,000. Balance = 140,000 − 65,000 = 75,000. Bank column: receipts 400,000 + 78,000 + 40,000 = 518,000. Payments 95,000 + 30,000 + 50,000 = 175,000. Balance = 518,000 − 175,000 = 343,000.

Debit					**Credit**				
Date	**Details**	**Disc. ₦**	**Cash ₦**	**Bank ₦**	**Date**	**Details**	**Disc. ₦**	**Cash ₦**	**Bank ₦**
1 Jun	Balances b/d		50,000	400,000	5 Jun	Chidi	5,000		95,000
3 Jun	Bisi	2,000		78,000	10 Jun	Wages		25,000	
8 Jun	Sales		60,000		12 Jun	Bank (C)		40,000	
12 Jun	Cash (C)			40,000	15 Jun	Cash (C)			30,000
15 Jun	Bank (C)		30,000		20 Jun	Rent			50,000
					30 Jun	Balances c/d		75,000	343,000
	Total	**2,000**	**140,000**	**518,000**		**Total**	**5,000**	**140,000**	**518,000**
1 Jul	Balances b/d		75,000	343,000					

The discount totals are posted: debit Discount Allowed ₦2,000; credit Discount Received ₦5,000. In the personal accounts, Bisi is credited ₦78,000 (bank) and ₦2,000 (discount), clearing her ₦80,000 debt; Chidi is debited ₦95,000 and ₦5,000, clearing his ₦100,000 account. The net effect on profit of the two discounts is a gain of 5,000 − 2,000 = ₦3,000.

Calculating a cash discount

WORKED EXAMPLE 3.2

Alhaji Musa sells goods worth ₦200,000 to a customer on terms of 2.5% cash discount if payment is made within 10 days. The customer pays on the eighth day. How much is received, and how is it recorded?

Solution: Discount = 2.5% × 200,000 = ₦5,000. Amount paid = 200,000 − 5,000 = ₦195,000. In the cash book: debit Bank ₦195,000 and debit Discount Allowed column ₦5,000 (together ₦200,000); credit the customer's account ₦195,000 (bank) and ₦5,000 (discount), a total of ₦200,000, which clears the debt.

Other items that pass through the bank

Besides ordinary receipts and payments, the bank account is affected by several other items that the business may not know about until the bank statement arrives:

- **Bank charges**: fees for maintaining the account, transfers, card and SMS alerts. They are debited to the Bank Charges account and credited to the bank column.
- **Interest**: interest charged on an overdraft (a payment) or credited on a deposit (a receipt).
- **Standing orders**: fixed payments the business instructs the bank to make regularly, for example insurance premiums or rent, until it cancels the instruction.
- **Direct debits**: payments that a creditor collects from the account, usually for varying amounts.
- **Direct credits (transfers)**: customers pay directly into the business account, for example by bank transfer, and the business learns of it only from the statement.
- **Dishonoured cheques**: a cheque received from a customer and banked may be returned unpaid because the drawer has insufficient funds. The bank deducts the amount from the account. The business credits Bank and debits the customer, who owes the money again.

3.4 Why the cash book and the bank statement differ

At intervals the bank sends the business a **bank statement**, a copy of the business's account in the bank's own books. Because the same money is recorded twice, once in the business's cash book and once by the bank, the two balances ought to agree. Very often they do not. The reasons are of two kinds, and the distinction decides what must be done about each.

Timing differences

These are items correctly recorded in the cash book that the bank has not yet processed. They will correct themselves in a few days, and no entry is made in the cash book.

- **Unpresented cheques** (outstanding cheques): cheques the business has written and entered in its cash book, but which the payee has not yet banked or the bank has not yet cleared. The cash book has already reduced the balance; the bank has not.
- **Uncredited deposits** (lodgements not yet credited): money paid into the bank, perhaps on the last day of the month, that is in the cash book but has not yet appeared on the statement.

Items not yet in the cash book

These are items the bank has recorded that the business has not yet entered in the cash book: bank charges, interest, standing orders, direct debits, direct credits and dishonoured cheques. They must now be written into the cash book. There may also be **errors** in the cash book, such as a cheque entered with its digits transposed, which must also be corrected in the cash book.

> **Key point.** The bank statement is written from the bank's point of view. If the business has ₦300,000 in its account, the bank *owes* the business that amount, so the statement shows a *credit* balance. This is the opposite of the cash book, where the same money is a debit balance. If the cash book has a debit balance (money in the bank), the statement should show a credit balance, and if the business is overdrawn, the cash book shows a credit balance while the statement shows a debit balance.

3.5 The procedure for bank reconciliation

The reconciliation proceeds in two steps.

1. **Update the cash book.** Compare the cash book with the bank statement entry by entry. Items on the statement that are missing from the cash book (charges, interest, standing orders, direct credits, dishonoured cheques) and any cash book errors are entered in the cash book. This gives the *adjusted cash book balance*, which is the true figure that goes into the balance sheet.
2. **Prepare the reconciliation statement.** Start from the balance shown by the bank statement. Add the uncredited deposits. Deduct the unpresented cheques. The result must equal the adjusted cash book balance.

The reconciliation can also be started from the cash book, in which case the signs are reversed: begin with the adjusted cash book balance, add the unpresented cheques, deduct the uncredited deposits, and the result is the statement balance. Both directions are acceptable; the choice depends on what the question asks.

The logic is simple. The bank statement lags behind the cash book. Cheques not yet presented have already been deducted by the business, so, to reach the business's figure from the bank's, they must be taken off. Deposits not yet credited have already been added by the business, so they must be added to the bank's figure.

WORKED EXAMPLE 3.3

On 31 March the bank column of Mrs Tolani Fashola's cash book shows a debit balance of ₦286,000. Her bank statement shows a credit balance of ₦340,000. On comparison she finds: cheques no. 0214 for ₦45,000 and no. 0216 for ₦32,000 have not been presented; a lodgement of ₦28,000 made on 31 March does not yet appear on the statement; bank charges of ₦3,000 and a standing order for insurance of ₦12,000 appear on the statement but not in the cash book; a customer paid ₦20,000 directly into the account and this appears only on the statement. Update the cash book and prepare a bank reconciliation statement.

Solution: Step 1, the corrected cash book balance. Start with the balance before adjustment, ₦286,000. Deduct bank charges 3,000 and the standing order 12,000; add the direct credit 20,000. Adjusted balance = 286,000 − 3,000 − 12,000 + 20,000 = **₦291,000**.

Step 2, the reconciliation. Unpresented cheques total 45,000 + 32,000 = 77,000.

Bank reconciliation statement at 31 March	**₦**	**₦**
Balance per bank statement (credit)		340,000
Add: uncredited deposit		28,000
		368,000
Less: unpresented cheques: no. 0214	45,000	
no. 0216	32,000	77,000
Adjusted cash book balance		**291,000**

Check: 340,000 + 28,000 = 368,000; 368,000 − 77,000 = 291,000, which equals the adjusted cash book balance. The reconciliation is complete.

WORKED EXAMPLE 3.4

The cash book of Mr Emmanuel Ogundipe shows a bank *overdraft* of ₦64,000 (credit balance) on 30 April. The bank statement is found to differ because: cheques of ₦40,000 are unpresented; a lodgement of ₦25,000 is not yet credited; bank charges of ₦4,000 are not in the cash book; and a customer's cheque for ₦18,000, banked earlier, has been dishonoured and the bank has debited it. Find the adjusted cash book balance and the balance on the bank statement.

Solution: Treat an overdraft as a negative balance. Start at −64,000. Bank charges reduce it further: −64,000 − 4,000 = −68,000. The dishonoured cheque is a payment out: −68,000 − 18,000 = −86,000. The adjusted cash book therefore shows an **overdraft of ₦86,000**.

To reach the statement balance, reverse the reconciliation: statement balance + uncredited deposits − unpresented cheques = adjusted cash book balance. So the statement balance = −86,000 − 25,000 + 40,000 = −71,000, an **overdraft of ₦71,000** on the bank statement.

Reconciliation at 30 April	**₦**
Overdraft per bank statement	(71,000)
Add: uncredited lodgement	25,000
	(46,000)
Less: unpresented cheques	(40,000)
Overdraft per adjusted cash book	(86,000)

Check: −71,000 + 25,000 = −46,000; −46,000 − 40,000 = −86,000. This agrees with the adjusted cash book overdraft.

Exam tip. Before you add or subtract, ask: "Is this item already in my cash book?" If not, it goes into the cash book adjustment (charges, interest, standing orders, direct credits, dishonoured cheques, errors). If it is in the cash book but not on the statement, it goes into the reconciliation (unpresented cheques and uncredited deposits). Never do both for the same item.

3.6 Correcting errors in the cash book

Mistakes made by the business when writing its own cash book are found during reconciliation. A common one is the **transposition** of digits: a cheque for ₦63,000 is entered as ₦36,000. The difference, 63,000 − 36,000 = 27,000, must be taken from the cash book balance, because the business has actually paid out more than it recorded. Other typical errors are entering a receipt on the wrong side, omitting an entry, and adding the column incorrectly. If a payment was entered as a receipt, the correction is twice the amount (once to remove the wrong receipt, once to record the payment).

WORKED EXAMPLE 3.5

The bank column of a cash book shows a balance of ₦512,000 (debit). The bank statement shows a different balance. The following are discovered: a cheque of ₦63,000 paid to a supplier was entered in the cash book as ₦36,000; cheques totalling ₦51,000 are unpresented; deposits of ₦70,000 are not yet credited; bank charges of ₦2,500 and interest credited by the bank of ₦1,500 are not in the cash book. Find the bank statement balance.

Solution: The error: the cash book understated the payment by 63,000 − 36,000 = 27,000, so the balance must be reduced by ₦27,000. Adjusted cash book balance = 512,000 − 27,000 − 2,500 + 1,500 = **₦484,000**. (512,000 − 27,000 = 485,000; 485,000 − 2,500 = 482,500; 482,500 + 1,500 = 484,000.)

Statement balance + uncredited deposits − unpresented cheques = adjusted balance, so statement balance = 484,000 − 70,000 + 51,000 = **₦465,000**.

Check: 465,000 + 70,000 = 535,000; 535,000 − 51,000 = 484,000. Correct.

3.7 Why reconciliation matters

A bank reconciliation is a control. It discovers errors made by the business or by the bank, such as an amount wrongly credited to another customer's account. It exposes fraud, for example a cheque altered after issue, a payment the business never authorised, or an employee who withdraws money and does not record it. It also ensures that the balance sheet shows the true bank balance, and that expenses such as bank charges are recorded in the period in which they arose. Good practice is to reconcile at least monthly, and a business with heavy cash flows may

do it weekly. The person who prepares the reconciliation should not be the person who writes the cheques or keeps the cash book, because separation of duties reduces the chance of undetected theft.

A cheque may also be **stale**, meaning that it was presented more than six months after the date on it, and the bank will refuse to pay it. A **post-dated cheque** carries a future date and cannot be paid before that date. In the reconciliation, a post-dated cheque received is held as unbanked until its date arrives. These details are useful for understanding why some cheques remain on the unpresented list for longer than expected.

Exercise 3.1

1. A debtor owing ₦120,000 pays ₦117,000 and is allowed a discount. State the amount of discount and the entries in the cash book and in the debtor's account.
2. Cash of ₦90,000 is paid into the bank. State the double entry and the name given to such an entry.
3. The bank column of a cash book shows a debit balance of ₦150,000. The bank statement shows bank charges of ₦3,500, a dishonoured customer's cheque of ₦20,000 and a standing order for ₦10,000, none of them yet in the cash book. Calculate the adjusted cash book balance.
4. A bank statement shows a credit balance of ₦220,000. Unpresented cheques total ₦60,000 and deposits not yet credited total ₦35,000. What is the correct cash book balance?
5. A cash book shows an overdraft of ₦40,000. The bank statement shows that the bank charged interest of ₦6,000 on the overdraft, not yet entered in the cash book. Calculate the adjusted overdraft.
6. Mr Danjuma starts the month with cash ₦30,000 and bank ₦200,000. He receives a cheque for ₦48,000 from a debtor, allowing ₦2,000 discount, and pays a creditor by cheque ₦57,000 after receiving ₦3,000 discount. Find the closing cash and bank balances and the total of each discount.
7. The adjusted cash book balance is ₦78,000 (debit). Unpresented cheques are ₦25,000 and uncredited lodgements ₦15,000. What does the bank statement show?
8. A cash book shows ₦340,000 and the bank statement ₦366,000. The only difference is unpresented cheques. How much are they?

9. During a year discount allowed was ₦12,400 and discount received was ₦8,900. State the net effect on profit.
10. A cheque for ₦54,000 paid to a supplier was entered in the cash book as ₦45,000. The cash book balance, before correction, is ₦410,000. What is the corrected balance?
11. State which of the following change the cash book and which appear only in the reconciliation statement: (a) bank charges; (b) cheque issued but not yet presented; (c) direct credit from a customer; (d) deposit made on the last day not yet credited; (e) dishonoured cheque.

Answers to Exercise 3.1

1. Discount = 120,000 − 117,000 = ₦3,000. Debit Bank 117,000 and debit Discount Allowed 3,000 in the cash book; credit debtor 117,000 and 3,000, clearing 120,000.
2. Debit Bank, credit Cash ₦90,000; a contra entry.
3. 150,000 − 3,500 − 20,000 − 10,000 = ₦116,500.
4. 220,000 + 35,000 − 60,000 = ₦195,000.
5. Overdraft = 40,000 + 6,000 = ₦46,000.
6. Cash ₦30,000; bank = 200,000 + 48,000 − 57,000 = ₦191,000; discount allowed ₦2,000, discount received ₦3,000.
7. 78,000 − 15,000 + 25,000 = ₦88,000.
8. Statement is greater than the cash book by 366,000 − 340,000 = ₦26,000, the unpresented cheques.
9. Received exceeds allowed by 8,900 − 12,400 = −3,500, so a net expense of ₦3,500 that reduces profit.
10. The payment was understated by 54,000 − 45,000 = 9,000; corrected balance = 410,000 − 9,000 = ₦401,000.
11. (a), (c) and (e) change the cash book; (b) and (d) are timing differences that appear only in the reconciliation statement.

Chapter summary

- The cash book is both a book of original entry and the Cash and Bank ledger accounts; receipts are debited and payments credited.

- The three-column cash book adds discount columns: discount allowed on the debit side and discount received on the credit side; they are memorandum columns whose totals are posted to the discount accounts.
- Discount allowed is an expense; discount received is an income. Trade discount is never recorded.
- A contra entry moves money between the cash and bank columns and needs no posting elsewhere.
- A credit balance in the bank column is an overdraft; the cash column cannot have a credit balance.
- The bank statement shows the business's money as a credit balance because the bank owes it to the business.
- Bank charges, interest, standing orders, direct debits, direct credits, dishonoured cheques and cash book errors are written into the cash book; unpresented cheques and uncredited deposits appear only in the reconciliation statement.
- Reconciliation: balance per statement + uncredited deposits − unpresented cheques = adjusted cash book balance.
- Reconciliation is a control against error and fraud and should be done regularly by someone other than the cashier.

Chapter 4: The Trial Balance, Errors and the Suspense Account

The double-entry system rests on a simple promise: for every debit there is an equal credit. If that promise has been kept throughout the year, then, when all the ledger balances are listed, the total of the debit balances must equal the total of the credit balances. This list is the **trial balance**. It is not a ledger account and it is not one of the final accounts; it is a working paper that checks the arithmetical accuracy of the books and supplies the figures from which the final accounts are drawn.

The trial balance is useful but limited. When it balances, the books are *probably* accurate, but several kinds of mistakes can hide behind a balanced total. When it does not balance, the difference tells the accountant that something is wrong, and a **suspense account** may be opened temporarily until the cause is found. This chapter explains how to prepare a trial balance, which errors it can and cannot reveal, and how every kind of error is corrected through the journal.

4.1 Preparing a trial balance

To prepare a trial balance, first balance every ledger account (as shown in Chapter 1), including the cash book and the personal accounts. Then list each balance in a two-column table, placing debit balances in the debit column and credit balances in the credit column. Finally, total each column. The classification of balances should become automatic.

Debit balances	Credit balances
Assets: premises, equipment, vehicles, inventory at the start, debtors, bank (if in funds), cash	Liabilities: creditors, loans, bank overdraft, accrued expenses
Expenses and losses: wages, rent, carriage inwards, discount allowed, bad debts	Incomes and gains: sales, discount received, commission received, rent received
Purchases and returns inward (sales returns)	Returns outward (purchases returns)
Drawings	Capital

The inventory (stock) that appears in a trial balance is the *opening* inventory carried forward from the previous year. The closing inventory is found by physical count at the year end and is not in the trial balance; it is dealt with in the final accounts (Chapter 5).

WORKED EXAMPLE 4.1

The following balances were extracted from the books of Mr Ahmed Lawal, a trader in Kaduna, on 31 December. Prepare the trial balance. All amounts are in naira.

Capital 1,870,000; Drawings 120,000; Purchases 1,450,000; Sales 2,310,000; Returns inward 30,000; Returns outward 50,000; Rent 180,000; Wages 360,000; Electricity 45,000; Premises 1,500,000; Equipment 400,000; Debtors 260,000; Creditors 210,000; Bank 295,000; Cash 25,000; Loan 600,000; Discount allowed 15,000; Discount received 20,000; Inventory at 1 January 380,000.

Solution: Each balance is placed in its proper column.

Account	Debit ₦	Credit ₦
Capital		1,870,000
Drawings	120,000	
Purchases	1,450,000	
Sales		2,310,000
Returns inward	30,000	
Returns outward		50,000
Rent	180,000	
Wages	360,000	
Electricity	45,000	
Premises	1,500,000	
Equipment	400,000	
Debtors	260,000	
Creditors		210,000
Bank	295,000	
Cash	25,000	
Loan		600,000
Discount allowed	15,000	
Discount received		20,000
Inventory at 1 January	380,000	
Totals	**5,060,000**	**5,060,000**

Check of totals. Debits: 120,000 + 1,450,000 + 30,000 + 180,000 + 360,000 + 45,000 + 1,500,000 + 400,000 + 260,000 + 295,000 + 25,000 + 15,000 + 380,000. Adding in order: 120,000 + 1,450,000 = 1,570,000; + 30,000 = 1,600,000; + 180,000 = 1,780,000; + 360,000 = 2,140,000; + 45,000 = 2,185,000; + 1,500,000 = 3,685,000; + 400,000 = 4,085,000; + 260,000 = 4,345,000; + 295,000 = 4,640,000; + 25,000 = 4,665,000; + 15,000 = 4,680,000; + 380,000 = **5,060,000**. Credits: 1,870,000 + 2,310,000 + 50,000 + 210,000 + 600,000 + 20,000 = **5,060,000**. The two columns agree, so the trial balance balances at ₦5,060,000.

Exam tip. Always add the columns twice. If the totals disagree by a number such as 9,000 or 90,000 (divisible by nine), suspect a transposition of digits, for example 54,000 written as 45,000. If the difference equals one balance, or exactly half of one, suspect a balance omitted or entered in the wrong column.

4.2 What the trial balance proves, and what it does not

If the trial balance agrees, it proves that the total debits in the ledger equal the total credits. It does *not* prove that every entry is right. It is a test of arithmetic, not a test of judgement. Seven classical errors can leave the totals equal, and an accountant must know all of them.

Error	Meaning	Example
Error of omission	A transaction is left out of the books entirely (both debit and credit)	A credit sale of ₦55,000 to Mr Ali is never recorded
Error of commission	An entry is made in the wrong account of the *same class*	A payment for electricity is debited to Water Rates account; a sale to Mr Ade is posted to Mr Ede
Error of principle	An item is entered in the wrong *class* of account, against accounting principles	The purchase of a van is debited to Purchases, or repairs are debited to Buildings
Error of original entry	The wrong figure is used in both the debit and the credit entry	A purchase of ₦72,000 is entered as ₦27,000 in both accounts
Compensating error	Two or more errors of equal size on opposite sides cancel each other	Sales overcast by ₦10,000 and Rent paid overcast by ₦10,000
Complete reversal of entries	The correct accounts and amount are used, but debit and credit are exchanged	A receipt from Bello is entered: Dr Bello, Cr Bank
Errors of transposition (a special case of original entry)	Digits of a figure are swapped in both entries	₦3,600 entered as ₦6,300 on both sides

Errors that **do** prevent the trial balance from agreeing include: a one-sided entry (a debit without a credit, or the reverse); an entry of different amounts on the two sides; two debits or two credits for one transaction; an addition error in an account; a balance carried to the wrong side or omitted from the trial balance; and a balance copied into the trial balance with the figures wrongly transcribed. When any of these occurs the difference is placed in a **suspense account**.

4.3 The suspense account

A **suspense account** is a temporary account opened to make the trial balance agree when the cause of the difference cannot be found immediately. It is not an asset, a liability or an expense; it is a holding account that must disappear once the errors are traced. The rule for the amount and side is straightforward.

- If the **credit** total is bigger than the debit total, the difference is *debited* to Suspense (placed in the debit column of the trial balance) so that the totals agree.
- If the **debit** total is bigger, the difference is *credited* to Suspense.

As each error is found, a journal entry corrects it, and one side of that entry is the Suspense account. When all errors that affected the trial balance have been corrected, the Suspense account must have a nil balance. If it does not, an error remains undiscovered.

Key point. To correct an error, ask three questions. (1) What was done? (2) What should have been done? (3) What entry puts it right? If the error affected only one side of the books, Suspense is used for the other half. If the error kept the trial balance balanced (omission, commission, principle, original entry, reversal, compensating), *no suspense account is involved*. Correcting entries are made in the journal with a narration.

4.4 Correcting errors through the journal

Errors that do not affect the trial balance

WORKED EXAMPLE 4.2

Write the journal entries to correct the following errors: (a) a credit sale of ₦55,000 to Mr Ali was completely omitted; (b) a payment of ₦25,000 for electricity was debited to Water Rates account; (c) a computer costing ₦150,000 bought on cash terms was debited to Purchases account; (d) a receipt of ₦18,000 from Bello was entered as a debit in Bello's account and a credit in the Bank account.

Solution:

Error	What was done	What should be done	Correcting entry
(a) Omission	Nothing	Dr Ali, Cr Sales 55,000	Dr Ali ₦55,000; Cr Sales ₦55,000
(b) Commission	Dr Water Rates 25,000	Dr Electricity 25,000	Dr Electricity ₦25,000; Cr Water Rates ₦25,000
(c) Principle	Dr Purchases 150,000	Dr Computer Equipment 150,000	Dr Computer Equipment ₦150,000; Cr Purchases ₦150,000
(d) Reversal	Dr Bello 18,000; Cr Bank 18,000	Dr Bank 18,000; Cr Bello 18,000	Dr Bank ₦36,000; Cr Bello ₦36,000

For the reversal, the entry must be *twice* the amount: 18,000 to cancel the wrong entry and another 18,000 to make the right one. After the correction, Bello's account is credited 36,000 against the wrongly made debit of 18,000, a net credit of 18,000, which is what the receipt should have been. The bank account is debited 36,000 against the wrong credit of 18,000, a net debit of 18,000.

Errors that affect the trial balance

WORKED EXAMPLE 4.3

The trial balance of Mrs Amaka Obi, a boutique owner in Onitsha, did not agree. The credit side exceeded the debit side by ₦33,000, and the difference was put to a suspense account. Net profit before corrections was ₦612,000. The following errors were later found: (1) a rent payment of ₦60,000 was credited correctly to Bank but debited to Rent account as ₦6,000; (2) a discount received of ₦21,000 was debited to the supplier's account but never credited to the Discount Received account; (3) a computer costing ₦150,000 was debited to Purchases account; (4) a credit sale of ₦40,000 to Mr Obi was omitted from the books. Prepare the journal entries, the suspense account and the corrected profit.

Solution: Errors (3) and (4) do not affect the trial balance. Errors (1) and (2) do.

(1) The debit to Rent was 54,000 too small (60,000 − 6,000 = 54,000). Correction: Dr Rent ₦54,000; Cr Suspense ₦54,000.

(2) The credit to Discount Received is missing. Correction: Dr Suspense ₦21,000; Cr Discount Received ₦21,000.

(3) Dr Computer Equipment ₦150,000; Cr Purchases ₦150,000.

(4) Dr Obi ₦40,000; Cr Sales ₦40,000.

Debit			**Credit**		
Details		**₦**	**Details**		**₦**
Balance from trial balance		33,000	Rent (error 1)		54,000
Discount Received (error 2)		21,000			
	Total	**54,000**		**Total**	**54,000**

The suspense account balances to nil (33,000 + 21,000 = 54,000), so every error has been found. Effect on profit: (1) rent is understated by 54,000, so profit falls by 54,000; (2) discount received increases profit by 21,000; (3) purchases fall by 150,000, so profit rises by 150,000; (4) sales rise by 40,000, so profit rises by 40,000. Corrected profit = 612,000 − 54,000 + 21,000 + 150,000 + 40,000 = **₦769,000**.

Compensating errors and errors of original entry

Errors of original entry and compensating errors do not disturb the totals, and are corrected only when discovered. For an error of original entry, the correcting entry equals the *difference* between the correct and the wrong amounts. If a purchase of ₦72,000 from Ike was entered as ₦27,000 in both Purchases and Ike's account, the difference is 72,000 − 27,000 = 45,000, and the correction is Dr Purchases ₦45,000; Cr Ike ₦45,000. For a compensating error such as Sales overcast by ₦10,000 and Wages overcast by ₦10,000, the corrections are Dr Sales ₦10,000; Cr Wages ₦10,000, so each account is reduced to its true figure.

WORKED EXAMPLE 4.4

A trader found the following errors after the trial balance agreed: (a) the Sales account was overcast by ₦10,000 and the Wages account was overcast by ₦10,000; (b) a purchase of ₦72,000 from Ike was entered as ₦27,000 in both books. Original profit was ₦480,000. State the type of each error, the correcting journal entries, and the corrected profit.

Solution: (a) is a compensating error. Sales were overstated by 10,000, so Dr Sales ₦10,000; Cr Wages ₦10,000. The effect on profit: sales fall by 10,000 (profit −10,000) and wages fall by 10,000 (profit +10,000), a net effect of nil. (b) is an error of original entry. Dr Purchases ₦45,000; Cr Ike ₦45,000. Purchases rise by 45,000, so profit falls by 45,000. Corrected profit = 480,000 − 10,000 + 10,000 − 45,000 = **₦435,000**.

4.5 Statement of corrected profit

When errors affect expense or income accounts, profit as first calculated is wrong, and an examiner may ask for a **statement of corrected profit**. The method is to begin with the original profit, add the items that increase profit (understated income, overstated expense), and deduct those that reduce it (overstated income, understated expense). Errors that concern only asset or liability accounts, or only the relationship between an asset and a liability, have no effect on profit; neither do errors of commission between two expense accounts.

Correction	Effect on profit
Expense understated (too little charged)	Profit falls
Expense overstated (too much charged)	Profit rises
Income or sales understated	Profit rises
Income or sales overstated	Profit falls
Capital expenditure wrongly charged as an expense (e.g. a machine in Purchases)	Profit rises
Revenue expenditure wrongly treated as an asset (e.g. repairs in Buildings)	Profit falls

Key point. A trial balance that agrees does not prove that the books are free from error: omission, commission, principle, original entry, compensating errors and complete reversal all leave the totals equal. A trial balance that fails to agree proves that at least one error exists, and the difference must be traced. Suspense is a temporary account, and a suspense balance should never appear in the final balance sheet if all errors are corrected.

4.6 Tracing a difference and a full suspense example

When a trial balance fails to agree, do not guess. Work through a fixed routine. First, re-add both columns. Second, divide the difference by two: if the result equals a balance in the list, that balance may have been placed in the wrong column. Third, test for a difference divisible by nine, which suggests transposed digits. Fourth, check that every ledger balance, including cash, bank and the opening inventory, has been listed, and that each carries the correct sign. Fifth, re-check the postings from the books of original entry, looking for one-sided entries and for totals posted to the wrong side. Only if the difference still cannot be found is it taken to suspense.

WORKED EXAMPLE 4.5

The trial balance of Mr Kunle Adeyemi, a hardware dealer in Ibadan, showed debit totals of ₦2,410,000 and credit totals of ₦2,400,000. The difference was put in a suspense account. Profit before correction was ₦300,000. Later it was found that (i) the Sales account had been undercast by ₦20,000 (the credit side was added up as ₦20,000 too little); and (ii) a wages payment of ₦14,000 had been credited to Bank correctly but debited to the Wages account as ₦4,000. Prepare the suspense account and the corrected profit.

Solution: Debits exceed credits by 2,410,000 − 2,400,000 = 10,000, so ₦10,000 is credited to Suspense. For (i), the Sales balance is too small by 20,000: Dr Suspense ₦20,000; Cr Sales ₦20,000. For (ii), the Wages debit is too small by 14,000 − 4,000 = 10,000: Dr Wages ₦10,000; Cr Suspense ₦10,000.

Debit		Credit	
Details	**₦**	**Details**	**₦**
Sales (error i)	20,000	Balance from trial balance	10,000
		Wages (error ii)	10,000
Total	**20,000**	**Total**	**20,000**

The suspense account is cleared. Corrected profit = 300,000 + 20,000 (sales increased) − 10,000 (wages increased) = **₦310,000**.

Exercise 4.1

1. State whether each of the following would be revealed by the trial balance: (a) a credit sale omitted from all books; (b) a debit entry with no corresponding credit; (c) a payment debited to the wrong expense account; (d) an addition error of ₦5,000 in the Sales account.
2. The debit side of a trial balance totals ₦640,000 and the credit side ₦655,000. State the amount, and the side, of the entry in the suspense account.
3. A payment of ₦12,000 for repairs was debited to Rent account. Write the correcting journal entry and name the error.
4. A payment of ₦22,000 to a supplier, Eke, was entered as: debit Bank, credit Eke. Write the correcting entry.

5. Net profit before correction is ₦450,000. Repairs of ₦30,000 on a building were debited to the Buildings account. Find the corrected profit.
6. A credit sale of ₦55,000 to Ali was omitted from all books. Write the correcting entry and state its effect on profit.
7. A suspense account was opened with a debit of ₦9,000 because a discount of ₦9,000 received was credited to the Discount Received account without the matching debit in the supplier's account. Write the correcting entry and state what happens to the suspense account.
8. Classify each error: (a) a sale to Mr Ade posted to the account of Mr Ede; (b) a vehicle debited to Purchases; (c) sales and purchases both overstated by ₦10,000; (d) a purchase of ₦8,100 entered as ₦1,800 in both accounts.
9. Prepare a trial balance from: Capital 500,000; Sales 900,000; Purchases 600,000; Wages 120,000; Rent 60,000; Equipment 300,000; Debtors 150,000; Creditors 90,000; Bank 260,000. State whether it balances.
10. A payment of ₦40,000 for rent was debited to Rent account as ₦4,000, with the credit to Bank correct. What is the entry required, and which side of the suspense account is used?
11. Explain why an agreed trial balance does not prove that the books are completely accurate.

Answers to Exercise 4.1

1. (a) No; (b) yes; (c) no; (d) yes, because only one account is wrong.
2. Credit is larger by 655,000 − 640,000 = 15,000, so Suspense is debited ₦15,000.
3. Error of commission. Dr Repairs ₦12,000; Cr Rent ₦12,000.
4. Complete reversal. Dr Eke ₦44,000; Cr Bank ₦44,000.
5. Repairs are an expense wrongly capitalised, so profit falls: 450,000 − 30,000 = ₦420,000.
6. Error of omission. Dr Ali ₦55,000; Cr Sales ₦55,000. Profit increases by ₦55,000.
7. Dr Supplier ₦9,000; Cr Suspense ₦9,000. The suspense account is cleared to nil.
8. (a) commission; (b) principle; (c) compensating; (d) original entry (transposition).
9. Debits: 600,000 + 120,000 + 60,000 + 300,000 + 150,000 + 260,000 = 1,490,000. Credits: 500,000 + 900,000 + 90,000 = 1,490,000. It balances.
10. Dr Rent ₦36,000; Cr Suspense ₦36,000; the suspense balance was a debit of ₦36,000 and is cleared.

11. Because errors such as omission, commission, principle, original entry, compensating errors and reversals keep debits equal to credits; the trial balance tests only the arithmetic equality of debit and credit totals.

Chapter summary

- A trial balance lists all ledger balances in debit and credit columns; if the double entry is complete, the totals agree.
- Assets, expenses, purchases, returns inward and drawings are debit balances; liabilities, capital, income, sales and returns outward are credit balances.
- An agreed trial balance does not prove accuracy: errors of omission, commission, principle, original entry, compensating errors and complete reversals are not disclosed.
- Errors that disturb the totals include one-sided entries, unequal amounts, addition mistakes and wrongly extracted balances.
- A suspense account holds a difference temporarily: debit it when credits exceed debits, credit it when debits exceed credits; it must be nil when all errors are corrected.
- Correct each error by asking what was done, what should have been done, and what entry puts it right; a reversal requires twice the amount.
- Corrections to income and expense accounts change profit, and a statement of corrected profit starts from the original profit and adjusts it.

Chapter 5: Final Accounts of a Sole Trader and Year-End Adjustments

Everything recorded so far, from the first source document to the trial balance, exists for one purpose: to tell the owner of a business how well the business has performed and what it is worth. The statements that answer these two questions are called the **final accounts**. They are prepared at the end of every accounting period, usually twelve months, and they consist of the trading account, the profit and loss account and the balance sheet (also called the statement of financial position).

The trial balance is only the raw material. Before final accounts can be drawn up, the accountant must correct the ledger balances for items that belong to the period but have not yet been recorded, such as wages owed, or that have been recorded but belong to a later period, such as insurance paid in advance. These year-end corrections are called **adjustments**, and they are among the most frequently examined topics in Principles of Accounts. This chapter builds the final accounts of a sole trader step by step, then introduces each adjustment, and ends with a complete example in which a trial balance and a list of adjustments are turned into finished statements.

5.1 The purpose and structure of final accounts

A **sole trader** is a person who owns and runs a business alone, is entitled to all its profit and is personally responsible for all its debts. In accounting, however, the business is treated as separate from its owner (the *business entity concept*), so the owner is a creditor of the business for the capital put in, and what the owner takes out is recorded as drawings.

The final accounts have three parts, each answering a different question.

- The **trading account** shows the **gross profit**, which is the profit made from buying and selling goods before any running expenses are considered.
- The **profit and loss account** starts with the gross profit, adds any other income, deducts the running expenses of the business, and shows the **net profit** (or net loss) for the period.
- The **balance sheet** lists the assets, liabilities and capital of the business at one date, the last day of the period. It is a snapshot, whereas the trading and profit and loss accounts describe what happened over a whole period.

The trading account and the profit and loss account are often combined and described together as the **income statement**. Both are, in effect, ledger accounts into which the balances of income and expense accounts are transferred at the end of the year. Once an income or expense balance has been moved there, the original account starts the next year at nil. The balances of assets, liabilities and capital are not closed in this way; they are carried down to the next period and are summarised in the balance sheet.

> **Key point.** The trading and profit and loss accounts cover a *period* and deal with income and expenses. The balance sheet is drawn up at a *date* and deals with assets, liabilities and capital. Never mix the two groups.

5.2 The trading account

The trading account compares what the business earned from selling goods with what those goods cost. Its parts are as follows.

- **Sales** are the selling price of goods sold, both for cash and on credit. **Returns inwards** (sales returns) are goods sent back by customers and are deducted from sales to give **net sales** (or net turnover).
- **Opening inventory** is the cost of goods unsold at the start of the period.
- **Purchases** are goods bought for resale. **Returns outwards** (purchases returns) are goods sent back to suppliers and are deducted from purchases.
- **Carriage inwards** is the cost of transporting purchased goods to the business. It is added to purchases, because it is part of what the goods cost to bring to a saleable condition. (**Carriage outwards**, the cost of delivering goods to customers, is a selling expense and belongs in the profit and loss account.)
- **Closing inventory** is the cost of goods still unsold at the end of the period. It is deducted from the goods available for sale, because only the goods actually sold should be charged against sales.

The **cost of goods sold** is therefore:

Cost of goods sold = Opening inventory + Net purchases + Carriage inwards − Closing inventory

and the **gross profit** is:

Gross profit = Net sales − Cost of goods sold

Closing inventory is valued at the **lower of cost and net realisable value**, an application of the prudence concept: if goods can now only be sold for less than they cost, they are valued at the lower selling figure so that the loss is recognised immediately. Net realisable value is the expected selling price less any cost needed to sell the goods.

WORKED EXAMPLE 5.1

From the books of Mr Bola Adeniyi, a provisions trader in Ibadan, the following figures are available for the year ended 31 December 2025: sales ₦150,000; returns inwards ₦6,000; opening inventory ₦12,000; purchases ₦84,000; returns outwards ₦4,000; carriage inwards ₦3,000; closing inventory ₦15,000. Prepare the trading account.

Solution: Net sales = 150,000 − 6,000 = ₦144,000. Net purchases = 84,000 − 4,000 = ₦80,000. Adding carriage inwards gives 80,000 + 3,000 = ₦83,000. Goods available for sale = 12,000 + 83,000 = ₦95,000. Cost of goods sold = 95,000 − 15,000 = ₦80,000. Gross profit = 144,000 − 80,000 = ₦64,000.

Trading account for the year ended 31 December 2025	**₦**	**₦**
Sales		150,000
Less returns inwards		(6,000)
Net sales		**144,000**
Opening inventory	12,000	
Purchases	84,000	
Less returns outwards	(4,000)	
	80,000	
Carriage inwards	3,000	
Goods available for sale	95,000	
Less closing inventory	(15,000)	
Cost of goods sold		(80,000)
Gross profit		**64,000**

5.3 The profit and loss account

The profit and loss account begins with the gross profit carried down from the trading account. To it are added **other income** such as discount received, rent received, commission received and interest received. From the total are deducted the **expenses** of running the business. Typical expenses are salaries and wages, rent and rates, electricity, insurance, carriage outwards, discount allowed, bad debts, telephone, advertising, repairs, depreciation and general expenses.

Expenses are often grouped by function (selling and distribution, administration, finance), but the examination at UTME level normally accepts a simple list. What matters is that every item is placed on the correct side: income adds to profit, expense reduces it.

An important distinction must be kept in mind. **Revenue expenditure** is spending on the day-to-day running of the business, such as wages, repairs and fuel, and is charged against the profit of the year. **Capital expenditure** is spending on acquiring or improving non-current assets, such as buying a delivery van or building an extension, and appears in the balance sheet. Wrongly treating capital spending as an expense understates both profit and assets; treating a running expense as an asset does the opposite. For example, ₦40,000 spent on petrol for the van is revenue expenditure, while the cost of the van itself is capital expenditure. Money spent to bring a new asset into use (delivery to the shop, installation, legal fees on a property) is capital expenditure too.

WORKED EXAMPLE 5.2

Continuing with Mr Adeniyi, the gross profit is ₦64,000. He also received discount of ₦2,500 and rent of ₦6,000 from subletting a store. His expenses were salaries ₦28,000; rent and rates ₦7,500; electricity ₦4,200; discount allowed ₦1,800; carriage outwards ₦2,500; sundry expenses ₦3,000. Prepare the profit and loss account.

Solution: Total income = 64,000 + 2,500 + 6,000 = ₦72,500. Total expenses = 28,000 + 7,500 + 4,200 + 1,800 + 2,500 + 3,000 = ₦47,000. Net profit = 72,500 − 47,000 = ₦25,500.

Profit and loss account for the year ended 31 December 2025	**₦**	**₦**
Gross profit		64,000
Discount received	2,500	
Rent received	6,000	8,500
		72,500
Salaries	28,000	
Rent and rates	7,500	
Electricity	4,200	
Discount allowed	1,800	
Carriage outwards	2,500	
Sundry expenses	3,000	(47,000)
Net profit		**25,500**

The net profit is then transferred to the capital account, where it is added to the owner's capital, and drawings are deducted. A net loss would be deducted instead.

5.4 The balance sheet

The balance sheet shows the financial position of the business on the last day of the period. It is built from the accounting equation: Assets − Liabilities = Capital. Its items are arranged in an accepted order.

- **Non-current assets** (fixed assets) are resources bought for long-term use in the business and not for resale: land and buildings, machinery, motor vehicles, furniture and fittings. They are

shown at cost less accumulated depreciation, which gives their **net book value**. By convention they are listed in order of permanence, with the most permanent (land) first.

- **Current assets** are cash or items expected to turn into cash within a year. They are listed in increasing order of liquidity: inventory, trade receivables (debtors), prepayments, bank and cash.
- **Current liabilities** are amounts due for payment within one year: trade payables (creditors), accrued expenses and bank overdraft.
- **Non-current liabilities** (long-term liabilities) are due after more than a year, such as a bank loan repayable over five years.
- **Working capital** is current assets minus current liabilities. It measures the funds available for day-to-day operations.
- **Capital** is the owner's investment in the business. The capital account is built up as: opening capital + net profit − drawings = closing capital.

The two sides of the balance sheet must agree: net assets (non-current assets + working capital − non-current liabilities) must equal closing capital. If they do not, there is an error somewhere in the statements.

> **Key point.** Drawings are not an expense. They are withdrawals of capital by the owner and are never charged in the profit and loss account. Goods taken by the owner for personal use are recorded as drawings at cost.

5.5 Adjustments: accruals and prepayments

The **accruals concept** (matching concept) requires that revenue and expenses be recognised in the period in which they are earned or incurred, regardless of when cash is received or paid. A trial balance records cash movements, so it often contains expenses that are too small or too large for the period. Two adjustments correct this.

Accrued expenses

An **accrued expense** (accrual) is an expense that has been incurred in the period but not yet paid. If wages for the last week of the year will be paid in January, the trial balance understates the year's wages. The correction is to *add* the accrual to the expense in the profit and loss account

and show the amount as a *current liability* in the balance sheet. In the ledger it appears as a credit balance carried down on the expense account.

Prepaid expenses

A **prepaid expense** (prepayment) is an amount paid in the period that relates to the next period, such as insurance paid for the whole of the following year. The correction is to *deduct* the prepayment from the expense in the profit and loss account and show it as a *current asset* in the balance sheet. In the ledger it appears as a debit balance carried down.

The same logic applies to income. **Income receivable** (accrued income) is income earned but not yet received, and is added to the income and shown as a current asset. **Income received in advance** is cash received for a later period, and is deducted from the income and shown as a current liability.

The expense charged to the profit and loss account can always be found with a simple formula:

Expense for the year = Cash paid + Accrual at end − Accrual at start + Prepayment at start − Prepayment at end

WORKED EXAMPLE 5.3

Alhaji Musa runs a printing shop in Kano. At 1 January 2025 he owed ₦800 for electricity. During 2025 he paid ₦9,600 for electricity, and at 31 December 2025 the unpaid bill was ₦1,200. Show the electricity account and state the charge to the profit and loss account.

Solution: The opening accrual of ₦800 was an amount owed from last year, so it is a credit balance brought down. Payments are debits. The closing accrual is a credit that is carried down. The transfer to profit and loss is the balancing figure.

Dr		**Electricity account**		**Cr**	
Date	**Details**	**₦**	**Date**	**Details**	**₦**
2025	Bank	9,600	1 Jan	Balance b/d	800
31 Dec	Balance c/d (accrued)	1,200	31 Dec	Profit and loss	10,000
		10,800			**10,800**
			1 Jan 2026	Balance b/d	1,200

Check with the formula: 9,600 + 1,200 − 800 = ₦10,000, the charge to profit and loss. The accrual of ₦1,200 is a current liability in the balance sheet.

WORKED EXAMPLE 5.4

Mrs Ngozi Eze pays rent annually. In the year ended 31 March 2026 she paid ₦18,000. At 31 March 2026, ₦3,000 of this related to the following year. Calculate the rent expense and state the balance sheet treatment.

Solution: Rent expense = 18,000 − 3,000 = ₦15,000, charged to the profit and loss account. The ₦3,000 prepaid is a debit balance carried down and is shown as a current asset (prepayments) in the balance sheet.

5.6 Adjustments: bad debts and the provision for doubtful debts

When goods are sold on credit, there is a risk that some customers will never pay. A debt that is definitely irrecoverable, because the customer has died, disappeared or been declared bankrupt, is a **bad debt**. It is written off by crediting the customer's account and debiting the bad debts account, which is an expense in the profit and loss account. Writing off recognises the loss when it becomes certain.

Prudence goes further. At the year end, some of the remaining receivables may still turn out to be bad, even though it is not yet known which. The business therefore creates a **provision for doubtful debts**, usually a percentage of the receivables remaining after bad debts have been written off. The provision does not reduce any customer's account; it is a separate credit balance that is deducted from receivables in the balance sheet, so that receivables are shown at the amount the business expects to collect.

Only the *change* in the provision is charged to the profit and loss account each year. If the new provision is larger than the old one, the **increase** is an expense. If it is smaller, the **decrease** is a gain, shown as other income or deducted from the expenses.

WORKED EXAMPLE 5.5

At 31 December 2024, Mr Tunde Salami's receivables were ₦60,000 and he made a provision of 5%. At 31 December 2025, receivables were ₦80,000 before writing off a bad debt of ₦2,000, and the provision is to remain at 5% of the receivables outstanding. Calculate the provision at each date and the effect on the 2025 profit and loss account.

Solution: Provision at 31 December 2024 = 5% × 60,000 = ₦3,000. Receivables at 31 December 2025 after the write-off = 80,000 − 2,000 = ₦78,000. New provision = 5% × 78,000 = ₦3,900. Increase = 3,900 − 3,000 = ₦900. The profit and loss account is charged with bad debts ₦2,000 and increase in provision ₦900, a total of ₦2,900. In the balance sheet, receivables are shown as 78,000 − 3,900 = ₦74,100.

Key point. Provisions are calculated on receivables *after* deducting bad debts written off, and the profit and loss account is charged only with the *change* in the provision. Writing off a bad debt reduces receivables; creating or increasing a provision does not.

5.7 Other year-end adjustments

Depreciation of non-current assets must be charged each year as an expense and deducted from the asset's cost in the balance sheet. The methods and ledger entries are treated fully in Chapter 6; here we simply apply a given rate.

Goods taken for own use. When the proprietor takes goods from the business, the purchases account is credited (reducing the cost of goods sold) and the drawings account is debited, both at cost. **Closing inventory** is also an adjustment: it is not in the trial balance, because the inventory account still shows the opening figure. It is entered twice, as a deduction in the trading account and as a current asset in the balance sheet.

Each adjustment is a double entry with one half in the income statement and the other in the balance sheet, which is why the balance sheet still balances afterwards.

5.8 A complete example

The following example brings every part of the chapter together. Study the working in full, because the examination often asks for a complete set of final accounts from a trial balance with adjustments.

WORKED EXAMPLE 5.6

The trial balance of Mrs Adaeze Okoro, who runs a general store in Enugu, at 31 December 2025 was as follows.

Account	Dr ₦	Cr ₦
Premises	400,000	
Furniture and fittings	90,000	
Motor van	150,000	
Inventory, 1 January 2025	60,000	
Purchases	520,000	
Returns inwards	10,000	
Carriage inwards	8,000	
Wages	95,000	
Rent and rates	24,000	
Insurance	9,000	
Electricity	14,000	
General expenses	12,000	
Trade receivables	80,000	
Bank	45,000	
Cash	5,000	
Drawings	70,000	
Bad debts	4,000	
Capital		500,000
Sales		935,000
Returns outwards		12,000
Trade payables		62,000
Provision for doubtful debts		3,000
Provision for depreciation: furniture and fittings		18,000
Provision for depreciation: motor van		60,000
Discount received		6,000
Totals	**1,596,000**	**1,596,000**

Adjustments at 31 December 2025: (a) closing inventory ₦75,000; (b) wages accrued ₦5,000; (c) insurance prepaid ₦1,500 and rent and rates prepaid ₦2,000; (d) electricity accrued ₦1,800; (e) a further bad debt of ₦2,000 is to be written off, and the provision is to be 5% of remaining receivables; (f) depreciate furniture and fittings at 10% and the motor van at 20%, both on cost; (g) Mrs Okoro took goods costing ₦3,000 for her own use, not yet recorded. Prepare the trading and profit and loss account and the balance sheet.

Solution: First deal with the adjustments one at a time.

1. Goods for own use: purchases = 520,000 − 3,000 = ₦517,000; drawings = 70,000 + 3,000 = ₦73,000.
2. Wages = 95,000 + 5,000 = ₦100,000. Rent and rates = 24,000 − 2,000 = ₦22,000. Insurance = 9,000 − 1,500 = ₦7,500. Electricity = 14,000 + 1,800 = ₦15,800.
3. Bad debts = 4,000 + 2,000 = ₦6,000. Receivables = 80,000 − 2,000 = ₦78,000. New provision = 5% × 78,000 = ₦3,900; the increase is 3,900 − 3,000 = ₦900.
4. Depreciation: furniture 10% × 90,000 = ₦9,000; van 20% × 150,000 = ₦30,000. Accumulated depreciation becomes 18,000 + 9,000 = ₦27,000 and 60,000 + 30,000 = ₦90,000 respectively.

Trading and profit and loss account for the year ended 31 December 2025	₦	₦
Sales		935,000
Less returns inwards		(10,000)
Net sales		925,000
Opening inventory	60,000	
Purchases (520,000 − 3,000 taken by owner)	517,000	
Less returns outwards	(12,000)	
	505,000	
Carriage inwards	8,000	
	573,000	
Less closing inventory	(75,000)	
Cost of goods sold		(498,000)
Gross profit		**427,000**
Discount received		6,000
		433,000
Wages	100,000	
Rent and rates	22,000	
Insurance	7,500	
Electricity	15,800	
General expenses	12,000	
Bad debts	6,000	
Increase in provision for doubtful debts	900	
Depreciation: furniture and fittings	9,000	
Depreciation: motor van	30,000	
Total expenses		(203,200)
Net profit		**229,800**

Cost of goods sold: 60,000 + 505,000 + 8,000 = 573,000; 573,000 − 75,000 = ₦498,000. Expenses: 100,000 + 22,000 + 7,500 + 15,800 + 12,000 + 6,000 + 900 + 9,000 + 30,000 = ₦203,200.

Balance sheet at 31 December 2025	Cost ₦	Depreciation ₦	Net ₦
Non-current assets			
Premises	400,000	0	400,000
Furniture and fittings	90,000	27,000	63,000
Motor van	150,000	90,000	60,000
	640,000	117,000	523,000
Current assets			
Inventory			75,000
Trade receivables (78,000 − provision 3,900)			74,100
Prepayments (1,500 + 2,000)			3,500
Bank			45,000
Cash			5,000
			202,600
Current liabilities			
Trade payables			62,000
Accruals (5,000 + 1,800)			6,800
			(68,800)
Working capital			133,800
Net assets			**656,800**
Financed by			
Capital at 1 January 2025			500,000
Add net profit			229,800
			729,800
Less drawings (70,000 + 3,000)			(73,000)
Capital at 31 December 2025			**656,800**

Check: non-current assets 523,000 + working capital 133,800 = 656,800, which equals closing capital. The balance sheet balances.

Exercise 5.1

1. Calculate the cost of goods sold: opening inventory ₦8,000; purchases ₦46,000; carriage inwards ₦1,500; returns outwards ₦2,500; closing inventory ₦9,000.
2. Using the cost of goods sold from question 1, with sales of ₦70,000 and returns inwards of ₦2,000, calculate the gross profit.
3. A business has gross profit of ₦24,000, discount received ₦1,000 and total expenses of ₦15,500. Find the net profit.
4. During the year a trader paid ₦24,000 for rent. At the start of the year ₦2,000 of rent was prepaid, and at the end ₦4,000 was prepaid. Calculate the rent charged to the profit and loss account.
5. Wages paid during the year were ₦52,000. Wages owing were ₦3,000 at the start and ₦4,500 at the end. Calculate the wages expense.
6. Receivables at the year end were ₦95,000 before writing off bad debts of ₦5,000. The provision is to be 4% of remaining receivables, and the opening provision was ₦2,500. Calculate the total charged to profit and loss for bad debts and the provision.
7. A trader's net profit before adjustment was ₦40,000. It was then discovered that the owner had taken goods costing ₦4,000 for personal use and the entry had not been made. Calculate the corrected net profit and state the effect on drawings.
8. Current assets: inventory ₦20,000, receivables ₦15,000, bank ₦6,000, prepayments ₦1,000. Current liabilities: payables ₦12,000, accruals ₦2,000. Calculate the working capital.
9. A sole trader began the year with capital of ₦120,000, made a net profit of ₦35,000 and took drawings of ₦18,000. Calculate the closing capital.
10. State where carriage inwards and carriage outwards are shown in the final accounts, and give the reason for the difference.

Answers to Exercise 5.1

1. ₦44,000: 8,000 + 46,000 − 2,500 + 1,500 − 9,000 = 44,000.
2. ₦24,000: net sales 70,000 − 2,000 = 68,000; 68,000 − 44,000 = 24,000.
3. ₦9,500: 24,000 + 1,000 − 15,500 = 9,500.
4. ₦22,000: 24,000 + 2,000 − 4,000 = 22,000.
5. ₦53,500: 52,000 − 3,000 + 4,500 = 53,500.

6. ₦6,100: receivables 95,000 − 5,000 = 90,000; provision 4% × 90,000 = 3,600; increase 3,600 − 2,500 = 1,100; total 5,000 + 1,100 = 6,100.
7. ₦44,000: removing ₦4,000 from purchases raises profit to 40,000 + 4,000 = 44,000; drawings increase by ₦4,000 and capital is reduced by the same amount.
8. ₦28,000: current assets 20,000 + 15,000 + 6,000 + 1,000 = 42,000; current liabilities 12,000 + 2,000 = 14,000; 42,000 − 14,000 = 28,000.
9. ₦137,000: 120,000 + 35,000 − 18,000 = 137,000.
10. Carriage inwards is added to purchases in the trading account, because it is part of the cost of bringing goods to a saleable state; carriage outwards is a selling expense and goes in the profit and loss account.

Chapter summary

- Final accounts consist of the trading account, the profit and loss account and the balance sheet; the first two cover a period and the last is drawn up at a date.
- Gross profit = net sales − cost of goods sold, where cost of goods sold = opening inventory + net purchases + carriage inwards − closing inventory.
- Net profit = gross profit + other income − expenses. Drawings are never an expense, and capital expenditure is never charged to profit.
- The balance sheet lists non-current assets, current assets in order of liquidity, current liabilities, and capital; working capital = current assets − current liabilities.
- Accruals are added to expenses and appear as current liabilities; prepayments are deducted from expenses and appear as current assets.
- Bad debts are written off as an expense; the provision for doubtful debts is a percentage of remaining receivables and only its change is charged to profit.
- Closing inventory, depreciation and goods taken by the owner are further adjustments, each of which affects both the income statement and the balance sheet.
- Net assets must equal closing capital, which is opening capital plus net profit less drawings.

Chapter 6: Depreciation, Disposal of Non-Current Assets and Control Accounts

A delivery van bought for ₦3,000,000 does not stay worth ₦3,000,000. Each year it is driven, it wears out, and a time comes when it must be sold or scrapped. If the whole cost of the van were charged as an expense in the year of purchase, that year would show a poor profit and every later year would look unfairly good, even though the van helped to earn income in all of them. Accountants solve this by spreading the cost of the asset over the years in which it is used. This spreading is called **depreciation**, and it is the first topic of this chapter.

The second topic is the **control account**. In a business with hundreds of customers and suppliers, the sales ledger and the purchases ledger contain many individual accounts. A control account gives the total of all those accounts in a single place and provides a check on the accuracy of the ledgers. Both topics are tested regularly, and both depend on accurate arithmetic and clear ledger layout.

6.1 The meaning and causes of depreciation

Depreciation is the systematic allocation of the cost of a non-current asset, less its expected residual value, over its useful life. It is a charge made in the profit and loss account each year, and the accumulated total is deducted from the asset's cost in the balance sheet. Three points are essential:

- Depreciation is *not* a cash payment. Cash is spent when the asset is bought; the annual depreciation charge is only an accounting entry that spreads that cost.
- Depreciation is *not* an attempt to show the market value of the asset. The net book value in the balance sheet may differ widely from what the asset could be sold for.
- Depreciation is an application of the **accruals (matching) concept**: the cost of an asset is matched with the income it helps to earn.

Depreciation arises for several reasons. **Wear and tear** from use reduces the physical condition of machines and vehicles. **Obsolescence** occurs when an asset becomes out of date because of new inventions, for example when a faster printing machine is introduced. **Passage of time**

(effluxion of time) reduces the value of assets with a fixed life, such as a lease of ten years. **Depletion** affects assets that are used up, such as a quarry or an oil well. Land is normally not depreciated, because it has an unlimited life.

To calculate depreciation we need three estimates: the **cost** of the asset (including delivery and installation), its **residual value** (scrap value) at the end of its life, and its **useful life** in years. Only the cost is known with certainty; the others are estimates, which is why depreciation figures are never exact.

6.2 The straight-line method

Under the **straight-line method** (equal instalment method) the same amount is charged every year:

Annual depreciation = (Cost − Residual value) ÷ Useful life

Alternatively, a fixed percentage of cost is used each year; if the rate is given as a percentage, the annual charge is simply rate × cost. Because the charge is the same every year, the net book value falls by equal steps until it reaches the residual value.

WORKED EXAMPLE 6.1

A bakery in Lagos buys an oven for ₦480,000. It is expected to last five years and then be sold as scrap for ₦30,000. Calculate the annual depreciation and the net book value after three years using the straight-line method.

Solution: Depreciable amount = 480,000 − 30,000 = ₦450,000. Annual depreciation = 450,000 ÷ 5 = ₦90,000. After three years accumulated depreciation = 3 × 90,000 = ₦270,000, so net book value = 480,000 − 270,000 = ₦210,000.

6.3 The reducing balance method

Under the **reducing balance method** (diminishing balance method) a fixed percentage is applied each year to the *net book value* at the start of the year, not to the original cost. The charge is therefore highest in the first year and falls steadily. This suits assets such as vehicles and computers that lose more of their value, and give more service, in their early years, and whose repair costs rise as they age. The total cost of using the asset (depreciation plus repairs) is then spread more evenly.

WORKED EXAMPLE 6.2

A machine costing ₦600,000 is depreciated at 20% per annum on the reducing balance. Calculate the depreciation charge and net book value for each of the first three years.

Solution:

Year	Net book value at start ₦	Depreciation (20%) ₦	Net book value at end ₦
1	600,000	120,000	480,000
2	480,000	96,000	384,000
3	384,000	76,800	307,200

Note that under this method the net book value never reaches zero, and the residual value is not used in the yearly calculation.

Key point. Straight line: the same amount every year, based on *cost*. Reducing balance: a falling amount, based on the *net book value* at the start of each year. Always read the question to see which base the percentage applies to. If it says "on cost" or does not name a method, use straight line.

Whichever method a business chooses should be applied **consistently** from year to year. A change of method must be justified and disclosed, because it affects profit and makes comparison between years misleading. This is the *consistency concept*.

6.4 Accounting entries for depreciation

Two approaches exist. In the older approach, depreciation is credited directly to the asset account, reducing the asset's recorded value each year. The modern and examination-preferred approach keeps the asset at cost and credits the depreciation to a separate **provision for depreciation account** (also called accumulated depreciation). Each year the entries are:

- Debit **Depreciation expense account** (or the profit and loss account) with the year's charge.
- Credit **Provision for depreciation account** with the same amount.

At the year end the expense account is closed by transferring its balance to the profit and loss account. The asset account continues to show cost, and the provision account shows the accumulated total. In the balance sheet, cost less accumulated depreciation gives the net book value.

WORKED EXAMPLE 6.3

Chukwuma Industries bought a machine on 1 January 2023 for ₦800,000 and depreciates it at 10% per annum on cost. The financial year ends on 31 December. Show the machine account and the provision for depreciation account for 2023 and 2024, and the balance sheet extract at the end of 2024.

Solution: Annual depreciation = 10% × 800,000 = ₦80,000. The asset account is unchanged at cost; only the provision grows.

Dr			Machine account			Cr
Date	**Details**	**₦**	**Date**	**Details**		**₦**
1 Jan 2023	Bank	800,000	31 Dec 2023	Balance c/d		800,000
1 Jan 2024	Balance b/d	800,000	31 Dec 2024	Balance c/d		800,000

Dr			Provision for depreciation: machine			Cr
Date	**Details**	**₦**	**Date**	**Details**		**₦**
31 Dec 2023	Balance c/d	80,000	31 Dec 2023	Profit and loss		80,000
31 Dec 2024	Balance c/d	160,000	1 Jan 2024	Balance b/d		80,000
			31 Dec 2024	Profit and loss		80,000
		160,000				**160,000**
			1 Jan 2025	Balance b/d		160,000

Balance sheet extract at 31 December 2024: machine at cost ₦800,000, less accumulated depreciation ₦160,000, net book value ₦640,000. The profit and loss account for each of 2023 and 2024 carries a charge of ₦80,000.

6.5 Depreciation for part of a year

Many assets are bought or sold during the year. Some businesses charge a full year in the year of purchase and none in the year of sale; others charge depreciation for the exact number of months of ownership. Unless a question states a policy, the safest course is to depreciate from the date of purchase, in months, and to say so.

WORKED EXAMPLE 6.4

A trader buys a truck on 1 April 2025 for ₦1,200,000 and depreciates trucks at 25% per annum on cost, charging depreciation for each month of ownership. The year ends on 31 December. Calculate the charge for 2025.

Solution: April to December is 9 months. Charge = 1,200,000 × 25% × 9/12 = 300,000 × 0.75 = ₦225,000.

6.6 Disposal of non-current assets

When an asset is sold or scrapped, its cost and its accumulated depreciation must be removed from the books, and any difference between its net book value and the sale price recorded as a **profit or loss on disposal**. These are accounting estimates being corrected: if the asset sells for more than its net book value, too much depreciation was charged in the past; if it sells for less, too little was charged.

A **disposal account** is used to collect all the figures. The three steps are:

1. Transfer the asset's cost to the disposal account: debit Disposal, credit Asset account.
2. Transfer the accumulated depreciation: debit Provision for depreciation, credit Disposal.
3. Record the sale proceeds: debit Bank (or the buyer), credit Disposal.

The balancing figure is then the profit or loss, which is carried to the profit and loss account. If the credit side exceeds the debit side before balancing, the balancing figure goes on the debit side and is a **profit**; if the debit side exceeds the credit side, the balancing figure goes on the credit side and is a **loss**.

WORKED EXAMPLE 6.5

The machine in Worked example 6.3 (cost ₦800,000, bought 1 January 2023, depreciated at 10% on cost) is sold on 31 December 2025 for ₦540,000, received in cash at bank. Show the disposal account and state the profit or loss.

Solution: Accumulated depreciation at 31 December 2025 is 3 × 80,000 = ₦240,000. Net book value = 800,000 − 240,000 = ₦560,000. Proceeds are ₦540,000, so there is a loss of 560,000 − 540,000 = ₦20,000.

Dr		Disposal account		Cr	
Date	**Details**	**₦**	**Date**	**Details**	**₦**
31 Dec 2025	Machine (cost)	800,000	31 Dec 2025	Provision for depreciation	240,000
			31 Dec 2025	Bank	540,000
			31 Dec 2025	Profit and loss (loss)	20,000
		800,000			**800,000**

The debit side (800,000) exceeds the credits (780,000), so the balancing figure of ₦20,000 is on the credit side and is a loss. It is charged to the profit and loss account as an expense.

WORKED EXAMPLE 6.6

A delivery vehicle that cost ₦1,200,000 has accumulated depreciation of ₦900,000. It is sold for ₦350,000. Find the profit or loss on disposal.

Solution: Net book value = 1,200,000 − 900,000 = ₦300,000. Proceeds ₦350,000 exceed this by ₦50,000, so there is a profit of ₦50,000. In the disposal account the debit side is 1,200,000 and the credit side is 900,000 + 350,000 = 1,250,000, so the balancing figure of ₦50,000 is on the debit side, which confirms a profit. It is credited to the profit and loss account as income.

Key point. Profit or loss on disposal = sale proceeds − net book value. A profit is added in the profit and loss account; a loss is deducted. It is never part of the trading account, since the asset was not bought for resale.

6.7 The purpose of control accounts

A trader with 300 customers keeps 300 accounts in the sales ledger. Finding an error among them when the books disagree would be slow. A **control account** (total account) summarises the ledger. It is a ledger account in which are recorded the *totals* of all the transactions that were entered in the individual accounts of the ledger. Its balance should equal the sum of the balances on all the individual accounts. If it does not, there is an error to be found.

Control accounts have four uses. They provide a check on the arithmetical accuracy of the ledgers. They help to locate errors, because the difference tells where to look. They reduce the risk of fraud, particularly when the person who keeps the control accounts is not the person who keeps the individual accounts. They also give the total receivables and payables immediately, which is helpful for preparing the balance sheet quickly. A control account is a memorandum check: when kept as part of the double-entry system it is a real ledger account, and the individual accounts are then only memorandum records.

6.8 The sales ledger control account

The **sales ledger control account** (total receivables account) summarises the customers' accounts. Items that increase the amount owed by customers go on the debit side, and items that reduce it go on the credit side.

Debit side (increases receivables)	Credit side (decreases receivables)
Opening balance of receivables	Cash and cheques received from customers
Credit sales (from the sales day book)	Discount allowed
Dishonoured cheques	Returns inwards (from the returns inwards book)
Interest charged to customers	Bad debts written off
Cash refunded to customers	Contra entries (set-off against payables)
	Closing balance carried down

Only *credit* sales appear. Cash sales do not go through the receivables ledger, so they must be excluded. Any credit balance on a customer's account (for example, from overpayment) is normally shown separately.

WORKED EXAMPLE 6.7

For the year 2025 the books of a Kano wholesaler show: receivables at 1 January ₦124,000; credit sales ₦860,000; cheques and cash received from customers ₦790,000; discount allowed ₦12,000; returns inwards ₦18,000; bad debts written off ₦6,000; cheques from customers dishonoured ₦5,000. Prepare the sales ledger control account and find the closing balance.

Solution: Debits = 124,000 + 860,000 + 5,000 = ₦989,000. Credits before the balance = 790,000 + 12,000 + 18,000 + 6,000 = ₦826,000. Balance c/d = 989,000 − 826,000 = ₦163,000.

Dr		**Sales ledger control account**		**Cr**	
Date	**Details**	**₦**	**Date**	**Details**	**₦**
1 Jan	Balance b/d	124,000	31 Dec	Bank (receipts)	790,000
31 Dec	Credit sales	860,000	31 Dec	Discount allowed	12,000
31 Dec	Bank (dishonoured cheques)	5,000	31 Dec	Returns inwards	18,000
			31 Dec	Bad debts	6,000
			31 Dec	Balance c/d	163,000
		989,000			**989,000**
1 Jan 2026	Balance b/d	163,000			

6.9 The purchases ledger control account

The **purchases ledger control account** (total payables account) is the mirror image. Items that increase what the business owes go on the credit side, and items that reduce it go on the debit side.

WORKED EXAMPLE 6.8

For the same wholesaler, the creditors' figures are: payables at 1 January ₦98,000; credit purchases ₦540,000; payments to suppliers ₦505,000; discount received ₦9,000; returns outwards ₦14,000. Prepare the purchases ledger control account.

Solution: Debits before the balance = 505,000 + 9,000 + 14,000 = ₦528,000. Credits = 98,000 + 540,000 = ₦638,000. Balance c/d = 638,000 − 528,000 = ₦110,000, which is a credit balance (amount owed).

Dr		**Purchases ledger control account**		**Cr**	
Date	**Details**	**₦**	**Date**	**Details**	**₦**
31 Dec	Bank (payments)	505,000	1 Jan	Balance b/d	98,000
31 Dec	Discount received	9,000	31 Dec	Credit purchases	540,000
31 Dec	Returns outwards	14,000			
31 Dec	Balance c/d	110,000			
		638,000			**638,000**
1 Jan 2026				Balance b/d	110,000

Contra entries

Some persons are both customers and suppliers. If Mr Okonkwo owes the business ₦7,000 for goods bought and the business owes him ₦9,000 for goods supplied, the two debts can be set off, and only the difference (₦2,000 owed to him) need be paid. In the control accounts, the contra of ₦7,000 is a credit in the sales ledger control account and a debit in the purchases ledger control account. In the individual ledgers, Okonkwo's receivable account is credited and his payable account is debited by ₦7,000.

6.10 Reconciling control accounts with the ledgers

At the end of the period, the balance on the control account is compared with the total of the list of balances taken from the individual ledger accounts. If the two figures differ, the cause is one or more errors. Errors that affect the individual accounts only (for example, a balance omitted

from the list or a posting to the wrong customer) require the list of balances to be corrected. Errors in the day books or in the control account totals (for example, a day book undercast) require the control account to be corrected. The steps are:

1. Establish the two totals: the control account balance and the total of the list of balances.
2. Decide for each error whether it affects the control account or the list.
3. Correct each total in turn, and show that the corrected totals agree.

WORKED EXAMPLE 6.9

On 31 December the sales ledger control account shows a balance of ₦163,000, but the list of individual customers' balances totals ₦162,800. Investigation finds: (a) the sales day book was undercast by ₦1,000; (b) a customer's balance of ₦1,200 was left out of the list. Reconcile the two figures.

Solution: Error (a) concerns the total posted to the control account. Corrected control balance = 163,000 + 1,000 = ₦164,000. Error (b) concerns the list. Corrected list total = 162,800 + 1,200 = ₦164,000. The two now agree at ₦164,000, which is the true amount owed by customers.

Note that agreement of the control account and the list does not prove everything is right. A sale posted to the wrong customer's account, or an entry omitted from both the day book and the ledger, would not cause any difference between them.

Exercise 6.1

1. Calculate the annual straight-line depreciation on a machine costing ₦360,000, with an expected residual value of ₦40,000 and a life of eight years.
2. A computer costing ₦400,000 is depreciated at 25% per annum on the reducing balance. Find its net book value after two years.
3. A machine costs ₦500,000, has a residual value of ₦50,000 and an expected life of five years. Using the straight-line method, find its net book value after three years.
4. Equipment costing ₦900,000 is bought on 1 October 2025 and depreciated at 10% per annum on cost, by months of ownership. Find the charge for the year ended 31 December 2025.
5. An asset that cost ₦250,000 has accumulated depreciation of ₦175,000 and is sold for ₦90,000. Calculate the profit or loss on disposal.

6. An asset that cost ₦1,000,000 has accumulated depreciation of ₦600,000 and is sold for ₦350,000. Calculate the profit or loss and state how it is treated in the profit and loss account.
7. Sales ledger: receivables at 1 January ₦45,000; credit sales ₦310,000; cash received from customers ₦280,000; discount allowed ₦4,000; returns inwards ₦6,000. Find the closing balance on the sales ledger control account.
8. Purchases ledger: payables at 1 January ₦30,000; credit purchases ₦180,000; payments ₦170,000; discount received ₦3,000; returns outwards ₦5,000. Find the closing balance on the purchases ledger control account.
9. A customer who owes ₦2,500 is also a supplier to whom the business owes ₦2,500 for goods. The two debts are set off. Using the closing balances from questions 7 and 8, state the control account balances after this contra.
10. Give three reasons why a business prepares control accounts.

Answers to Exercise 6.1

1. ₦40,000: (360,000 − 40,000) ÷ 8 = 320,000 ÷ 8 = 40,000.
2. ₦225,000: year 1 depreciation 25% × 400,000 = 100,000, NBV 300,000; year 2 depreciation 75,000, NBV 225,000.
3. ₦230,000: annual charge (500,000 − 50,000) ÷ 5 = 90,000; three years = 270,000; 500,000 − 270,000 = 230,000.
4. ₦22,500: 900,000 × 10% × 3/12 = 90,000 × 0.25 = 22,500.
5. Profit of ₦15,000: NBV = 250,000 − 175,000 = 75,000; proceeds 90,000 − 75,000 = 15,000.
6. Loss of ₦50,000: NBV = 1,000,000 − 600,000 = 400,000; proceeds 350,000; loss 50,000, charged as an expense in the profit and loss account.
7. ₦65,000: 45,000 + 310,000 = 355,000; less 280,000 + 4,000 + 6,000 = 290,000; 355,000 − 290,000 = 65,000.
8. ₦32,000: credits 30,000 + 180,000 = 210,000; debits 170,000 + 3,000 + 5,000 = 178,000; 210,000 − 178,000 = 32,000.
9. Sales ledger control account becomes 65,000 − 2,500 = ₦62,500 (debit); purchases ledger control account becomes 32,000 − 2,500 = ₦29,500 (credit).
10. Any three: they check the arithmetical accuracy of the ledgers; they help locate errors; they reduce the risk of fraud; they give total receivables and payables quickly for the balance sheet.

Chapter summary

- Depreciation spreads the cost of a non-current asset over its useful life; it is a non-cash expense and does not measure market value.
- Causes include wear and tear, obsolescence, passage of time and depletion; land is not normally depreciated.
- Straight line: (cost − residual value) ÷ life, equal each year. Reducing balance: a fixed percentage of the net book value at the start of each year.
- The asset account remains at cost; the annual charge is debited to the profit and loss account and credited to the provision for depreciation account.
- For part-year ownership, depreciation is time-apportioned: cost × rate × months/12.
- On disposal, transfer the cost and the accumulated depreciation to the disposal account, record the proceeds, and take the balancing figure to profit and loss as a profit or loss.
- Profit or loss on disposal = proceeds − net book value.
- A sales ledger control account records total credit sales, receipts, discounts, returns and bad debts; a purchases ledger control account is its mirror image, and contras reduce both.
- Reconciliation corrects the control account for errors in day books and totals, and the list of balances for errors in individual accounts, until both agree.

Chapter 7: Partnership Accounts

One person alone seldom has all the money, skill and time that a growing business requires. Two doctors may open a clinic together, three friends may start a transport business, and a trader may bring in a partner who has capital. Whenever two or more people agree to carry on business together with a view to making profit, they form a **partnership**. In Nigeria such businesses are regulated by the Partnership Act, which applies to the extent that the partners have not agreed otherwise in writing.

The accounting for a partnership follows the same principles as for a sole trader: a trading account, a profit and loss account and a balance sheet. What is new is the need to share the profit between the partners, to keep a separate record for each partner's capital and drawings, and to deal with changes in the membership of the firm. This chapter covers each of these in turn.

7.1 The nature of a partnership and the partnership agreement

A partnership has between two and twenty members in most ordinary trading firms (professional firms may have more). Its main features are these:

- Partners contribute capital and often share in management.
- Profits and losses are shared in an agreed ratio.
- Every partner is personally liable for the debts of the firm, without limit, and each can bind the others by acts done in the ordinary course of business.
- The firm is not a separate legal person from its partners, although for accounting purposes it is treated as a business entity separate from the owners.

The partners should draw up a **partnership agreement** (deed of partnership) stating the matters that affect the accounts: the capital each will contribute, the profit-sharing ratio, any interest on capital, any interest on drawings, any partners' salaries, and the rules for admission and retirement of partners.

If there is no agreement, the Partnership Act provides the following rules:

1. Profits and losses are shared **equally**, whatever the capital contributed.
2. No interest is allowed on partners' capital.
3. No interest is charged on partners' drawings.

4. No partner is entitled to a salary.
5. A partner who lends money to the firm beyond the agreed capital is entitled to interest on the loan at 5% per annum.

> **Key point.** If a question says nothing about how profits are shared, share them equally and make none of the other appropriations. A partner's *loan* interest is an expense of the business charged before the net profit is found; it is not part of the appropriation of profit.

7.2 The partners' capital and current accounts

A sole trader has a single capital account. A partnership keeps, for each partner, a capital account and usually also a current account.

Under the **fixed capital** method, each partner's capital account holds only the capital agreed in the partnership agreement and changes only when the capital is formally altered. All other items go through a separate **current account**: the partner's share of profit, interest on capital and salary are credited, and drawings and interest on drawings are debited. A credit balance on the current account means the firm owes the partner; a debit balance means the partner has drawn more than has been earned.

Under the **fluctuating capital** method there is no current account, and the capital account itself is increased by profit and reduced by drawings, so its balance changes every year. The fixed capital method is preferred because it separates capital, which is a long-term investment, from the yearly movement of profit and drawings.

In the balance sheet, the capital and current accounts of every partner are shown, and their total equals the net assets of the firm.

7.3 The appropriation account

After the net profit has been found in the profit and loss account, it is carried to the **profit and loss appropriation account**, which shows how the profit is divided among the partners. It is a continuation of the profit and loss account. The usual items are:

- **Interest on capital.** A reward to partners for investing money. It is calculated as capital × rate, usually on the fixed capital balance. It is deducted in the appropriation account and credited to each partner's current account.

- **Partners' salaries.** A reward for work done for the firm, deducted in the appropriation account and credited to the partner's current account. Salaries here are an appropriation of profit and not an expense of the business.
- **Interest on drawings.** A charge made to discourage partners from withdrawing money too early. It is *added* to the profit available for sharing and debited to the partner's current account. It is calculated as drawings × rate × time outstanding (from the date of withdrawal to the year end).
- **Share of the remaining profit.** What is left after the above is divided in the agreed profit-sharing ratio.

The profit-sharing ratio is often expressed as a ratio, for example 3:2, which means that the first partner takes 3/5 and the second 2/5 of the residual profit. A loss after the appropriations is shared in the same ratio.

WORKED EXAMPLE 7.1

Adewale and Bisi are in partnership in Abeokuta and share profits and losses in the ratio 3:2. Their fixed capitals are Adewale ₦400,000 and Bisi ₦200,000. The partnership agreement provides for interest on capital at 10% per annum, a salary of ₦30,000 per year for Bisi, and interest on drawings at 5% per annum. Net profit for the year ended 31 December 2025 was ₦180,000. Drawings for the whole year were Adewale ₦60,000 and Bisi ₦40,000, taken at the start of the year. The partners' current accounts had no balances at 1 January. Prepare the appropriation account and the current accounts.

Solution:

- Interest on capital: Adewale 10% × 400,000 = ₦40,000; Bisi 10% × 200,000 = ₦20,000; total ₦60,000.
- Interest on drawings (full year at 5%): Adewale 5% × 60,000 = ₦3,000; Bisi 5% × 40,000 = ₦2,000; total ₦5,000.
- Profit available for sharing = 180,000 + 5,000 − 60,000 − 30,000 = ₦95,000.
- Shares of ₦95,000 in the ratio 3:2: one part = 95,000 ÷ 5 = 19,000; Adewale 3 × 19,000 = ₦57,000; Bisi 2 × 19,000 = ₦38,000.

Appropriation account for the year ended 31 December 2025	**₦**	**₦**
Net profit		180,000
Interest on drawings: Adewale	3,000	
Interest on drawings: Bisi	2,000	5,000
		185,000
Interest on capital: Adewale	40,000	
Interest on capital: Bisi	20,000	
Salary: Bisi	30,000	(90,000)
Profit available for sharing		95,000
Share of profit: Adewale (3/5)	57,000	
Share of profit: Bisi (2/5)	38,000	(95,000)

Dr	Current accounts			Cr	
Details	**Adewale ₦**	**Bisi ₦**	**Details**	**Adewale ₦**	**Bisi ₦**
Drawings	60,000	40,000	Interest on capital	40,000	20,000
Interest on drawings	3,000	2,000	Salary		30,000
Balance c/d	34,000	46,000	Share of profit	57,000	38,000
	97,000	**88,000**		**97,000**	**88,000**

Check: Adewale's credits 40,000 + 57,000 = 97,000, debits 60,000 + 3,000 = 63,000, balance 34,000 credit. Bisi's credits 20,000 + 30,000 + 38,000 = 88,000, debits 40,000 + 2,000 = 42,000, balance 46,000 credit. In the balance sheet, the partners' interests are: capital accounts 400,000 + 200,000 = 600,000; current accounts 34,000 + 46,000 = 80,000; total ₦680,000.

WORKED EXAMPLE 7.2

Efe, Gbenga and Hauwa are partners with no written agreement. Hauwa lent the firm ₦80,000 two years ago. The profit for the year before charging her loan interest was ₦124,000. Calculate each partner's share of the profit and the total Hauwa receives from the firm for the year.

Solution: With no agreement, loan interest is 5% per annum: 5% × 80,000 = ₦4,000. It is an expense, so profit for sharing = 124,000 − 4,000 = ₦120,000. Profit is shared equally: 120,000 ÷ 3 = ₦40,000 each. Hauwa receives 40,000 (profit share) + 4,000 (loan interest) = ₦44,000.

7.4 The partnership balance sheet

The balance sheet of a partnership is laid out like that of a sole trader, except that the financing section lists each partner's capital account and current account. Capital is usually presented in a table of columns, one for each partner. A debit balance on a current account is deducted from the partner's capital.

7.5 Goodwill

Goodwill is the value of a business over and above the fair value of its net assets. It exists because a business may have a good reputation, a loyal group of customers, a favourable location or skilled staff. A well-known shop in Balogun Market, for instance, may earn much more profit than the value of its stock and fittings alone would suggest. Goodwill is an intangible non-current asset. Under the prudence concept it is recorded in the books only when it has been *purchased*, for example when a new partner pays for a share in the firm's reputation; goodwill built up over years by the business itself is not normally shown in the balance sheet.

Goodwill needs to be valued when there is a change in the partnership, because the old partners built up the goodwill, and it would be unfair if a newcomer shared in the future profits without paying for it. Valuation methods include a number of years' purchase of average profits, or the excess of expected profit over a normal return.

7.6 Admission of a new partner

When a new partner is admitted, the old partnership comes to an end and a new one begins, with a new profit-sharing ratio. The old partners give up part of their share of profit to the new partner. The proportion each one gives up is the **sacrifice**, and the ratio of the sacrifices is the **sacrificing ratio**. The new partner compensates them for the goodwill, either by paying cash directly or through an adjustment in the capital accounts.

The usual method is to raise goodwill in the books and write it off immediately:

1. Debit Goodwill and credit the old partners' capital accounts with the value of goodwill, in the *old* profit-sharing ratio.
2. Write off goodwill by debiting all partners' capital accounts, in the *new* profit-sharing ratio, and crediting Goodwill.

WORKED EXAMPLE 7.3

Ade and Bola share profits in the ratio 3:1. Their capital balances are Ade ₦300,000 and Bola ₦200,000. They admit Chidi as a partner with a one-fifth share of profits. Chidi pays ₦150,000 into the bank as his capital, and the remaining profit is to be shared between Ade and Bola in their old ratio. Goodwill is valued at ₦200,000 and is not to remain in the books. Calculate the new profit ratio and prepare the capital accounts.

Solution: Chidi takes 1/5, leaving 4/5. Ade gets 3/4 of 4/5 = 3/5 and Bola gets 1/4 of 4/5 = 1/5. The new ratio is therefore 3:1:1, in fifths. Sacrifice: Ade 3/4 − 3/5 = 15/20 − 12/20 = 3/20; Bola 1/4 − 1/5 = 5/20 − 4/20 = 1/20; the sacrificing ratio is 3:1. Chidi's share of goodwill is 1/5 × 200,000 = ₦40,000, which compensates Ade and Bola in the ratio 3:1 (₦30,000 and ₦10,000).

Step 1: Goodwill created, shared in the old ratio 3:1: Ade 150,000; Bola 50,000. Step 2: Goodwill written off in the new ratio 3:1:1: Ade 120,000; Bola 40,000; Chidi 40,000.

Dr	Capital accounts			Cr			
Details	**Ade ₦**	**Bola ₦**	**Chidi ₦**	**Details**	**Ade ₦**	**Bola ₦**	**Chidi ₦**
Goodwill written off	120,000	40,000	40,000	Balance b/d	300,000	200,000	
Balance c/d	330,000	210,000	110,000	Goodwill created	150,000	50,000	
				Bank			150,000
	450,000	**250,000**	**150,000**		**450,000**	**250,000**	**150,000**

Check: the closing capitals total 330,000 + 210,000 + 110,000 = ₦650,000, which equals the old net assets of 500,000 plus the new cash of 150,000. Goodwill has been created and written off with no effect on the total.

7.7 Retirement and death of a partner

When a partner retires or dies, the partnership agreement is ended and a new one begins between the remaining partners. The same ideas apply in reverse. Goodwill is valued, and the outgoing partner (or the estate of a deceased partner) is credited with a share of it, to be paid for by the remaining partners through their capital accounts in the ratio in which they gain the retiring

partner's share (the *gaining ratio*). Assets may be revalued first, and any profit or loss on revaluation is shared in the old ratio. The amount due to the outgoing partner is then paid in cash or transferred to a loan account to be repaid in instalments.

7.8 Dissolution of a partnership

A partnership is **dissolved** when it comes to an end, for example by agreement, through the bankruptcy of a partner, or by court order. The assets are sold or taken over, the liabilities are paid, and the balance is divided among the partners according to their capital account balances. The accounts used are the **realisation account**, the partners' capital accounts and the bank account.

The steps are as follows. The book values of all assets (except cash and bank) are debited to the realisation account, and the liabilities to be settled are credited to it. The cash received from selling the assets is credited to the realisation account, the payments to settle liabilities and the expenses of dissolution are debited, and the balancing figure is the profit or loss on realisation, shared among the partners in the profit-sharing ratio. Finally, the balance on each partner's capital account is paid out of the bank so that the bank account is cleared.

WORKED EXAMPLE 7.4

Peter and Queen, who share profits equally, dissolve their partnership. At that date the balances were: premises ₦300,000; inventory ₦80,000; receivables ₦50,000; bank ₦40,000; payables ₦70,000; capital accounts: Peter ₦250,000 and Queen ₦150,000. The premises were sold for ₦340,000; inventory for ₦60,000; receivables paid ₦46,000; payables were settled for ₦66,000; the expenses of dissolution were ₦4,000. Prepare the realisation account, the capital accounts and the bank account.

Solution: Net assets agree with capital: assets 300,000 + 80,000 + 50,000 + 40,000 = 470,000; less payables 70,000 = 400,000 = 250,000 + 150,000.

Dr	**Realisation account**		**Cr**
Details	**₦**	**Details**	**₦**
Premises	300,000	Payables	70,000
Inventory	80,000	Bank: premises sold	340,000
Receivables	50,000	Bank: inventory sold	60,000
Bank: payables paid	66,000	Bank: receivables collected	46,000
Bank: expenses	4,000		
Profit on realisation: Peter	8,000		
Profit on realisation: Queen	8,000		
	516,000		**516,000**

Debits before profit: 300,000 + 80,000 + 50,000 + 66,000 + 4,000 = 500,000. Credits: 70,000 + 340,000 + 60,000 + 46,000 = 516,000. Profit = 16,000, shared equally at 8,000 each. Capital accounts after the profit: Peter 250,000 + 8,000 = ₦258,000; Queen 150,000 + 8,000 = ₦158,000. Bank account: opening 40,000 + 340,000 + 60,000 + 46,000 = 486,000; payments 66,000 + 4,000 = 70,000; balance 416,000. This equals 258,000 + 158,000 = ₦416,000, so Peter receives ₦258,000 and Queen ₦158,000 and the bank account is closed.

Exercise 7.1

1. State four rules of the Partnership Act that apply when partners have no agreement.

2. Tade and Uche share profits and losses 2:1. If the profit to be shared is ₦60,000, how much does each receive?
3. Calculate the interest on capital at 8% per annum for partners with capitals of ₦500,000 and ₦300,000.
4. A firm has net profit of ₦150,000. Partner A is entitled to a salary of ₦20,000, and interest on capital is A ₦18,000 and B ₦12,000. The balance is shared equally. Find the total amount credited to each partner.
5. A partner drew ₦90,000 on 1 July and the year ends on 31 December. Interest on drawings is 8% per annum. Calculate the interest charged.
6. Explain the difference between the fixed capital method and the fluctuating capital method.
7. Ola and Pam share profits equally and admit Quincy for a one-quarter share, the remaining three quarters being shared equally by Ola and Pam. Goodwill is ₦120,000. Find the new ratio, each old partner's sacrifice, and the amount Quincy brings in for goodwill.
8. A and B share profits 3:1. C is admitted with a one-fifth share and the rest is divided between A and B in their old ratio. Find the new profit ratio.
9. A partnership with two equal partners made a profit of ₦90,000 before charging interest on a ₦50,000 loan from one partner at 6% per annum. Find the profit each partner shares.
10. On dissolution, an asset with book value ₦80,000 is sold for ₦72,000. The partners share profit 3:1. Calculate the loss each bears.

Answers to Exercise 7.1

1. Any four: profits and losses shared equally; no interest on capital; no interest on drawings; no partnership salaries; interest at 5% on partners' loans.
2. Tade ₦40,000 (2/3 × 60,000) and Uche ₦20,000 (1/3 × 60,000).
3. 8% × 500,000 = ₦40,000 and 8% × 300,000 = ₦24,000.
4. Balance to share = 150,000 − 20,000 − 18,000 − 12,000 = 100,000, so each gets 50,000. A: 20,000 + 18,000 + 50,000 = ₦88,000; B: 12,000 + 50,000 = ₦62,000; total 150,000.
5. ₦3,600: 90,000 × 8% × 6/12 = 3,600.
6. Under fixed capital, capital accounts do not change and all profit items and drawings pass through current accounts; under fluctuating capital, there are no current accounts and the

capital balance changes with profit and drawings.

7. New ratio Ola:Pam:Quincy = 3/8 : 3/8 : 1/4 = 3:3:2. Each sacrifices 1/2 − 3/8 = 1/8. Quincy's share of goodwill is 1/4 × 120,000 = ₦30,000, credited to Ola and Pam at ₦15,000 each.
8. Remaining 4/5 shared 3:1 gives A 3/5 and B 1/5; new ratio A:B:C = 3:1:1.
9. Loan interest = 6% × 50,000 = 3,000; profit shared = 90,000 − 3,000 = 87,000; each partner gets ₦43,500 (the lending partner also receives the ₦3,000 interest).
10. Loss = 80,000 − 72,000 = ₦8,000; the first partner bears 3/4 × 8,000 = ₦6,000 and the second 1/4 × 8,000 = ₦2,000.

Chapter summary

- A partnership is formed by two or more persons carrying on business together to make profit; partners are personally liable for the firm's debts.
- Without an agreement, profits are shared equally, with no interest on capital or drawings and no salaries, but loans earn 5% interest.
- Under the fixed capital method, each partner has a capital account that does not change and a current account for profit share, interest, salary and drawings.
- The appropriation account divides the net profit: deduct interest on capital and salaries, add interest on drawings, and share the balance in the profit ratio.
- Interest on a partner's loan is an expense charged before net profit; salary and interest on capital are appropriations.
- Goodwill is the excess value of a business over its net assets; it is recorded only when purchased, and is usually created and written off on a change of partners.
- On admission, old partners sacrifice part of their shares; goodwill is credited to them in the old ratio and written off in the new ratio.
- On dissolution, the realisation account collects book values and proceeds; its profit or loss is shared by the partners, and the capital balances are then paid out of the bank.

Chapter 8: Company Accounts

A sole trader and a partnership are limited by the wealth and the lifetime of the people who own them. A business that wishes to raise large sums of money, to survive the death of its founders and to protect its owners from personal ruin needs a different structure: the **company**. In Nigeria, companies are formed and regulated under the Companies and Allied Matters Act (CAMA), and are registered with the Corporate Affairs Commission. Banks, breweries, cement manufacturers, telecommunications firms and many small family enterprises in Lagos, Kano and Port Harcourt operate as companies.

This chapter explains what makes a company different, how share capital is raised and recorded, how reserves arise, and how the final accounts of a company are prepared. The central ideas are simple. The owners are called **shareholders**; their capital is divided into **shares**; the profit they are entitled to is paid as **dividends**; and the profit retained is shown as **reserves**. The income statement is extended by an **appropriation account**, and the balance sheet shows the owners' interest under the heading of share capital and reserves.

8.1 Features of a limited liability company

A company is a **separate legal person**. It can own property, borrow money, sign contracts, sue and be sued in its own name. The assets belong to the company, not to the shareholders, and the debts are the debts of the company. Three consequences follow.

- **Limited liability.** If the company cannot pay its debts, a shareholder can lose only the amount he has paid, or has agreed to pay, on his shares. His house, car and savings cannot be seized to settle the company's debts.
- **Perpetual succession.** The company continues even when shareholders die, sell their shares or leave. Ownership changes hands, but the company carries on.
- **Transferability of shares.** Shares in a public company can be bought and sold, usually through the Nigerian Exchange, without disturbing the business.

A **private company** restricts the right to transfer its shares and may not offer them to the public. A **public company** may invite the public to subscribe for its shares and debentures, and its shares may be quoted on the stock exchange. A company is managed by **directors** elected by shareholders, and its accounts must be presented to the shareholders each year at the annual general meeting.

Key point. In a company the owners are shareholders, the managers are directors, and the business is a separate legal person. Shareholders receive *dividends*; they do not take "drawings", and they do not hold a "capital account" that moves with profit and loss, as the partners do in Chapter 7.

8.2 Share capital

The capital of a company is divided into shares of a fixed **nominal** (par or face) value, such as ₦1 or 50 kobo. The following terms must be used exactly.

Term	Meaning
Authorised (registered or nominal) capital	The maximum share capital the company is permitted to issue, as stated in its memorandum
Issued capital	The nominal value of the shares actually issued to shareholders
Called-up capital	The part of the nominal value of the issued shares which the company has asked shareholders to pay
Paid-up capital	The part of the called-up capital which has actually been received
Calls in arrear	Called-up money not yet paid by shareholders
Uncalled capital	The part of the issued nominal value not yet asked for

For example, a company may be authorised to issue 2,000,000 shares of ₦1 each, which is an authorised capital of ₦2,000,000. If it issues 1,500,000 shares, issued capital is ₦1,500,000. If it has called 80 kobo on each issued share, called-up capital is 1,500,000 × 0.80 = ₦1,200,000. If shareholders of 50,000 shares have not paid their 80 kobo, calls in arrear are 50,000 × 0.80 = ₦40,000 and paid-up capital is 1,200,000 − 40,000 = ₦1,160,000. Only issued capital appears in the balance sheet; the authorised capital is shown by way of a note.

Ordinary shares and preference shares

Ordinary shares carry the main risk of the business. Ordinary shareholders have voting rights, and they receive a dividend only after all other claims have been met. The dividend varies with the profit: it may be large in a good year and nil in a bad one. If the company is wound up, they are paid last, but they also enjoy all the surplus.

Preference shares carry a fixed rate of dividend, for example 10 per cent of the nominal value, which must be paid before any ordinary dividend. In the event of winding up, preference shareholders are repaid before ordinary shareholders. They usually have no vote. *Cumulative* preference shares carry forward any unpaid dividend to the following years; *non-cumulative* shares do not. *Redeemable* preference shares may be bought back by the company on a stated date.

Debentures

A **debenture** is a loan to the company, evidenced by a certificate and usually secured on its assets. A debenture holder is a *creditor*, not an owner. Debenture interest is paid at a fixed rate whether or not the company makes a profit, and it is charged as an expense in the income statement *before* arriving at profit. The debenture itself is shown in the balance sheet as a non-current liability when repayable after more than one year. This is the key difference from a dividend, which is paid out of profit and is shown in the appropriation account.

Feature	Ordinary share	Preference share	Debenture
Holder is	Owner	Owner	Creditor
Return	Variable dividend	Fixed-rate dividend	Fixed interest
Paid if no profit?	No	No	Yes
Return charged	Appropriation of profit	Appropriation of profit	Expense in income statement
Vote	Yes	Usually no	No
Repayment on winding up	Last	Before ordinary	First (before shareholders)

8.3 Issuing shares

A company that wishes to raise capital issues a **prospectus** inviting the public to apply for shares. The shares may be payable in one sum on application, or by instalments: an amount on **application**, a further amount on **allotment** (when the directors decide who receives shares), and later amounts on **calls**. Every receipt is debited to the Bank account. The credit is to Share Capital for the nominal value, and to **Share Premium** for any excess over nominal value.

A share is issued at a **premium** when the issue price exceeds the nominal value. A share of nominal value ₦1 issued at ₦1.40 carries a premium of 40 kobo. A share is never issued below its nominal value in Nigeria, so a discount on shares does not arise at this level. The premium is a

capital reserve; it cannot be distributed as dividend, but it may be used to pay up bonus shares.

WORKED EXAMPLE 8.1

Adamu Foods Plc issues 200,000 ordinary shares of ₦1 each at ₦1.40 per share, payable as follows: 50 kobo on application, 60 kobo on allotment (including the premium) and 30 kobo on the first and final call. All the shares are applied for and allotted, and all the money is received. Show the journal entries and the Bank account.

Solution: The amounts received are: application 200,000 × 0.50 = ₦100,000; allotment 200,000 × 0.60 = ₦120,000; call 200,000 × 0.30 = ₦60,000. Total received = 100,000 + 120,000 + 60,000 = ₦280,000, which equals 200,000 × 1.40. Share capital = 200,000 × 1 = ₦200,000; share premium = 200,000 × 0.40 = ₦80,000; and 200,000 + 80,000 = ₦280,000.

Stage	Debit	Credit	₦
Application money received (200,000 × 0.50)	Bank	Application	100,000
Application money transferred to capital	Application	Ordinary Share Capital	100,000
Allotment money due (200,000 × 0.60)	Allotment	Ordinary Share Capital (200,000 × 0.20) Share Premium (200,000 × 0.40)	40,000 80,000
Allotment money received	Bank	Allotment	120,000
Call due (200,000 × 0.30)	Call	Ordinary Share Capital	60,000
Call money received	Bank	Call	60,000

The Bank account is debited 100,000 + 120,000 + 60,000 = ₦280,000. Share capital is credited 100,000 + 40,000 + 60,000 = ₦200,000, which is 200,000 shares at ₦1. Share premium is credited ₦80,000. The Application, Allotment and Call accounts each close with a nil balance because every sum due was received. Note that the 60 kobo due on allotment is split into 20 kobo of nominal value and 40 kobo of premium; the shares reach their full nominal value of ₦1 only when the call of 30 kobo is made (0.50 + 0.20 + 0.30 = 1.00).

Oversubscription

When the public applies for more shares than are on offer, the issue is **oversubscribed**. The directors may reject some applications, or allot fewer shares than applied for. Money received for shares not allotted must be returned to the applicants, or may be kept and credited against the amount due on allotment.

WORKED EXAMPLE 8.2

Bola Textiles Ltd offers 100,000 ordinary shares of ₦1 at ₦1.20, payable in full on application. Applications are received for 150,000 shares. Shares are allotted to the applicants of 100,000 shares and the money for the remainder is refunded. Calculate the cash received, the refund, the share capital and the share premium.

Solution: Cash received = 150,000 × 1.20 = ₦180,000. Refund = 50,000 × 1.20 = ₦60,000. Cash retained = 180,000 − 60,000 = ₦120,000. Share capital = 100,000 × 1 = ₦100,000. Share premium = 100,000 × 0.20 = ₦20,000. Check: 100,000 + 20,000 = ₦120,000, which agrees with the cash retained.

8.4 Reserves

Shareholders' funds consist of share capital plus **reserves**. Reserves are profits or gains that belong to the shareholders but have not been paid out as dividends. There are two kinds.

- **Capital reserves** arise from gains that are not trading profits, such as share premium and the surplus on revaluing land and buildings. They cannot be paid out as dividends.
- **Revenue reserves** arise from retained trading profits, such as the *general reserve* and the balance on the income statement, usually called *retained earnings* (or retained profit). They may be distributed as dividends, subject to the law, but the directors keep them in the business to finance growth.

Exam tip. A reserve is *not* a sum of cash set aside. A "general reserve" of ₦100,000 does not mean that ₦100,000 sits in the bank. The reserve is a claim of shareholders, and the money has been used in the business to buy assets such as inventory and machinery.

Bonus issues and rights issues

A **bonus issue** (or scrip or capitalisation issue) gives existing shareholders extra shares free of charge, in proportion to their holdings. The company receives no cash. It converts reserves into share capital: the reserves are debited and share capital is credited. Total shareholders' funds are unchanged.

A **rights issue** offers new shares to existing shareholders, in proportion to their holdings, usually at a price below the market price. The company receives cash, so total shareholders' funds increase.

WORKED EXAMPLE 8.3

The equity of Chukwu Engineering Plc is: ordinary share capital (1,000,000 shares of ₦1) ₦1,000,000; share premium ₦200,000; general reserve ₦100,000; retained earnings ₦300,000. The company makes (a) a bonus issue of 1 share for every 4 held, using the share premium first and the balance from the general reserve; then (b) a rights issue of 1 share for every 5 held at ₦1.50 per share, all taken up. Show the equity after each step.

Solution: (a) Bonus shares = 1,000,000 ÷ 4 = 250,000 shares, with nominal value ₦250,000. Share premium of ₦200,000 is used in full, and the remaining 250,000 − 200,000 = ₦50,000 comes from the general reserve. After the bonus issue: share capital 1,250,000; share premium nil; general reserve 100,000 − 50,000 = 50,000; retained earnings 300,000. Total = 1,250,000 + 50,000 + 300,000 = ₦1,600,000, unchanged from the original 1,000,000 + 200,000 + 100,000 + 300,000 = ₦1,600,000.

(b) Rights shares = 1,250,000 ÷ 5 = 250,000 shares. Cash received = 250,000 × 1.50 = ₦375,000. Nominal value = ₦250,000 and premium = 250,000 × 0.50 = ₦125,000. After the rights issue: share capital 1,250,000 + 250,000 = 1,500,000; share premium 125,000; general reserve 50,000; retained earnings 300,000. Total = 1,500,000 + 125,000 + 50,000 + 300,000 = **₦1,975,000**. Check: 1,600,000 + 375,000 = ₦1,975,000.

8.5 The final accounts of a company

A company prepares the same trading and income statement as a sole trader, with changes in layout and with extra items. The income statement is continued below profit by a **profit and loss appropriation account** that shows how profit is shared out between the government, the shareholders and the reserves.

Items special to companies

- **Directors' remuneration** and **auditors' fees** are expenses charged before profit.
- **Debenture interest** is an expense. If only part has been paid, the unpaid part is an accrual shown among current liabilities.
- **Company income tax** is charged on profit. The estimated tax for the year is deducted from profit before tax to give profit after tax, and the unpaid amount is shown as a current liability.
- **Dividends** are appropriations of profit. An *interim* dividend is paid during the year; a *final* dividend is proposed after the year end. A proposed dividend not yet paid is shown as a current liability, so it appears in the balance sheet of the year to which the profit relates.
- **Transfers to reserves**, such as to the general reserve, are appropriations. They do not leave the business; they simply move profit from one part of shareholders' funds to another.
- **Retained earnings brought forward** from the previous year are added to the profit available for appropriation, and the unappropriated balance is carried forward.

> **Key point.** Dividend on shares is calculated on the *nominal value*, unless the question says otherwise. A 10 per cent dividend on ₦200,000 of preference shares is ₦20,000. A dividend stated in kobo per share is multiplied by the number of shares. Divide the total dividend by the number of shares to find the dividend per share.

Layout of the income statement and appropriation account

Line	Content
Revenue (sales)	Net of returns inward
Less: cost of sales	Opening inventory + purchases − closing inventory
Gross profit	Revenue − cost of sales
Less: operating expenses	Wages, administration, distribution, depreciation, directors' fees
Operating profit	Gross profit − operating expenses
Less: debenture interest	The full year's charge
Profit before tax	
Less: income tax	
Profit after tax (profit for the year)	
Add: retained earnings brought forward	
Less: dividends, transfers to reserves	Appropriations
Retained earnings carried forward	Shown in the balance sheet

Layout of the balance sheet

The balance sheet (statement of financial position) lists non-current assets first, at cost less accumulated depreciation, then current assets, then current liabilities, then non-current liabilities. Finally the **equity** section shows share capital and each reserve. The two halves must agree: net assets equal shareholders' funds.

WORKED EXAMPLE 8.4

The following balances were taken from the books of Ogunleye Industries Ltd, Abeokuta, at 31 December. Prepare the income statement with appropriation account for the year, and the balance sheet. Amounts are in naira.

Account	Debit ₦	Credit ₦
Ordinary share capital (₦1 shares)		1,000,000
10% preference share capital (₦1 shares)		200,000
Share premium		100,000
General reserve		80,000
Retained earnings at 1 January		150,000
12% debentures		300,000
Sales		3,500,000
Provision for depreciation: machinery		240,000
Creditors		260,000
Inventory at 1 January	250,000	
Purchases	2,000,000	
Wages	300,000	
Administration expenses	180,000	
Distribution costs	120,000	
Debenture interest paid	18,000	
Land and buildings at cost	1,200,000	
Machinery at cost	800,000	
Debtors	400,000	
Bank	532,000	
Interim ordinary dividend paid	30,000	
Totals	**5,830,000**	**5,830,000**

Adjustments: (i) closing inventory is ₦300,000; (ii) depreciate machinery at 10% of cost; (iii) debenture interest for the second half-year, ₦18,000, is unpaid; (iv) provide ₦250,000 for company income tax; (v) propose the full preference dividend and a final ordinary dividend of ₦60,000; (vi) transfer ₦20,000 to the general reserve.

Solution: First verify the trial balance. Debits: 250,000 + 2,000,000 + 300,000 + 180,000 + 120,000 + 18,000 + 1,200,000 + 800,000 + 400,000 + 532,000 + 30,000 = 5,830,000. Credits: 1,000,000 + 200,000 + 100,000 + 80,000 + 150,000 + 300,000 + 3,500,000 + 240,000 + 260,000 = 5,830,000. The totals agree.

Workings: Cost of sales = 250,000 + 2,000,000 − 300,000 = ₦1,950,000. Depreciation = 10% × 800,000 = ₦80,000. Full-year debenture interest = 12% × 300,000 = ₦36,000 (18,000 paid + 18,000 accrued). Preference dividend = 10% × 200,000 = ₦20,000.

Ogunleye Industries Ltd: Income statement for the year ended 31 December		
Item	**₦**	**₦**
Sales		3,500,000
Less: cost of sales (250,000 + 2,000,000 − 300,000)		1,950,000
Gross profit		**1,550,000**
Wages	300,000	
Administration expenses	180,000	
Distribution costs	120,000	
Depreciation: machinery	80,000	680,000
Operating profit		**870,000**
Less: debenture interest		36,000
Profit before tax		**834,000**
Less: income tax		250,000
Profit after tax		**584,000**
Add: retained earnings brought forward		150,000
		734,000
Less: interim ordinary dividend paid	30,000	
Preference dividend proposed	20,000	
Final ordinary dividend proposed	60,000	
Transfer to general reserve	20,000	130,000
Retained earnings carried forward		**604,000**

Check: 1,550,000 − 680,000 = 870,000; 870,000 − 36,000 = 834,000; 834,000 − 250,000 = 584,000; 584,000 + 150,000 = 734,000; 30,000 + 20,000 + 60,000 + 20,000 = 130,000; 734,000 − 130,000 = 604,000.

Balance sheet as at 31 December			
Item	**Cost ₦**	**Depreciation ₦**	**Net ₦**
Non-current assets			
Land and buildings	1,200,000	nil	1,200,000
Machinery	800,000	320,000	480,000
	2,000,000	320,000	**1,680,000**
Current assets			
Inventory			300,000
Debtors			400,000
Bank			532,000
			1,232,000
Current liabilities			
Creditors		260,000	
Debenture interest accrued		18,000	
Income tax payable		250,000	
Proposed dividends (20,000 + 60,000)		80,000	608,000
Net current assets (1,232,000 − 608,000)			**624,000**
Total assets less current liabilities			**2,304,000**
Non-current liability: 12% debentures			300,000
Net assets			**2,004,000**
Financed by (equity)			
Ordinary share capital			1,000,000
10% preference share capital			200,000
Share premium			100,000
General reserve (80,000 + 20,000)			100,000
Retained earnings			604,000
Shareholders' funds			**2,004,000**

The accumulated depreciation is 240,000 + 80,000 = ₦320,000. Net assets 2,304,000 − 300,000 = 2,004,000, which equals shareholders' funds: 1,000,000 + 200,000 + 100,000 + 100,000 + 604,000 = ₦2,004,000. The balance sheet balances.

8.6 Dividends: calculations

Dividends are usually examined as calculations. The preference dividend is calculated first, because it ranks before the ordinary dividend. The ordinary dividend is what the directors choose to propose, and it is usually expressed as a rate, as kobo per share, or as a total.

WORKED EXAMPLE 8.5

Dangote Brothers Stores Ltd has 600,000 ordinary shares of ₦1 and 200,000 8% preference shares of ₦1. Profit after tax is ₦210,000, and retained earnings brought forward are ₦40,000. The directors propose the preference dividend, an ordinary dividend of 15 kobo per share and a transfer of ₦30,000 to general reserve. Find the dividends, the retained earnings carried forward and the dividend as a percentage of the ordinary nominal capital.

Solution: Preference dividend = 8% × 200,000 = ₦16,000. Ordinary dividend = 600,000 × 0.15 = ₦90,000, which is 90,000 ÷ 600,000 = 15% of the nominal value. Profit available = 210,000 + 40,000 = ₦250,000. Appropriations = 16,000 + 90,000 + 30,000 = ₦136,000. Retained earnings carried forward = 250,000 − 136,000 = **₦114,000**.

Exam tip. Do not confuse the three uses of the word "interest" and "dividend". Debenture *interest* comes out of profit before tax; *tax* comes out before dividends; *dividends* and transfers to reserves come last. The order of the calculation is: profit before interest, interest, tax, dividends, transfers, balance carried forward.

Exercise 8.1

1. A company has an authorised capital of 5,000,000 ordinary shares of 50 kobo each and has issued 3,000,000 of them. Find the authorised capital and the issued capital in naira.
2. A company issues 400,000 ordinary shares of ₦1 at ₦1.25, payable in full on application, and all applications are accepted. Calculate the cash received, the share

capital and the share premium.

3. Calculate the annual dividend on ₦300,000 of 10% preference shares.
4. A company has ₦500,000 of 8% debentures. Find the annual interest and state where it appears in the final accounts.
5. A company with 900,000 ordinary shares of ₦1 makes a bonus issue of 1 share for every 3 held. How many bonus shares are issued, and what is the amount transferred from reserves?
6. A company has 800,000 ordinary shares of ₦1 and makes a rights issue of 1 share for every 4 held at ₦2.00 per share. Calculate the number of new shares, the cash raised and the amount credited to share premium.
7. A company offers 200,000 ordinary shares at ₦1.50, payable in full on application. It receives applications for 320,000 shares and allots 200,000. Find the refund and the cash retained.
8. Profit after tax is ₦400,000; retained earnings brought forward are ₦100,000. Appropriations are: preference dividend ₦24,000, ordinary dividend ₦150,000, transfer to general reserve ₦50,000. Find the retained earnings carried forward.
9. Equity is: share capital ₦1,000,000, share premium ₦150,000, general reserve ₦200,000 and retained earnings ₦350,000. Find the total shareholders' funds, and the share capital and funds after a bonus issue of 1 for 5 made wholly out of the general reserve.
10. A company has 600,000 ordinary shares of ₦1 and proposes an ordinary dividend of ₦90,000. Find the dividend per share in kobo and as a percentage.
11. Give three differences between a debenture holder and an ordinary shareholder.
12. A company is authorised to issue 1,000,000 shares of ₦1. It issues 800,000 shares and calls 75 kobo per share. Holders of 20,000 shares fail to pay the call. Find the called-up capital, the calls in arrear and the paid-up capital.

Answers to Exercise 8.1

1. Authorised: 5,000,000 × 0.50 = ₦2,500,000. Issued: 3,000,000 × 0.50 = ₦1,500,000.
2. Cash: 400,000 × 1.25 = ₦500,000. Share capital ₦400,000. Premium 400,000 × 0.25 = ₦100,000.
3. 10% × 300,000 = ₦30,000.

4. 8% × 500,000 = ₦40,000. It is an expense in the income statement, charged before profit before tax.
5. 900,000 ÷ 3 = 300,000 bonus shares; ₦300,000 is transferred from reserves to share capital.
6. 800,000 ÷ 4 = 200,000 new shares; cash 200,000 × 2.00 = ₦400,000; premium 200,000 × 1.00 = ₦200,000.
7. Refund (320,000 − 200,000) × 1.50 = 120,000 × 1.50 = ₦180,000. Retained: 200,000 × 1.50 = ₦300,000. Check: 320,000 × 1.50 = 480,000, and 480,000 − 180,000 = 300,000.
8. 400,000 + 100,000 = 500,000; deduct 24,000 + 150,000 + 50,000 = 224,000; carried forward ₦276,000.
9. Funds = 1,000,000 + 150,000 + 200,000 + 350,000 = ₦1,700,000. Bonus shares = 1,000,000 ÷ 5 = 200,000, nominal ₦200,000. Share capital becomes ₦1,200,000; general reserve falls to nil; total funds remain ₦1,700,000.
10. 90,000 ÷ 600,000 = ₦0.15, which is 15 kobo per share or 15%.
11. Any three: the debenture holder is a creditor, the shareholder an owner; the debenture holder receives fixed interest even in a loss; the shareholder's dividend varies and depends on profit; debenture interest is an expense but a dividend is an appropriation; the debenture holder is repaid first on winding up; the debenture holder has no vote.
12. Called-up: 800,000 × 0.75 = ₦600,000. Calls in arrear: 20,000 × 0.75 = ₦15,000. Paid-up: 600,000 − 15,000 = ₦585,000.

Chapter summary

- A company is a separate legal person with limited liability, perpetual succession and transferable shares; its owners are shareholders and its managers are directors.
- Authorised capital is the legal maximum; issued capital is the nominal value of shares issued; called-up capital is the amount demanded; paid-up capital is the amount actually received.
- Ordinary shareholders receive a variable dividend and carry the most risk; preference shareholders receive a fixed rate first; debenture holders are creditors who receive interest as an expense.
- Shares issued above nominal value create a share premium, a capital reserve; oversubscribed money for shares not allotted is refunded.
- Reserves are shareholders' claims, not cash. Capital reserves (share premium, revaluation surplus) cannot be paid as dividends; revenue reserves (general reserve, retained earnings) arise from profit.

- A bonus issue turns reserves into share capital without cash and leaves total funds unchanged; a rights issue raises new cash from existing shareholders.
- The appropriation account runs in this order: profit after interest and tax, retained earnings brought forward, dividends, transfers to reserves, retained earnings carried forward.
- In the balance sheet, tax payable and proposed dividends are current liabilities, debentures are non-current liabilities, and net assets equal shareholders' funds.

Chapter 9: Non-Profit Organisations and Manufacturing Accounts

Not every organisation exists to make a profit for its owners. Social clubs, sports clubs, old students' associations, town unions, churches and mosques, cooperative societies and charities exist to serve their members or the community. They still receive and spend money, own property and owe debts, so they must keep accounts. Because they have no owner who takes a share of profit, they do not prepare a trading and income statement in the ordinary way. Instead they prepare a **receipts and payments account**, an **income and expenditure account** and a **balance sheet**.

The second half of this chapter turns to a very different kind of business: the manufacturer. A trader buys goods and resells them; a manufacturer buys raw materials and converts them into finished goods. To find the true cost of what it makes, a manufacturer prepares a **manufacturing account** before the trading account. Both topics need careful classification of items, and both are favourite sources of UTME questions.

9.1 Features of non-profit organisations

A non-profit organisation is run for the benefit of its members or for a cause. Any excess of income over expenditure is called a **surplus**, not a profit, and a shortfall is a **deficit**, not a loss. The surplus is not shared among members. It is added to the club's capital, which is called the **accumulated fund**.

The ordinary sole trader's capital is calculated as assets minus liabilities. The accumulated fund is calculated in exactly the same way. When a club has kept no proper books and wishes to find its accumulated fund at the start of a year, it prepares a **statement of affairs**: it lists all assets and all liabilities, and the difference is the accumulated fund.

> **Key point.** Accumulated fund = total assets − total liabilities. Each year it increases by the surplus (or decreases by the deficit) shown in the income and expenditure account, and by items taken directly to capital such as entrance fees and life membership fees where the club's rules require.

9.2 The receipts and payments account

A small club often keeps only a cash book. At the year end the cash book is summarised as a **receipts and payments account**. It is simply a cash account, with receipts on the left (debit) and payments on the right (credit), beginning with the opening cash and bank balance and ending with the closing balance.

Receipts and payments account	Income and expenditure account
A summary of the cash book	A statement of income earned and expenses incurred for the year
Records cash actually received and paid	Uses the accruals principle: income earned and expenses incurred, whether paid or not
Includes capital receipts and payments, such as the purchase of equipment	Includes only revenue items, plus depreciation
Begins and ends with cash and bank balances	Ends with a surplus or deficit
Does not include depreciation or accruals	Includes depreciation, accruals and prepayments

9.3 The income and expenditure account

The **income and expenditure account** is the equivalent of the income statement. Expenditure is shown on the left and income on the right in the traditional form, or income is listed first and then expenditure is deducted in the modern form. To prepare it from a receipts and payments account, follow these rules.

1. **Remove capital items.** Purchase of equipment, the building of a pavilion and the repayment of a loan are not expenses. They appear in the balance sheet.
2. **Adjust for accruals and prepayments.** Expenses are brought to the amount belonging to the year: add expenses owing at the end, deduct those paid in advance.
3. **Adjust subscriptions** (see 9.4) to the amount earned during the year.
4. **Add depreciation** of equipment and fittings as an expense.
5. **Show the profit or loss on special activities** such as the bar, the canteen or the annual dinner, as a single net figure.
6. **Treat capital receipts correctly.** Entrance fees and donations may be taken to income or to the accumulated fund, according to the club's policy; in examination questions follow any instruction given, and otherwise treat entrance fees as income and special legacies or donations for buildings as capital.

9.4 Subscriptions

Subscriptions are the main income of most clubs, and adjustments to them are the commonest source of errors. The amount credited to the income and expenditure account is the subscription *earned* for the year. It is found by adjusting the cash received.

- **Arrears** are subscriptions due for the year (or earlier years) but not yet paid. At the end of the year they are a current asset. Arrears at the start have been earned in a previous year, so they are deducted from this year's cash receipts.
- **Advance (prepaid) subscriptions** have been received for a future period. They are a current liability. Advance subscriptions brought forward were received last year, so they are added to this year's cash receipts.

Step	Effect on subscriptions income
Cash received in the year	Starting figure
Less: arrears at the beginning	Deduct (earned last year)
Add: arrears at the end	Add (earned this year, not yet received)
Add: advance at the beginning	Add (received last year for this year)
Less: advance at the end	Deduct (relates to next year)

The same result is obtained from a **subscriptions account**: opening arrears and closing advance on the debit side, opening advance, cash received and closing arrears on the credit side, and the balancing figure transferred to the income and expenditure account. The method is the same as the accrued and prepaid expenses of Chapter 5.

Entrance fees, life membership and donations

An **entrance fee** is paid once by a new member. Many clubs treat it as income of the year; some take it to the accumulated fund because it is not a recurring receipt. A **life membership fee** is a single payment that frees a member from paying subscriptions for the rest of his life. Because it pays for many years, it is credited to a **life membership account** (a liability or a fund) and a fixed fraction, such as one-tenth, is transferred to income each year. **Donations** for general use are income; a donation for a specific capital purpose, such as building a clubhouse, is credited to a fund.

Trading activities

When a club runs a bar, a canteen or a social event, it prepares a small trading account for it. For a bar: sales less cost of sales (opening inventory plus purchases less closing inventory) gives the **bar profit**, and bar wages and expenses are deducted if they are separately known. Only the net result is placed in the income and expenditure account. For an event such as an annual dinner, income from ticket sales less the cost of the catering gives a profit or loss on the dinner.

WORKED EXAMPLE 9.1

The following is the receipts and payments account of the Ikeja Youth Sports Club for the year ended 31 December. Amounts are in naira.

Receipts		Payments	
Details	**₦**	**Details**	**₦**
Balance b/d (cash and bank)	40,000	Bar purchases	110,000
Subscriptions	310,000	Wages	90,000
Entrance fees	20,000	Rent	60,000
Donations	50,000	Equipment purchased	100,000
Bar sales	180,000	Printing	15,000
		Balance c/d	225,000
Total	**600,000**	**Total**	**600,000**

Additional information: (a) subscriptions in arrears were ₦20,000 on 1 January and ₦30,000 on 31 December; subscriptions paid in advance were ₦10,000 on 1 January and ₦15,000 on 31 December; (b) bar inventory was ₦12,000 on 1 January and ₦18,000 on 31 December; (c) rent owing at 31 December was ₦5,000; (d) equipment had a book value of ₦200,000 on 1 January, and is to be depreciated at 10% per year on its book value at the year end before depreciation (including the new purchase). Prepare the income and expenditure account, a statement of the accumulated fund at 1 January, and the balance sheet at 31 December.

Solution: **Subscriptions earned** = 310,000 − 20,000 + 30,000 + 10,000 − 15,000 = ₦315,000. Check with the subscriptions account: debit side 20,000 (arrears b/d) + 315,000 (income and expenditure) + 15,000 (advance c/d) = 350,000; credit side 10,000 (advance b/d) + 310,000 (cash) + 30,000 (arrears c/d) = 350,000.

Bar profit: cost of sales = 12,000 + 110,000 − 18,000 = ₦104,000; bar profit = 180,000 − 104,000 = ₦76,000. **Rent** for the year = 60,000 + 5,000 = ₦65,000. **Depreciation** = 10% × (200,000 + 100,000) = ₦30,000.

Ikeja Youth Sports Club: Income and expenditure account for the year ended 31 December		
Item	**₦**	**₦**
Income		
Subscriptions	315,000	
Entrance fees	20,000	
Donations	50,000	
Profit on bar	76,000	461,000
Expenditure		
Wages	90,000	
Rent	65,000	
Printing	15,000	
Depreciation of equipment	30,000	200,000
Surplus of income over expenditure		**261,000**

Accumulated fund at 1 January: assets were cash and bank 40,000 + equipment 200,000 + subscription arrears 20,000 + bar inventory 12,000 = ₦272,000. The only liability was subscriptions received in advance of ₦10,000. Accumulated fund = 272,000 − 10,000 = ₦262,000.

Balance sheet as at 31 December		
Item	**₦**	**₦**
Non-current asset		
Equipment (200,000 + 100,000 − 30,000)		270,000
Current assets		
Bar inventory	18,000	
Subscriptions in arrears	30,000	
Cash and bank	225,000	273,000
Total assets		**543,000**
Current liabilities		
Subscriptions in advance	15,000	
Rent owing	5,000	20,000
Net assets		**523,000**
Accumulated fund		
Balance at 1 January	262,000	
Add: surplus for the year	261,000	523,000

The balance sheet balances at ₦523,000.

Exam tip. Never include the purchase of equipment, or the full amount paid for a building, in the income and expenditure account. Only depreciation goes there. Also, a bar or canteen appears as a net profit, not as separate sales and purchases, in the income and expenditure account. Remember that opening arrears of subscriptions are a *deduction*, closing arrears an *addition*, and the opposite for subscriptions in advance.

9.5 Why manufacturers need a manufacturing account

A trader’s cost of sales is simply opening inventory plus purchases minus closing inventory. A manufacturer does not buy finished goods; it buys timber, flour, cloth or steel, pays workers to process them, and uses machinery and power. The cost of the finished goods is the sum of all

these things, and it must be calculated in a separate **manufacturing account**. The balance of this account, the **cost of production** (or factory cost of goods completed), then takes the place of purchases in the trading account.

9.6 Classifying manufacturing costs

Class	Meaning	Examples
Direct materials	Materials that become part of the product and can be traced to it	Timber in a chair, flour in bread, cloth in a shirt
Direct labour	Wages of workers who make the product	Wages of carpenters or machine operators
Direct expenses	Other expenses traceable to a specific product	Royalties per unit, hire of a special tool for one job
Factory overheads (indirect costs)	Factory costs that cannot be traced to units	Factory supervisor's salary, factory rent, power, indirect materials such as lubricants, depreciation of machinery
Non-factory expenses	Costs outside production	Office salaries, sales commission, delivery to customers

Prime cost is the total of direct materials, direct labour and direct expenses. **Factory overheads** are added to prime cost. A manufacturer usually has **work in progress**: goods that have been started but are not finished at the year end. The opening work in progress is added and the closing work in progress is deducted, to find the cost of goods *completed* in the year.

> **Key point.** Cost of raw materials used = opening inventory of raw materials + purchases + carriage inwards − closing inventory of raw materials. Prime cost = raw materials used + direct labour + direct expenses. Cost of production = prime cost + factory overheads + opening work in progress − closing work in progress.

9.7 The manufacturing account

The manufacturing account is prepared in the same period as the trading account. Its final figure, cost of production, is carried to the trading account. There, the cost of sales becomes: opening inventory of finished goods + cost of production − closing inventory of finished goods. Depreciation of factory machinery and factory rent are charged in the manufacturing account;

depreciation of office equipment and the manager’s salary are charged in the income statement. Where an expense such as rent or electricity serves both factory and office, it is divided in the ratio given in the question.

WORKED EXAMPLE 9.2

The following information relates to Adebisi Furniture Works, Ibadan, for the year ended 31 December. Prepare the manufacturing account, the trading and income statement, and find the cost per unit if 5,125 chairs were completed.

Raw materials inventory: 1 January ₦40,000; 31 December ₦50,000. Purchases of raw materials ₦520,000. Carriage inwards ₦10,000. Direct wages ₦300,000. Indirect factory wages ₦80,000. Factory rent and rates ₦60,000. Factory power ₦45,000. Depreciation of factory machinery ₦30,000. Work in progress: 1 January ₦25,000; 31 December ₦35,000. Finished goods inventory: 1 January ₦60,000; 31 December ₦80,000. Sales ₦1,500,000. Administration expenses ₦120,000. Selling and distribution expenses ₦90,000.

Solution: Raw materials used = 40,000 + 520,000 + 10,000 − 50,000 = ₦520,000. Prime cost = 520,000 + 300,000 = ₦820,000. Factory overheads = 80,000 + 60,000 + 45,000 + 30,000 = ₦215,000.

Adebisi Furniture Works: Manufacturing account for the year ended 31 December		
Item	**₦**	**₦**
Raw materials: opening inventory	40,000	
Add: purchases	520,000	
Carriage inwards	10,000	
	570,000	
Less: closing inventory	50,000	
Raw materials used		520,000
Direct wages		300,000
Prime cost		**820,000**
Factory overheads: indirect wages	80,000	
Rent and rates	60,000	
Power	45,000	
Depreciation of machinery	30,000	215,000
		1,035,000
Add: opening work in progress		25,000
		1,060,000
Less: closing work in progress		35,000
Cost of production		**1,025,000**

Trading and income statement for the year ended 31 December		
Item	**₦**	**₦**
Sales		1,500,000
Opening inventory of finished goods	60,000	
Add: cost of production	1,025,000	
	1,085,000	
Less: closing inventory of finished goods	80,000	
Cost of sales		1,005,000
Gross profit		**495,000**
Administration expenses	120,000	
Selling and distribution expenses	90,000	210,000
Net profit		**285,000**

Cost per unit = 1,025,000 ÷ 5,125 = **₦200** per chair (5,125 × 200 = 1,025,000). In the balance sheet, the closing inventories appear under current assets as raw materials ₦50,000, work in progress ₦35,000 and finished goods ₦80,000, a total of ₦165,000.

WORKED EXAMPLE 9.3

For a certain year a factory has prime cost of ₦600,000, factory overheads of ₦150,000, opening work in progress of ₦20,000 and a cost of production of ₦740,000. Find the closing work in progress. If the factory made 3,700 units, find the unit cost.

Solution: Cost of production = prime cost + overheads + opening WIP − closing WIP. So 740,000 = 600,000 + 150,000 + 20,000 − closing WIP = 770,000 − closing WIP. Closing WIP = 770,000 − 740,000 = **₦30,000**. Unit cost = 740,000 ÷ 3,700 = **₦200**.

Manufacturing profit (beyond the basic course)

Some firms transfer the goods from the factory to the trading account at a **transfer price** above cost, to see whether the factory is efficient compared with buying from outside suppliers. The difference between the transfer price and the cost of production is the **manufacturing profit**. In Worked example 9.2, if goods are transferred at cost plus 20%, the transfer price is 1,025,000 × 1.20 = ₦1,230,000, and the manufacturing profit is 1,230,000 − 1,025,000 = ₦205,000. Because this profit is not yet earned on finished goods still in stock, a **provision for unrealised profit** is

made on closing inventory valued at the transfer price: if closing finished goods stand at ₦96,000 at transfer price, the profit included is 96,000 × 20/120 = ₦16,000, and this is deducted from gross profit and from the inventory in the balance sheet (96,000 − 16,000 = ₦80,000 at cost).

Exercise 9.1

1. State two differences between a receipts and payments account and an income and expenditure account.
2. A club has the following assets and liabilities at 1 January: cash ₦30,000; premises ₦400,000; equipment ₦100,000; subscriptions in arrears ₦10,000; creditors ₦25,000; subscriptions in advance ₦5,000. Calculate the accumulated fund.
3. A club received ₦240,000 in subscriptions during the year. Arrears were ₦15,000 at the start and ₦20,000 at the end; advance subscriptions were ₦8,000 at the start and ₦12,000 at the end. Calculate the subscriptions income for the year.
4. A club bar had sales of ₦90,000, opening inventory ₦6,000, purchases ₦55,000 and closing inventory ₦9,000. Calculate the bar profit.
5. Tickets for the annual dinner raised ₦75,000, and the dinner cost ₦82,000. State the profit or loss.
6. A club's income was ₦310,000 and its expenditure ₦265,000. If the accumulated fund at the start was ₦400,000, find the surplus and the new accumulated fund.
7. A club received entrance fees of ₦30,000 and its policy is to capitalise them. Explain where the fees are recorded.
8. Raw materials: opening inventory ₦15,000, purchases ₦120,000, closing inventory ₦20,000. Direct wages ₦90,000. Calculate the raw materials used and the prime cost.
9. Prime cost is ₦205,000, factory overheads ₦55,000, opening work in progress ₦10,000 and closing work in progress ₦14,000. Find the cost of production.
10. If 640 units were made at the cost of production in Question 9, find the cost per unit.
11. Classify as direct cost, factory overhead or non-factory expense: (a) wages of machine operators; (b) salary of a factory supervisor; (c) electricity used by factory machines; (d) commission to salesmen.
12. Opening finished goods were ₦30,000, the cost of production was ₦256,000 and closing finished goods were ₦36,000. Sales were ₦340,000. Calculate the cost of sales and the gross profit.

Answers to Exercise 9.1

1. Any two: the first is a summary of the cash book while the second uses accruals; the first includes capital receipts and payments but the second only revenue items; the first has no depreciation while the second does; the first ends with a cash balance, the second with a surplus or deficit.
2. Assets = 30,000 + 400,000 + 100,000 + 10,000 = 540,000. Liabilities = 25,000 + 5,000 = 30,000. Accumulated fund = ₦510,000.
3. 240,000 − 15,000 + 20,000 + 8,000 − 12,000 = ₦241,000.
4. Cost of sales = 6,000 + 55,000 − 9,000 = 52,000. Profit = 90,000 − 52,000 = ₦38,000.
5. 75,000 − 82,000 = ₦7,000 loss.
6. Surplus = 310,000 − 265,000 = ₦45,000. New accumulated fund = 400,000 + 45,000 = ₦445,000.
7. It is credited directly to the accumulated fund (capital) instead of to the income and expenditure account; the cash is debited in the receipts and payments account.
8. Used = 15,000 + 120,000 − 20,000 = ₦115,000. Prime cost = 115,000 + 90,000 = ₦205,000.
9. 205,000 + 55,000 + 10,000 − 14,000 = ₦256,000.
10. 256,000 ÷ 640 = ₦400 per unit.
11. (a) direct cost (direct labour); (b) factory overhead; (c) factory overhead; (d) non-factory expense (selling expense).
12. Cost of sales = 30,000 + 256,000 − 36,000 = ₦250,000. Gross profit = 340,000 − 250,000 = ₦90,000.

Chapter summary

- Non-profit organisations record a surplus or deficit, not a profit or loss, and their capital is the accumulated fund (assets minus liabilities).
- A receipts and payments account is a summary of the cash book; an income and expenditure account applies accruals and shows only revenue items and depreciation.
- Subscriptions earned = cash received − opening arrears + closing arrears + opening advance − closing advance.
- Bars, canteens and dinners appear as a net profit or loss; capital items such as equipment purchases never appear in the income and expenditure account.
- Entrance fees may be capitalised or treated as income; life membership fees are spread over several years.

- A manufacturing account finds prime cost (direct materials, labour and expenses), adds factory overheads and adjusts for work in progress to give the cost of production.
- Cost of production replaces purchases in the trading account; cost of sales = opening finished goods + cost of production − closing finished goods.
- A transfer price above cost creates a manufacturing profit, and unrealised profit in closing inventory must be removed.

Chapter 10: Interpretation of Accounts and Incomplete Records

Preparing final accounts is only half of the accountant's task. A set of figures that nobody can understand helps no one, and the owner of a shop in Kano who learns that she made a net profit of ₦86,400 still wants to know whether that is good, whether it is better than last year, and whether the business can pay its bills. The first half of this chapter shows how to answer such questions with **accounting ratios**, which turn raw figures into comparisons. The second half deals with the opposite problem: the many small traders in markets across Nigeria who never keep a full double-entry system, so that the accountant must build the final accounts from **incomplete records**. Both topics draw on everything in earlier chapters, in particular the trading account, the profit and loss account and the balance sheet of Chapter 5, the control accounts of Chapter 6 and the accounting equation of Chapter 1.

10.1 Why accounts are interpreted

A single figure has little meaning on its own. A profit of ₦500,000 is excellent for a small tailoring business and disappointing for a large wholesaler. Interpretation therefore compares one figure with another: profit with sales, current assets with current liabilities, this year with last year, or one business with another of the same kind. The result of such a comparison, expressed as a percentage, a proportion (such as 2:1) or a number of times or days, is called a **ratio**.

Several groups use ratios. The *owner* wants to know whether the business is rewarding the capital invested in it. A *bank manager* deciding on a loan wants to know whether the business can pay its debts. A *supplier* offering credit wants to know whether the business will pay on time. *Investors* and the *tax authorities* also read the accounts. Because each group asks a different question, ratios are grouped into three families:

- **Profitability ratios** show how well the business earns profit from its sales and its capital.
- **Liquidity ratios** show whether the business can meet its short-term debts.
- **Efficiency ratios** show how well the business uses its inventory and manages its debtors and creditors.

Key point. A ratio is meaningful only when it is compared with something: the same business in earlier years, another business in the same trade, or an accepted standard. Always state the formula you are using, show the working, and then say what the answer means.

10.2 Profitability ratios

The **gross profit margin** expresses gross profit as a percentage of net sales:

Gross profit margin = (gross profit ÷ net sales) × 100.

The **net profit margin** does the same with net profit, which is gross profit less all expenses:

Net profit margin = (net profit ÷ net sales) × 100.

The two margins together tell a story. A healthy gross margin but a weak net margin means that expenses are too high. A weak gross margin points to selling prices that are too low, or to the cost of goods being too high, or to losses such as theft or wastage of inventory.

The **mark-up** is closely related, but it is based on cost and not on sales:

Mark-up = (gross profit ÷ cost of sales) × 100.

Students often confuse margin and mark-up, so remember the base: *margin is a percentage of selling price; mark-up is a percentage of cost price*. If goods cost ₦100 and are sold for ₦125, the gross profit is ₦25, the mark-up is 25% and the margin is 25/125 = 20%. A mark-up of 25% therefore corresponds to a margin of 20%, and the two are linked by cost + profit = sales.

The **return on capital employed** (ROCE) measures the profit earned on the long-term funds invested. For a sole trader, capital employed is usually the owner's capital (closing capital, or opening capital plus long-term loans):

ROCE = (net profit ÷ capital employed) × 100.

ROCE should be compared with the return available elsewhere. If the owner could earn 10% by simply placing the money in a safe investment, a ROCE of 6% suggests that the effort and risk of running the business are not being rewarded.

WORKED EXAMPLE 10.1

In the year ended 31 December 2025 Mrs Adaeze Obi, a provisions dealer in Onitsha, had net sales of ₦480,000 and cost of sales of ₦312,000. Her expenses were ₦96,000 and her capital employed was ₦600,000. Calculate the gross profit margin, the net profit margin, the mark-up and the return on capital employed.

Solution: Gross profit = 480,000 − 312,000 = ₦168,000. Gross profit margin = 168,000/480,000 × 100 = 35%. Net profit = 168,000 − 96,000 = ₦72,000. Net profit margin = 72,000/480,000 × 100 = 15%. Mark-up = 168,000/312,000 × 100 = 53.8% (to one decimal place). ROCE = 72,000/600,000 × 100 = 12%.

10.3 Liquidity ratios

A business can be profitable and still collapse because it cannot find cash to pay wages and suppliers on the due date. Liquidity ratios warn of this danger. In Chapter 5 you met **working capital**, which is current assets minus current liabilities. The two ratios below express the same idea as a proportion.

The **current ratio** (working capital ratio) is:

Current ratio = current assets : current liabilities.

A ratio of 2:1 means that the business has ₦2 of current assets for every ₦1 it owes in the short term. A ratio between 1.5:1 and 2:1 is traditionally considered comfortable, although the proper level depends on the trade. A supermarket, which receives cash from customers every day, can operate safely on a lower ratio than a manufacturer that waits for months to be paid.

Inventory is the least liquid current asset because it must first be sold, and perhaps sold on credit, before it becomes cash. The **liquid (acid test or quick) ratio** therefore leaves it out:

Liquid ratio = (current assets − closing inventory) : current liabilities.

A liquid ratio of about 1:1 is usually regarded as satisfactory, because it shows that the business could pay its short-term creditors from its debtors and cash alone, without having to sell any inventory.

Key point. A very high current ratio is not always good. It may mean that the business is holding too much idle cash, allowing customers too long to pay, or piling up unsold inventory. A very low ratio suggests the danger of *overtrading*, which means expanding sales faster than the funds available to finance them.

10.4 Efficiency ratios

Inventory turnover shows how many times the average inventory was sold and replaced during the year:

Inventory turnover = cost of sales ÷ average inventory, where average inventory = (opening inventory + closing inventory) ÷ 2.

Dividing 365 days by the turnover gives the average number of days that inventory is held. A fast turnover means that goods do not stay long on the shelves, so less money is tied up, and perishable goods such as tomatoes or bread must turn over very quickly. A slow turnover may indicate over-buying, goods that are out of fashion, or inventory that is damaged or obsolete.

The **trade receivables collection period** measures how long customers take to pay:

Collection period = (trade receivables ÷ credit sales) × 365 days.

The **trade payables payment period** measures how long the business takes to pay its suppliers:

Payment period = (trade payables ÷ credit purchases) × 365 days.

If credit sales or credit purchases are not given separately, the total sales or purchases may be used, and the assumption should be stated. A business generally prefers a short collection period and can safely accept a longer payment period, but a payment period that grows steadily may signal that the business is short of cash and is using its suppliers as an unwilling bank. It may also cost the business its early settlement discounts or its reputation.

WORKED EXAMPLE 10.2

The following relate to Mr Kayode Balogun, a hardware dealer in Ibadan, for 2025: cost of sales ₦540,000; opening inventory ₦80,000; closing inventory ₦90,000; credit sales ₦511,000; credit purchases ₦550,000. At 31 December 2025 his trade receivables were ₦70,000, bank balance ₦40,000 and trade payables ₦80,000, and he owed ₦20,000 for expenses. Calculate (a) the current ratio, (b) the liquid ratio, (c) the inventory turnover and the days of inventory held, (d) the collection period, and (e) the payment period.

Solution: Current assets = inventory 90,000 + receivables 70,000 + bank 40,000 = ₦200,000. Current liabilities = payables 80,000 + accrued expenses 20,000 = ₦100,000.

(a) Current ratio = 200,000 : 100,000 = 2:1.

(b) Liquid ratio = (200,000 − 90,000) : 100,000 = 110,000 : 100,000 = 1.1:1.

(c) Average inventory = (80,000 + 90,000) ÷ 2 = 85,000. Turnover = 540,000 ÷ 85,000 = 6.35 times (6.4 to one decimal place, or about 6 times). Days held = 365 ÷ 6.35 = 57.5 days, about 57 to 58 days.

(d) Collection period = 70,000/511,000 × 365 = 50 days (511,000 ÷ 365 = 1,400 of sales per day, and 70,000 ÷ 1,400 = 50).

(e) Payment period = 80,000/550,000 × 365 = 53.1 days, about 53 days.

Comment: the business is liquid and pays its suppliers a little more slowly (53 days) than its customers pay it (50 days), which is a comfortable position.

10.5 Interpreting a set of ratios

The skill the examiner rewards is not the arithmetic but the explanation of what changed and why it matters. A good answer first states the movement, then the likely cause, then the consequence for the business. The next example shows the method.

WORKED EXAMPLE 10.3

Mrs Fatima Sani runs a general store in Kano. Extracts from her accounts are given below.

Item	2024 (₦)	2025 (₦)
Net sales	800,000	960,000
Gross profit	240,000	249,600
Expenses	144,000	163,200
Net profit	96,000	86,400
Closing inventory	90,000	130,000
Current assets (including inventory)	180,000	210,000
Current liabilities	120,000	168,000

Calculate the gross profit margin, the net profit margin, the current ratio and the liquid ratio for each year, and comment on the performance of the business.

Solution:

Ratio	2024	2025
Gross profit margin	240,000/800,000 × 100 = 30%	249,600/960,000 × 100 = 26%
Net profit margin	96,000/800,000 × 100 = 12%	86,400/960,000 × 100 = 9%
Current ratio	180,000 : 120,000 = 1.5:1	210,000 : 168,000 = 1.25:1
Liquid ratio	(180,000 − 90,000) : 120,000 = 0.75:1	(210,000 − 130,000) : 168,000 = 0.48:1

Comment: Sales rose by 20% (160,000 ÷ 800,000), but net profit actually fell by ₦9,600. The reason is the fall in the gross margin from 30% to 26%: she probably cut selling prices to win more customers, or her suppliers raised their prices, or some goods were lost. Expenses grew more slowly than sales (18% of sales in 2024, 17% in 2025), so expense control was not the cause of the problem. Liquidity has weakened badly. The current ratio fell from 1.5:1 to 1.25:1 and the liquid ratio from 0.75:1 to 0.48:1, which means that she can no longer pay her short-term debts without selling inventory. Inventory rose by ₦40,000 (from 90,000 to 130,000), which suggests that she bought more than she could sell. She should review her pricing, reduce her purchases and collect any money owed to her.

10.6 Limitations of ratio analysis

Ratios are a guide to questions to be asked, not an answer in themselves. Their main limitations are these.

- They are based on historical cost and take no account of inflation, which in Nigeria can make last year's figures and this year's figures hard to compare.
- The balance sheet is drawn up on one day. If the date is just after a seasonal peak, such as after the Christmas trade, the inventory and receivables will not be typical of the year.
- Businesses use different accounting policies, for example different depreciation methods or inventory valuations, so ratios of two firms may not be comparable.
- Ratios ignore non-financial matters such as the quality of the staff, the reputation of the brand, or the state of the economy.
- Figures can be arranged by management to produce good-looking ratios, a practice called *window dressing*, for instance by paying creditors just before the year-end to improve the liquid ratio.

Exercise 10.1

1. A trader's net sales were ₦250,000 and her cost of sales ₦175,000. Calculate the gross profit margin and the mark-up.
2. A wholesaler adds a mark-up of 40% to the cost of all goods. Goods that cost ₦350,000 are sold in the year. Calculate the sales and the gross profit, and express the gross profit as a margin on sales.
3. At the year-end a business has inventory ₦48,000, trade receivables ₦36,000 and bank ₦16,000, and its only current liability is trade payables of ₦40,000. Calculate the current ratio and the liquid ratio.
4. A business made credit sales of ₦730,000 in the year and had trade receivables of ₦120,000 at the year-end. Calculate the collection period in days.
5. A trader's opening inventory was ₦40,000, purchases ₦260,000 and closing inventory ₦60,000. Calculate the cost of sales, the inventory turnover, and the average number of days for which inventory was held (to the nearest day).
6. A business earned a net profit of ₦54,000 on net sales of ₦540,000 with capital employed of ₦450,000. Calculate the net profit margin and the ROCE. Business A has a gross margin of 40% and an inventory turnover of 3 times a year, while Business B

has a gross margin of 15% and an inventory turnover of 20 times. Suggest, with a reason, what kind of goods each might sell.

Answers to Exercise 10.1

1. Gross profit = 250,000 − 175,000 = ₦75,000. Margin = 75,000/250,000 × 100 = 30%. Mark-up = 75,000/175,000 × 100 = 42.9%.
2. Gross profit = 40% × 350,000 = ₦140,000, so sales = 350,000 + 140,000 = ₦490,000. Margin = 140,000/490,000 × 100 = 28.6%.
3. Current assets = 48,000 + 36,000 + 16,000 = ₦100,000. Current ratio = 100,000 : 40,000 = 2.5:1. Liquid ratio = (100,000 − 48,000) : 40,000 = 52,000 : 40,000 = 1.3:1.
4. Collection period = 120,000/730,000 × 365 = 60 days.
5. Cost of sales = 40,000 + 260,000 − 60,000 = ₦240,000. Average inventory = (40,000 + 60,000) ÷ 2 = 50,000. Turnover = 240,000 ÷ 50,000 = 4.8 times. Days = 365 ÷ 4.8 = 76 days.
6. Net margin = 54,000/540,000 × 100 = 10%. ROCE = 54,000/450,000 × 100 = 12%. Business A earns a high margin on goods that sell slowly, like furniture or jewellery; Business B earns a thin margin on goods that sell very quickly, like foodstuffs in a supermarket.

10.7 Incomplete records: the problem

Many small traders, farmers and artisans do not keep a full double-entry system. They may only have a cash box, a bank statement, a bundle of invoices and a notebook of what customers owe. Records of this kind are called **incomplete records** or **single-entry records**. This does not stop the owner needing to know the profit, nor stop the tax office, a bank or a prospective partner from asking for proper accounts. The accountant's task is to rebuild the missing figures from what is available. Two basic methods are used: the **statement of affairs** method, which gives only the profit, and the **reconstruction** method, which produces full trading, profit and loss accounts.

10.8 The statement of affairs and profit from the change in capital

The accounting equation states that capital = assets − liabilities. A list of all the assets and liabilities of a business on a particular date, from which the capital is deduced, is called a **statement of affairs**. It has the layout of a balance sheet, but because it is built from lists and not from a trial balance, the capital is a *balancing figure*, not a ledger balance.

If the statement of affairs is drawn up at the start and at the end of the year, the profit can be found without any trading account. The owner's capital increases when the business makes a profit and when the owner puts in more money, and decreases when the owner takes out drawings or when the business makes a loss. Therefore:

Net profit = closing capital − opening capital + drawings − additional capital introduced.

> **Key point.** Net profit = increase in capital + drawings − new capital introduced. Capital introduced during the year must be *deducted*, because it increases closing capital without being earned. Drawings must be *added back*, because they reduced closing capital although they were not an expense. Depreciation and other year-end adjustments should be put into the closing statement of affairs before the closing capital is found.

WORKED EXAMPLE 10.4

Mama Nkechi, a trader in Aba, keeps no proper books. On 1 January 2025 she had shop equipment ₦250,000, inventory ₦120,000, trade receivables ₦45,000 and cash ₦30,000, and she owed suppliers ₦60,000. On 31 December 2025 she had shop equipment ₦250,000 (before depreciation of ₦25,000 for the year), inventory ₦150,000, trade receivables ₦62,000 and bank ₦58,000, and she owed suppliers ₦41,000. During the year she took drawings of ₦96,000 and introduced ₦20,000 of additional capital. Find the net profit for the year.

Solution: Opening statement of affairs: assets = 250,000 + 120,000 + 45,000 + 30,000 = ₦445,000; liabilities = ₦60,000; opening capital = 445,000 − 60,000 = ₦385,000.

Closing statement of affairs: equipment after depreciation = 250,000 − 25,000 = 225,000; assets = 225,000 + 150,000 + 62,000 + 58,000 = ₦495,000; liabilities = ₦41,000; closing capital = 495,000 − 41,000 = ₦454,000.

Net profit = 454,000 − 385,000 + 96,000 − 20,000 = 69,000 + 96,000 − 20,000 = ₦145,000.

This method is quick, but it gives only one figure. It cannot show the gross profit, the expenses, or how much was sold, and these details are usually needed. For them we must reconstruct the accounts.

10.9 Reconstructing the trading and profit and loss accounts

The reconstruction method works by *finding the missing figures from the information that survives*. The usual steps are as follows.

1. Draw up the **opening statement of affairs** to find the opening capital.
2. Prepare a **summary of the cash and bank receipts and payments** for the year. Opening balances, closing balances and the receipts and payments themselves are known, and the summary is checked by balancing it.
3. Find **credit sales** using a trade receivables account (or total debtors account): opening receivables + credit sales − cash received from customers = closing receivables, so credit sales are the missing figure. Total sales are credit sales plus cash sales.
4. Find **credit purchases** in the same way with a trade payables account: opening payables + credit purchases − cash paid to suppliers = closing payables.

5. Find the **expense for the year** for each expense from the cash paid, adjusted for the amounts owing or prepaid at the start and the end of the year. Add depreciation.
6. Prepare the **trading account and profit and loss account** and the **balance sheet**. The new closing capital should equal opening capital + net profit − drawings. This is a useful check on the arithmetic.

Sometimes the figures for sales or purchases are not available at all, but the trader knows the usual profit percentage. Two relationships are then very useful. A **mark-up** on cost of x% means that sales = cost of sales × (100 + x)/100. A **margin** of y% on sales means that cost of sales = sales × (100 − y)/100. They allow a missing item in the trading account, such as closing inventory lost in a fire or by theft, to be estimated, because the expected closing inventory = opening inventory + purchases − cost of sales.

WORKED EXAMPLE 10.5

Mr Chukwudi Eze, a trader in Enugu, does not keep a ledger. On 1 January 2025 his assets and liabilities were: fittings ₦100,000; inventory ₦60,000; trade receivables ₦40,000; cash and bank ₦30,000; trade payables ₦30,000. During 2025 he received ₦520,000 from credit customers and banked ₦140,000 of cash sales. He paid suppliers ₦380,000, wages ₦62,000, rent ₦24,000 and other expenses ₦18,000, and took drawings of ₦120,000. At 31 December 2025 trade receivables were ₦55,000, trade payables ₦35,000 and inventory ₦70,000; wages of ₦4,000 were owing, and the fittings are to be depreciated by 10% of their opening value (₦10,000). Prepare the trading and profit and loss account for 2025 and the balance sheet at 31 December 2025.

Solution:

Step 1. Opening capital = assets (100,000 + 60,000 + 40,000 + 30,000 = 230,000) − liabilities 30,000 = ₦200,000.

Step 2. Cash and bank summary: opening balance 30,000; receipts 520,000 + 140,000 = 660,000; payments 380,000 + 62,000 + 24,000 + 18,000 + 120,000 = 604,000. Closing balance = 30,000 + 660,000 − 604,000 = ₦86,000.

Step 3. Credit sales: 40,000 + credit sales − 520,000 = 55,000, so credit sales = 55,000 + 520,000 − 40,000 = ₦535,000. Total sales = 535,000 + 140,000 = ₦675,000.

Step 4. Purchases: 30,000 + purchases − 380,000 = 35,000, so purchases = 35,000 + 380,000 − 30,000 = ₦385,000.

Step 5. Wages for the year = 62,000 paid + 4,000 owing = ₦66,000.

Step 6. Final accounts:

Trading and profit and loss account for the year ended 31 December 2025	₦	₦
Sales		675,000
Opening inventory	60,000	
Purchases	385,000	
Goods available for sale	445,000	
Less closing inventory	(70,000)	
Cost of goods sold		(375,000)
Gross profit		**300,000**
Wages	66,000	
Rent	24,000	
Other expenses	18,000	
Depreciation of fittings	10,000	
Total expenses		(118,000)
Net profit		**182,000**

Balance sheet at 31 December 2025	₦	₦
Fittings (100,000 − 10,000)		90,000
Inventory	70,000	
Trade receivables	55,000	
Cash and bank	86,000	
Current assets		211,000
Trade payables	(35,000)	
Accrued wages	(4,000)	
Current liabilities		(39,000)
Net assets		**262,000**
Opening capital		200,000
Add net profit		182,000
Less drawings		(120,000)
Closing capital		**262,000**

Check: net assets of ₦262,000 equal closing capital of ₦262,000, so the reconstruction is consistent. The gross margin is 300,000/675,000 = 44.4%, and the net margin is 182,000/675,000 = 27.0%.

WORKED EXAMPLE 10.6

A fire at the store of Mr Tunde Adebayo in Lagos destroyed part of his inventory. The records show that he began the year with inventory of ₦90,000, made purchases of ₦600,000, and made sales of ₦800,000 to the date of the fire. He always sells at a mark-up of 25% on cost. Find the cost of the inventory destroyed if goods worth ₦20,000 at cost were saved.

Solution: With a mark-up of 25%, sales = 125% of cost, so cost of sales = 800,000 × 100/125 = ₦640,000. (Equivalently, the margin is 20%, and cost of sales = 80% × 800,000 = 640,000.) Goods available for sale = 90,000 + 600,000 = ₦690,000. Inventory that should have been on hand before the fire = 690,000 − 640,000 = ₦50,000. Inventory saved = ₦20,000, so inventory destroyed = 50,000 − 20,000 = ₦30,000. This figure is the basis of a claim on the insurance company.

10.10 Practical points for incomplete-records questions

Several features of incomplete-records problems recur in examinations. First, *always begin with the opening capital*, because the opening statement of affairs is needed both for the balance sheet and for checking the answer. Second, *read each cash payment carefully*: a payment to suppliers is a payment for purchases, not an expense, and a payment by the owner for private use is a drawing. Third, goods taken by the owner for personal use are drawings of inventory, so they are deducted from purchases in the trading account and added to drawings in the capital account. Fourth, bad debts and discounts allowed must be inserted into the trade receivables account as credits, and discounts received into the trade payables account as debits, together with the cash, before the balancing figure is found. Finally, cheques and cash may be intermingled, so keep the cash and bank summary tidy and total it before making any decision.

Exercise 10.2

1. On 1 January a trader had assets of ₦380,000 and liabilities of ₦95,000. On 31 December her assets were ₦520,000 and her liabilities ₦110,000. She took drawings of

₦70,000 and introduced no new capital. Calculate her net profit for the year.

2. A business had opening capital of ₦150,000 and closing capital of ₦210,000. The owner withdrew ₦80,000 for private use and brought in ₦30,000 of additional capital during the year. Calculate the net profit.
3. (a) At the start of the year trade receivables were ₦24,000 and at the end ₦31,000. Customers paid ₦296,000 during the year, and no bad debts or discounts occurred. Calculate the credit sales. (b) Trade payables were ₦18,000 at the start and ₦22,000 at the end, and ₦205,000 was paid to suppliers. Calculate the credit purchases.
4. A trader sells all goods at a mark-up of 50% on cost. His opening inventory was ₦30,000, his purchases ₦210,000 and his closing inventory ₦40,000. Calculate the cost of sales, the sales and the gross profit.
5. Thieves broke into a store and removed inventory. Sales to the date of the theft were ₦400,000, at a margin of 25% on sales. Opening inventory was ₦50,000, purchases ₦320,000, and the inventory remaining after the theft was counted at cost as ₦20,000. Calculate the cost of the inventory stolen.

Answers to Exercise 10.2

1. Opening capital = 380,000 − 95,000 = 285,000. Closing capital = 520,000 − 110,000 = 410,000. Profit = 410,000 − 285,000 + 70,000 = ₦195,000.
2. Profit = 210,000 − 150,000 + 80,000 − 30,000 = 60,000 + 80,000 − 30,000 = ₦110,000.
3. (a) Credit sales = 31,000 + 296,000 − 24,000 = ₦303,000. (b) Credit purchases = 22,000 + 205,000 − 18,000 = ₦209,000.
4. Cost of sales = 30,000 + 210,000 − 40,000 = ₦200,000. Mark-up of 50% gives sales = 200,000 × 150/100 = ₦300,000. Gross profit = 300,000 − 200,000 = ₦100,000.
5. Cost of sales = 75% × 400,000 = ₦300,000. Inventory that should have been present = 50,000 + 320,000 − 300,000 = ₦70,000. Inventory stolen = 70,000 − 20,000 = ₦50,000.

Chapter summary

- Ratios compare one figure with another so that performance can be judged. They are useful only when compared with earlier years, other businesses or accepted standards.
- Profitability: gross margin = gross profit ÷ sales × 100; net margin = net profit ÷ sales × 100; mark-up = gross profit ÷ cost of sales × 100; ROCE = net profit ÷ capital

employed × 100.

- Margin is based on selling price and mark-up on cost, so a mark-up of 25% equals a margin of 20%.
- Liquidity: current ratio = current assets : current liabilities; liquid ratio = (current assets − inventory) : current liabilities. About 2:1 and 1:1 are traditionally satisfactory, but the standard depends on the trade.
- Efficiency: inventory turnover = cost of sales ÷ average inventory; collection period = receivables ÷ credit sales × 365; payment period = payables ÷ credit purchases × 365.
- Good interpretation states the change, the likely cause and its effect on the business. Ratios are limited by inflation, year-end timing, differing policies, non-financial factors and window dressing.
- Incomplete records are records kept without full double entry. A statement of affairs lists assets and liabilities and finds capital as the balancing figure.
- Net profit from changes in capital = closing capital − opening capital + drawings − capital introduced.
- To reconstruct the accounts, find the opening capital, summarise cash and bank, use receivables and payables accounts to find credit sales and credit purchases, adjust expenses for accruals and prepayments, and prepare final accounts; net assets must equal closing capital.
- When sales or inventory are unknown, mark-up or margin lets the cost of sales be found, and missing or stolen inventory = opening inventory + purchases − cost of sales − inventory remaining.